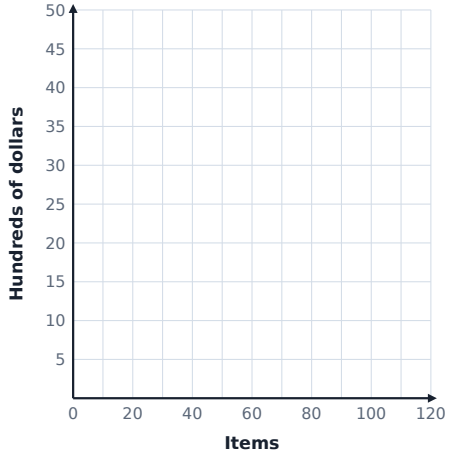
ANSWER KEY Reduced copy of the question page, with its answer

A company's cost is $C = 2{,}000 + 15x$ dollars and its revenue is $R = 40x$ dollars for x items.

Graph both equations for $0 \le x \le 120$ (the vertical axis is in hundreds of dollars), label each line, and mark the break-even point.

Break-even point: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q1

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q1/5

801

QUESTION ID

S22-F1-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: On the grid (in hundreds of dollars): the cost line from (0, 20) to (120, 38) and the revenue line from (0, 0) to (120, 48), labeled, meeting at (80, 32)

Written: (80, 3,200): 80 items, where cost and revenue are both $3,200

GRADING NOTE Grade both: the two labeled lines, and the break-even point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q1

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q1/5

A theater sells tickets at p dollars each. The number of tickets sold is n = 600 − 20p.

Write an equation for the revenue R in terms of p.

R = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q2

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q2/5

802

QUESTION ID

S22-F1-Q2

TYPE

Short answer

CORRECT ANSWER

p(600 − 20p)

GRADING NOTE 600p − 20p² is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q2

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q2/5

The revenue is $R = p(600 - 20p)$ dollars for a ticket price of p dollars.

Which price gives the greatest revenue?

A. $15

B. $20

C. $30

D. $600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q3

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q3/5

803

QUESTION ID

S22-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q3

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q3/5

True or false?

At the break-even point of C = 2,000 + 15x and R = 40x, cost and revenue are both $3,200.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q4

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q4/5

804

QUESTION ID

S22-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q4

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q4/5

True or false?

$R = p(600 - 20p)$ is a linear equation in p.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q5

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q5/5

805

QUESTION ID

S22-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has a p^2 term, so it is quadratic.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q5

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q5/5

SET 22 • 8.EE.C.8.c

FORWARD 2

Represent constraints with a system of inequalities • High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3

A bakery has 30 pounds of flour. Each batch of bread uses 2 pounds and each batch of muffins uses 0.5 pound.

Write an inequality for b batches of bread and m batches of muffins.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q1

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q1/5

807

QUESTION ID

S22-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$2b + 0.5m \leq 30$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q1

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q1/5

A bakery has at most 20 hours of oven time. Each batch of bread takes 1 hour and each batch of muffins takes 2 hours.

Write an inequality for b batches of bread and m batches of muffins.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q2

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q2/5

808

QUESTION ID

S22-F2-Q2

TYPE

Short answer

CORRECT ANSWER

b + 2m ≤ 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q2

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q2/5

A bakery's limits are $2b + 0.5m \leq 30$ (flour) and $b + 2m \leq 20$ (oven hours).

Can it make 10 batches of bread and 5 batches of muffins?

A. No; both limits are broken.

B. No; the flour runs out.

C. Yes; both limits are met.

D. No; the oven time runs out.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q3

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q3/5

809

QUESTION ID

S22-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Yes; both limits are met.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q3

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q3/5

True or false?

In a model of batches of bread b and muffins m, the constraints $b \geq 0$ and $m \geq 0$ make sense because batches cannot be negative.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q4

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q4/5

QUESTION ID

S22-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q4

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q4/5

True or false?

The point (12, 6) meets both limits $2b + 0.5m \leq 30$ and $b + 2m \leq 20$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q5

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q5/5

811

QUESTION ID
S22-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $12 + 2(6) = 24$, which is more than 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q5

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q5/5

SET 23 • 8.F.A.1 • FUNCTIONS

Decide whether a relation is a function

MAIN	Grade 8 main question	**8.F.A.1**
BACKWARD 1	Find the output of a rule	**6.EE.C.9**
BACKWARD 2	Read points in the first quadrant	**5.G.A.2**
FORWARD 1	Use function notation and find the domain and range	**HSF-IF.A.1**
FORWARD 2	Find the inverse of a linear function	**HSF-BF.B.4.a**

SET 23 • 8.F.A.1

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.A.1

True or false?

The table shows y as a function of x.

x	1	2	3	4
y	5	5	7	9

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q1

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q1/5

814

QUESTION ID

S23-M-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q1

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q1/5

Is the relation in the table a function? Explain.

x	2	3	3	5
y	4	6	8	10

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q2

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q2/5

815

QUESTION ID

S23-M-Q2

TYPE

Short answer

CORRECT ANSWER

No. The input 3 has two different outputs (6 and 8).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q2

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q2/5

Which set of ordered pairs is a function?

A. (5, 1), (6, 2), (5, 3)

B. (0, 0), (0, 1), (1, 1)

C. (1, 2), (1, 3), (2, 4)

D. (1, 4), (2, 4), (3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q3

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q3/5

QUESTION ID

S23-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (1, 4), (2, 4), (3, 4)

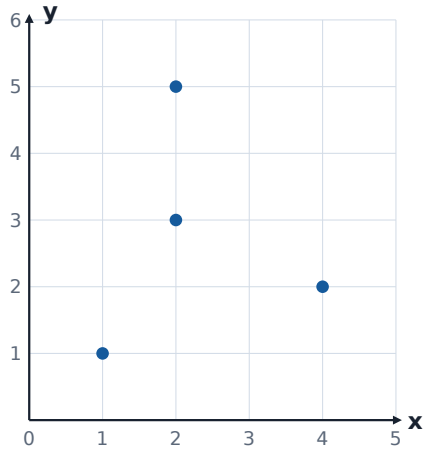
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q3

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q3/5

Do the points on the graph show y as a function of x? Explain.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q4

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q4/5

817

QUESTION ID

S23-M-Q4

TYPE

Short answer

CORRECT ANSWER

No. The input x = 2 has two outputs (3 and 5).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q4

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q4/5

True or false?

A rule takes a birth month as its input and gives one student born in that month as its output. Ana and Ben were both born in January. This rule is a function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q5

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q5/5

818

QUESTION ID

S23-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The input January is paired with two different outputs, Ana and Ben, so the rule is not a function.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q5

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q5/5

SET 23 • 8.F.A.1

BACKWARD 1

Find the output of a rule • Grade 6 • Expressions & Equations • 6.EE.C.9

The rule is y = 2x + 1.

What is y when x = 4?

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q1/5

820

QUESTION ID

S23-B1-Q1

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q1/5

The table follows a pattern.

What is y when x = 4?

x	1	2	3
y	4	7	10

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q2/5

821

QUESTION ID

S23-B1-Q2

TYPE

Short answer

CORRECT ANSWER

13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q2/5

The rule is: output = input × 3.

What is the output for an input of 7?

A. 4

B. 73

C. 21

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q3/5

822

QUESTION ID

S23-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q3/5

True or false?

For $y = x - 5$, an input of $x = 9$ gives $y = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q4/5

823

QUESTION ID

S23-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q4/5

True or false?

For y = 4x, an input of x = 3 gives y = 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q5/5

824

QUESTION ID

S23-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

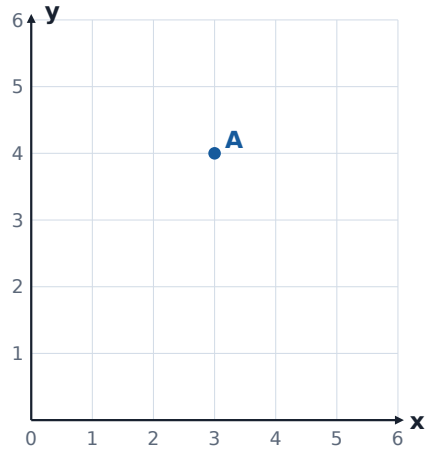
Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q5/5

SET 23 • 8.F.A.1

BACKWARD 2

Read points in the first quadrant • Grade 5 • Geometry • 5.G.A.2

What are the coordinates of point A?



Coordinates: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q1/5

826

QUESTION ID

S23-B2-Q1

TYPE

Short answer

CORRECT ANSWER

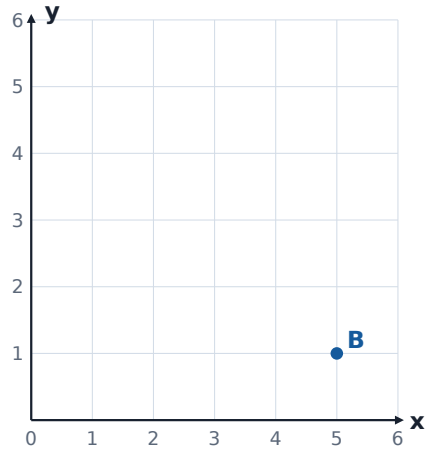
(3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q1/5

What are the coordinates of point B?



Coordinates: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q2/5

827

QUESTION ID

S23-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(5, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q2/5

Which point is 2 units to the right of the origin and 5 units up?

A. (5, 5)

B. (5, 2)

C. (2, 2)

D. (2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q3/5

828

QUESTION ID

S23-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q3/5

True or false?

The point (0, 3) is on the y-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q4/5

829

QUESTION ID

S23-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q4/5

True or false?

The point (4, 0) is 4 units up from the origin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q5/5

QUESTION ID

S23-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q5/5

SET 23 • 8.F.A.1

FORWARD 1

Use function notation and find the domain and range • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1

A function f is made of the pairs (1, 3), (2, 5), and (4, 9).

What is its domain?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q1/5

832

QUESTION ID

S23-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1, 2, and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q1/5

A function f is made of the pairs (1, 3), (2, 5), and (4, 9).

What is its range?

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q2/5

833

QUESTION ID

S23-F1-Q2

TYPE

Short answer

CORRECT ANSWER

3, 5, and 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q2/5

The output of f is 7 when the input is 2.

Which statement says this in function notation?

A. $f = 2 \times 7$

B. $f(7) = 2$

C. $2f = 7$

D. $f(2) = 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q3/5

834

QUESTION ID

S23-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. f(2) = 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q3/5

True or false?

If $f(x) = x^2$, then $f(-3) = 9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q4/5

835

QUESTION ID

S23-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q4/5

True or false?

A function can assign two different outputs to the same input.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q5/5

836

QUESTION ID

S23-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q5/5

SET 23 • 8.F.A.1

FORWARD 2

Find the inverse of a linear function • High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a

$f(x) = 2x + 3$

Write the inverse function.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q1/5

838

QUESTION ID

S23-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(x − 3)/2

GRADING NOTE $\frac{x}{2} - \frac{3}{2}$ is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F2-Q1**

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q1/5

$f(x) = 2x + 3$

Solve $f(x) = 11$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q2/5

839

QUESTION ID

S23-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q2/5

$g(x) = \frac{x}{5}$

What is the inverse function?

A. $g^{-1}(x) = x + 5$

B. $g^{-1}(x) = x - 5$

C. $g^{-1}(x) = 5x$

D. $g^{-1}(x) = \frac{5}{x}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q3/5

840

QUESTION ID

S23-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $g^{-1}(x) = 5x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q3/5

True or false?

If $f(2) = 7$ and f has an inverse, then $f^{-1}(7) = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q4/5

841

QUESTION ID

S23-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q4/5

True or false?

Every function has an inverse function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q5/5

842

QUESTION ID

S23-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $f(x) = x^2$ gives the same output for 2 and -2, so it has no inverse on all real numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q5/5

SET 24 • 8.F.A.1 • FUNCTIONS

Graph a function as the set of ordered pairs of inputs and outputs

MAIN	Grade 8 main question	**8.F.A.1**
BACKWARD 1	Make ordered pairs from an equation	**6.EE.C.9**
BACKWARD 2	Graph points from a pattern	**5.G.A.2**
FORWARD 1	Choose a domain that fits a function's situation	**HSF-IF.B.5**
FORWARD 2	Graph quadratic functions and show their key features	**HSF-IF.C.7.a**

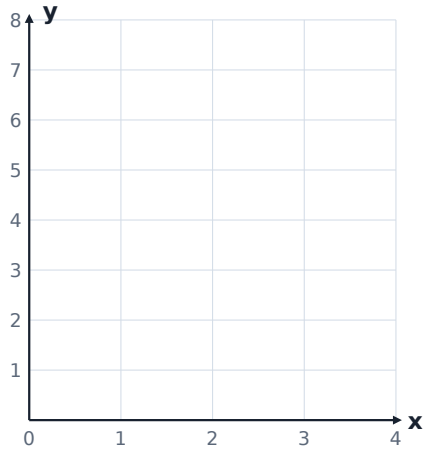
SET 24 • 8.F.A.1

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.A.1

The function y = 2x + 1 has the inputs 0, 1, 2, and 3.

Plot the ordered pairs (input, output) on the grid.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q1

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q1/5

845

QUESTION ID

S24-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Points at (0, 1), (1, 3), (2, 5), and (3, 7)

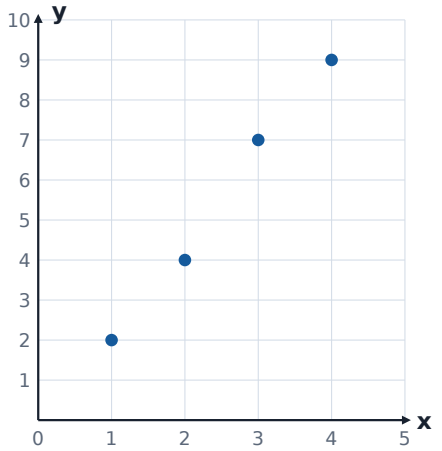
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q1

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q1/5

The points show a function.

What is the output when the input is 3?



Output: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q2

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q2/5

846

QUESTION ID

S24-M-Q2

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q2

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q2/5

The graph of a function contains the point (4, 10).

Which statement is true?

A. When the input is 4, the output is 10.

B. When the input is 10, the output is 4.

C. The graph has only one point.

D. The function adds 4 to every input.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q3

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q3/5

847

QUESTION ID

S24-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. When the input is 4, the output is 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q3

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q3/5

The rule is "output = input squared."

Write the ordered pairs (input, output) for the inputs −2, −1, 0, 1, and 2.

Ordered pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q4

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q4/5

848

QUESTION ID

S24-M-Q4

TYPE

Short answer

CORRECT ANSWER

(−2, 4), (−1, 1), (0, 0), (1, 1), (2, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q4

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q4/5

True or false?

The graph of a function can contain both (2, 5) and (2, 8).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q5

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q5/5

849

QUESTION ID

S24-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The input 2 cannot have two outputs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q5

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q5/5

SET 24 • 8.F.A.1

BACKWARD 1

Make ordered pairs from an equation • Grade 6 • Expressions & Equations • 6.EE.C.9

For y = 3x, what are the values of y when x = 0, 1, and 2?

Values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q1/5

851

QUESTION ID

S24-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0, 3, and 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

For y = x + 4, what is y when x = 5?

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q2/5

852

QUESTION ID

S24-B1-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q2/5

Which pair (x, y) fits y = 5x?

A. (2, 7)

B. (10, 2)

C. (2, 10)

D. (5, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q3/5

853

QUESTION ID

S24-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (2, 10)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q3/5

True or false?

The pair (3, 9) fits y = 3x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q4/5

854

QUESTION ID

S24-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q4/5

True or false?

The pair (4, 6) fits y = x + 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q5/5

855

QUESTION ID

S24-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q5

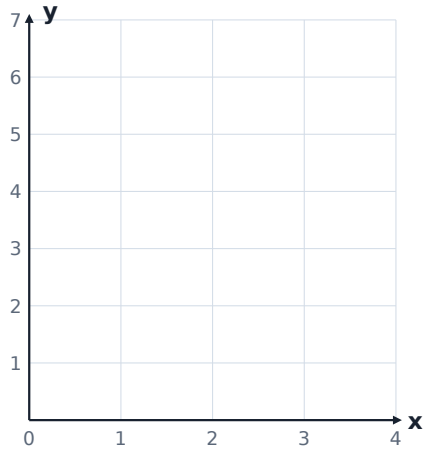
Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q5/5

SET 24 • 8.F.A.1

BACKWARD 2

Graph points from a pattern • Grade 5 • Geometry • 5.G.A.2

Plot the points (1, 2), (2, 4), and (3, 6) on the grid.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q1/5

857

QUESTION ID

S24-B2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Points at (1, 2), (2, 4), and (3, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q1/5

The y-values start at 1 and increase by 2: (1, 1), (2, 3), (3, 5), ...

What is the point for x = 4?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q2/5

858

QUESTION ID

S24-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(4, 7)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q2/5

Which point is 3 units to the right of the origin and 1 unit up?

A. (3, 1)

B. (3, 3)

C. (1, 3)

D. (1, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q3/5

859

QUESTION ID

S24-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (3, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q3/5

True or false?

The point (2, 6) is 2 units to the right of the origin and 6 units up.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q4/5

860

QUESTION ID

S24-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q4/5

True or false?

To plot (5, 2), you move 5 units up and 2 units to the right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q5/5

861

QUESTION ID

S24-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q5/5

SET 24 • 8.F.A.1

FORWARD 1

Choose a domain that fits a function's situation • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5

A function gives the cost of n concert tickets at $12 each. A buyer may buy at most 8 tickets.

What is a sensible domain?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q1/5

863

QUESTION ID

S24-F1-Q1

TYPE

Short answer

CORRECT ANSWER

The whole numbers 0 through 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q1/5

h(t) gives the height of a ball t seconds after it is thrown, until it lands at t = 3.

What is a sensible domain?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q2/5

QUESTION ID

S24-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$0 \leq t \leq 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q2/5

p is the number of people in a car that seats 5.

Which domain makes sense for p?

A. $-5 \leq p \leq 5$

B. 0, 1, 2, 3, 4, 5

C. All real numbers

D. $p \geq 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q3/5

865

QUESTION ID

S24-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0, 1, 2, 3, 4, 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q3/5

True or false?

For the area of a square as a function of its side length, the domain contains only positive numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q4/5

866

QUESTION ID

S24-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q4/5

True or false?

A function that gives the number of buses needed for n students should have all real numbers as its domain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q5/5

867

QUESTION ID

S24-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q5

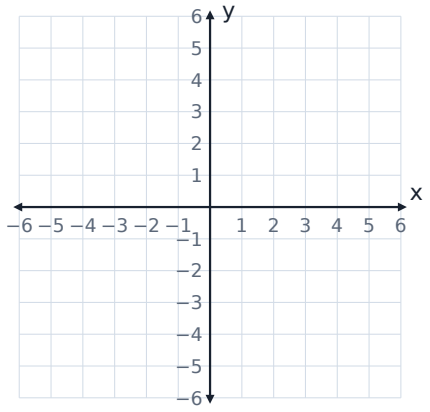
High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q5/5

SET 24 • 8.F.A.1

FORWARD 2

Graph quadratic functions and show their key features • High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a

Graph $y = x^2 - 4$ on the coordinate plane. Mark the vertex and the x-intercepts.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q1/5

869

QUESTION ID

S24-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A U-shaped parabola with its vertex at (0, −4), crossing the x-axis at (−2, 0) and (2, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q1/5

What is the vertex of the graph of $y = (x - 1)^2 + 2$?

Vertex: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q2/5

870

QUESTION ID

S24-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(1, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q2/5

Where does the graph of $y = x^2 - 9$ cross the x-axis?

A. At (3, 0) and (−3, 0)

B. At (9, 0) and (−9, 0)

C. At (0, −9)

D. At (3, 0) only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q3/5

871

QUESTION ID

S24-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. At (3, 0) and (−3, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q3/5

True or false?

The graph of $y = -x^2$ opens downward.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q4/5

872

QUESTION ID

S24-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q4/5

True or false?

The graph of $y = 2x + 3$ has a vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q5/5

873

QUESTION ID

S24-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q5/5

SET 25 • 8.F.A.2 • FUNCTIONS

Compare two functions represented in different ways

MAIN	Grade 8 main question	**8.F.A.2**
BACKWARD 1	Find a unit rate from a table, graph, or equation	**7.RP.A.2.b**
BACKWARD 2	Find the starting value of a relationship	**6.EE.C.9**
FORWARD 1	Compare average rates of change of linear and nonlinear function…	**HSF-IF.B.6**
FORWARD 2	Describe the effect of transformations on the graph of a function	**HSF-BF.B.3**

SET 25 • 8.F.A.2

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.A.2

f(x) = 3x + 2. The table shows g.

Which function has the greater rate of change?

x	0	2	4
g(x)	1	9	17

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q1

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q1/5

876

QUESTION ID

S25-M-Q1

TYPE

Short answer

CORRECT ANSWER

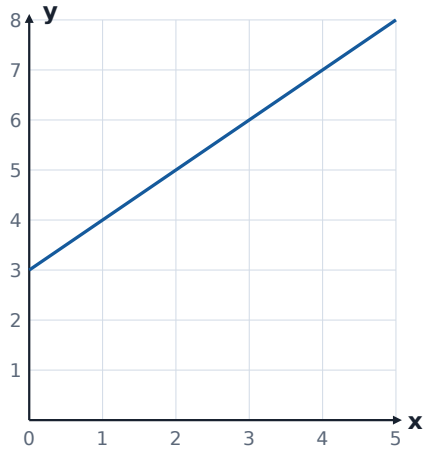
g (its rate is 4; f's is 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q1

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q1/5

The graph shows f.
$g(x) = -2x + 5$.

Which function has the greater initial value?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q2

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q2/5

877

QUESTION ID

S25-M-Q2

TYPE

Short answer

CORRECT ANSWER

g (5 is greater than 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q2

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q2/5

Function A is y = 0.5x + 10. Function B is a line through (0, 4) and (2, 8).

Which statement is true?

A. B has the greater rate of change; A has the greater initial value.

B. B has the greater rate of change and the greater initial value.

C. A has the greater rate of change and the greater initial value.

D. They have the same rate of change.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q3

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q3/5

878

QUESTION ID

S25-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. B has the greater rate of change; A has the greater initial value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q3

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q3/5

Car A travels d = 45t miles in t hours. The table shows Car B.

Which car is faster, and by how much?

Hours	1	2	3
Miles	50	100	150

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q4

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q4/5

879

QUESTION ID

S25-M-Q4

TYPE

Short answer

CORRECT ANSWER

Car B, by 5 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q4

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q4/5

True or false?

A function with x-values 0, 1, 2 and y-values 6, 4, 2 decreases faster than $y = -x + 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q5

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q5/5

880

QUESTION ID

S25-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q5

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q5/5

SET 25 • 8.F.A.2

BACKWARD 1

Find a unit rate from a table, graph, or equation • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

The table shows a proportional relationship.

What is the unit rate?

x	2	4	6
y	5	10	15

Unit rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q1/5

882

QUESTION ID

S25-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q1/5

What is the unit rate in y = 7x?

Unit rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q2/5

883

QUESTION ID

S25-B1-Q2

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q2/5

A proportional graph passes through (0, 0) and (3, 12).

What is the unit rate?

A. 36

B. 12

C. 3

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q3/5

884

QUESTION ID

S25-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q3/5

True or false?

A proportional table with the pairs (1, 9) and (2, 18) has a unit rate of 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q4/5

885

QUESTION ID

S25-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q4/5

True or false?

The unit rate in y = 0.5x is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q5/5

886

QUESTION ID

S25-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q5/5

SET 25 • 8.F.A.2

BACKWARD 2

Find the starting value of a relationship • Grade 6 • Expressions & Equations • 6.EE.C.9

For y = 4x + 7, what is y when x = 0?

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q1/5

888

QUESTION ID

S25-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q1/5

For the table, what is y when x = 0?

x	0	1	2
y	9	11	13

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q2/5

889

QUESTION ID

S25-B2-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q2/5

Which equation gives y = 5 when x = 0?

A. y = 5x + 2

B. y = 2x + 5

C. y = 5x

D. y = x − 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q3/5

890

QUESTION ID

S25-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. y = 2x + 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q3/5

True or false?

For y = 3x + 1, y = 1 when x = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q4/5

891

QUESTION ID

S25-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q4/5

True or false?

For y = 6x, y = 6 when x = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q5/5

892

QUESTION ID

S25-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q5/5

SET 25 • 8.F.A.2

FORWARD 1

Compare average rates of change of linear and nonlinear functions over intervals • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6

$f(x) = x^2$. The table shows g.

Find the average rate of change of each function from $x = 0$ to $x = 3$.

x	0	1	2	3
g(x)	0	3	6	9

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q1/5

894

QUESTION ID

S25-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Both are 3: f goes from 0 to 9 and g goes from 0 to 9 over 3 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q1/5

f(x) = 2^x and g(x) = 3x + 5.

Find the average rate of change of each function from x = 0 to x = 4. Which is greater?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q2/5

895

QUESTION ID

S25-F1-Q2

TYPE

Short answer

CORRECT ANSWER

f: (16 − 1) ÷ 4 = 3.75; g: 3. f has the greater average rate of change.

GRADING NOTE Both rates and the comparison are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q2/5

$f(x) = x^2$ and $g(x) = 4x$.

Which statement about the average rates of change from $x = 1$ to $x = 3$ is true?

A. f's rate is greater.

B. f's rate is 9 and g's rate is 12.

C. g's rate is greater.

D. They are equal (both are 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q3/5

896

QUESTION ID

S25-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. They are equal (both are 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q3/5

True or false?

The average rate of change of $f(x) = -x^2 + 4$ from $x = 0$ to $x = 2$ is -2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q4/5

897

QUESTION ID

S25-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q4/5

True or false?

The average rate of change of a linear function depends on which interval you use.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q5/5

898

QUESTION ID

S25-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A linear function has the same rate over every interval.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q5/5

SET 25 • 8.F.A.2

FORWARD 2

Describe the effect of transformations on the graph of a function • High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3

How does the graph of g(x) = f(x) + 3 compare with the graph of f?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q1/5

900

QUESTION ID

S25-F2-Q1

TYPE

Short answer

CORRECT ANSWER

It is the graph of f shifted up 3 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q1/5

How does the graph of $g(x) = (x - 2)^2$ compare with the graph of $y = x^2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q2/5

901

QUESTION ID

S25-F2-Q2

TYPE

Short answer

CORRECT ANSWER

It is shifted 2 units to the right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q2/5

Which function shifts the graph of y = |x| down 4 units?

A. y = |x| − 4

B. y = |x + 4|

C. y = 4|x|

D. y = |x − 4|

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q3/5

902

QUESTION ID

S25-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. y = |x| − 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q3/5

True or false?

The graph of $y = -f(x)$ is the reflection of the graph of $y = f(x)$ across the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q4/5

903

QUESTION ID

S25-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q4/5

True or false?

The graph of y = f(x + 5) is the graph of y = f(x) shifted 5 units to the right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q5/5

904

QUESTION ID

S25-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is shifted 5 units to the left.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q5/5

SET 26 • 8.F.A.3 • FUNCTIONS

Decide whether a function is linear or nonlinear

MAIN	Grade 8 main question	**8.F.A.3**
BACKWARD 1	Decide whether a table shows a proportional relationship	**7.RP.A.2.a**
BACKWARD 2	Find how much y changes for each change in x	**6.EE.C.9**
FORWARD 1	Show that linear functions grow by equal differences and expone..	**HSF-LE.A.1.a**
FORWARD 2	Use factoring to find the zeros and maximum of a quadratic functi..	**HSF-IF.C.8.a**

SET 26 • 8.F.A.3

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.A.3

Which equation is a linear function?

A. $y = 4x - 7$

B. $y = x^2 + 1$

C. $y = 2^x$

D. $y = \frac{3}{x}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q1

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q1/5

907

QUESTION ID

S26-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. y = 4x − 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q1

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q1/5

Is the function in the table linear? Explain.

x	0	1	2	3
y	1	2	5	10

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q2

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q2/5

908

QUESTION ID

S26-M-Q2

TYPE

Short answer

CORRECT ANSWER

No. y increases by 1, then 3, then 5, so the rate of change is not constant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q2

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q2/5

When n people each shake hands once with every other person, the number of handshakes is h = n(n − 1) ÷ 2.

Is h a linear function of n? Use the values for n = 2, 3, and 4 to support your answer.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q3

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q3/5

909

QUESTION ID

S26-M-Q3

TYPE

Short answer

CORRECT ANSWER

No. The values are 1, 3, and 6, which increase by 2 and then by 3, so the rate of change is not constant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q3

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q3/5

Is the function in the table linear? Explain.

x	0	2	4	6
y	7	4	1	−2

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q4

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q4/5

910

QUESTION ID

S26-M-Q4

TYPE

Short answer

CORRECT ANSWER

Yes. y decreases by 3 each time x increases by 2, a constant rate of change of −1.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q4

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q4/5

True or false?

y = 3(x + 2) is a linear function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q5

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q5/5

911

QUESTION ID

S26-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q5

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q5/5

SET 26 • 8.F.A.3

BACKWARD 1

Decide whether a table shows a proportional relationship • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

True or false?

The table shows a proportional relationship.

x	1	2	3
y	4	8	12

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q1/5

913

QUESTION ID

S26-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q1/5

Is the relationship in the table proportional? Explain.

x	1	2	3
y	3	5	7

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q2/5

914

QUESTION ID

S26-B1-Q2

TYPE

Short answer

CORRECT ANSWER

No. The ratios 3, 2.5, and about 2.33 are not equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q2/5

Which table shows a proportional relationship?

A. x: 0, 1, 2 and y: 1, 2, 3

B. x: 1, 2, 3 and y: 1, 4, 9

C. x: 1, 2, 3 and y: 2, 3, 4

D. x: 2, 4, 6 and y: 3, 6, 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q3/5

915

QUESTION ID

S26-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x: 2, 4, 6 and y: 3, 6, 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q3/5

The pairs (5, 2) and (10, 4) are from a proportional relationship.

What is the constant of proportionality (y per x)?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q4/5

916

QUESTION ID

S26-B1-Q4

TYPE

Short answer

CORRECT ANSWER

0.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q4/5

True or false?

y = x + 1 is a proportional relationship.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q5/5

917

QUESTION ID

S26-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q5/5

SET 26 • 8.F.A.3

BACKWARD 2

Find how much y changes for each change in x • Grade 6 • Expressions & Equations • 6.EE.C.9

How much does y increase each time x increases by 1?

x	0	1	2	3
y	5	8	11	14

Increase: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q1/5

919

QUESTION ID

S26-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q1/5

How much does y increase each time x increases by 2?

x	0	2	4
y	1	5	9

Increase: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q2/5

920

QUESTION ID

S26-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q2/5

In y = 6x, how much does y increase when x increases by 1?

A. 1

B. 7

C. 6

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q3/5

921

QUESTION ID

S26-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q3/5

True or false?

In the table with x-values 1, 2, 3 and y-values 10, 20, 30, y increases by 10 each time x increases by 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q4/5

922

QUESTION ID

S26-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q4/5

True or false?

In y = x + 9, y increases by 9 each time x increases by 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q5/5

923

QUESTION ID

S26-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q5/5

SET 26 • 8.F.A.3

FORWARD 1

Show that linear functions grow by equal differences and exponential functions by equal factors • High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a

Show that f(x) = 4x + 1 grows by the same amount over every interval of length 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q1

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q1/5

925

QUESTION ID

S26-F1-Q1

TYPE

Short answer

CORRECT ANSWER

f(x + 2) − f(x) = 4(x + 2) + 1 − (4x + 1) = 8, for every x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q1

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q1/5

Show that g(x) = 3^x grows by the same factor over every interval of length 1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q2

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q2/5

926

QUESTION ID

S26-F1-Q2

TYPE

Short answer

CORRECT ANSWER

g(x + 1) ÷ g(x)
= 3^(x + 1) ÷ 3^x
= 3, for every x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q2

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q2/5

Which function grows by equal factors over equal intervals?

A. $y = 2x^2$

B. $y = x + 5$

C. $y = 5x + 2$

D. $y = 5(2)^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q3

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q3/5

927

QUESTION ID

S26-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. y = 5(2)^x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q3

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q3/5

True or false?

The linear function $f(x) = mx + b$ changes by $m(x_2 - x_1)$ from x_1 to x_2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q4

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q4/5

928

QUESTION ID

S26-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q4

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q4/5

True or false?

$y = x^2$ grows by equal differences over equal intervals.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q5

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q5/5

929

QUESTION ID

S26-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q5

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q5/5

SET 26 • 8.F.A.3

FORWARD 2

Use factoring to find the zeros and maximum of a quadratic function • High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a

ANSWER KEY Reduced copy of the question page, with its answer

A ball's height is $h(t) = -16t^2 + 32t = -16t(t - 2)$ feet after t seconds.

Use the factored form to find when the ball is at height 0.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q1/5

931

QUESTION ID

S26-F2-Q1

TYPE

Short answer

CORRECT ANSWER

t = 0 and t = 2 seconds

GRADING NOTE Both times are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q1/5

A ball's height is h(t) = −16t(t − 2) feet after t seconds. Its greatest height is halfway between the zeros.

When does it reach its greatest height, and what is that height?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q2/5

932

QUESTION ID

S26-F2-Q2

TYPE

Short answer

CORRECT ANSWER

At t = 1 second; 16 feet

GRADING NOTE Both the time and the height are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q2/5

$f(x) = x^2 - 6x + 8 = (x - 2)(x - 4)$

What are the zeros of f?

A. 2 and 4

B. −2 and −4

C. 8 only

D. 6 and 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q3/5

933

QUESTION ID

S26-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2 and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q3/5

True or false?

The graph of a quadratic function is symmetric about a vertical line through its vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q4/5

934

QUESTION ID

S26-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q4/5

True or false?

The zeros of f(x) = (x − 3)(x + 5) are 3 and 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q5/5

935

QUESTION ID

S26-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The zeros are 3 and −5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q5/5

SET 27 • 8.F.B.4 • FUNCTIONS

Find the rate of change and initial value from two points, a table, or a graph

MAIN	Grade 8 main question	**8.F.B.4**
BACKWARD 1	Find a constant rate from a table	**7.RP.A.2.b**
BACKWARD 2	Subtract integers	**7.NS.A.1.c**
FORWARD 1	Construct linear and exponential functions from two points	**HSF-LE.A.2**
FORWARD 2	Write arithmetic and geometric sequences recursively and explicit..	**HSF-BF.A.2**

SET 27 • 8.F.B.4

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.B.4

A linear function passes through (2, 7) and (5, 16).

What are its rate of change and its initial value?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q1

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q1/5

938

QUESTION ID

S27-M-Q1

TYPE

Short answer

CORRECT ANSWER

Rate of change 3; initial value 1

GRADING NOTE Grade both values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q1

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q1/5

The table shows a linear function.

What are its rate of change and its initial value?

x	1	3	5
y	10	16	22

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q2

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q2/5

939

QUESTION ID

S27-M-Q2

TYPE

Short answer

CORRECT ANSWER

Rate of change 3; initial value 7

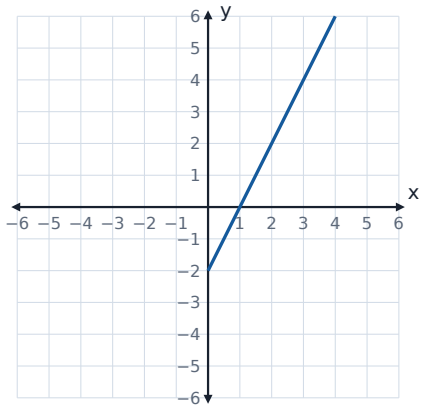
GRADING NOTE Grade both values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q2

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q2/5

The graph shows a linear function.

What are its rate of change and its initial value?



A. Rate 4, initial value −2

B. Rate −2, initial value 2

C. Rate 2, initial value −2

D. Rate 2, initial value 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q3

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q3/5

940

QUESTION ID

S27-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Rate 2, initial value −2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q3

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q3/5

A line passes through (−1, 8) and (3, 0).

Write its equation in the form y = mx + b.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q4

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q4/5

941

QUESTION ID

S27-M-Q4

TYPE

Short answer

CORRECT ANSWER

y = −2x + 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q4

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q4/5

True or false?

A linear function with f(0) = 5 and f(2) = 1 has a rate of change of −2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q5

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q5/5

942

QUESTION ID

S27-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q5

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q5/5

SET 27 • 8.F.B.4

BACKWARD 1

Find a constant rate from a table • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

The table shows a proportional relationship.

What is the constant rate (y per x)?

x	3	6	9
y	12	24	36

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q1/5

944

QUESTION ID

S27-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q1/5

The table shows a proportional relationship.

What is the constant rate (y per x)?

Hours	2	5	8
Miles	7	17.5	28

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q2/5

945

QUESTION ID

S27-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3.5 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q2/5

A proportional table contains the pair (4, 10).

What is the constant rate (y per x)?

A. 0.4

B. 40

C. 2.5

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q3/5

946

QUESTION ID

S27-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q3/5

True or false?

A proportional table with the pairs (2, 1) and (6, 3) has a constant rate of 0.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q4/5

947

QUESTION ID

S27-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q4/5

True or false?

A proportional table with the pairs (5, 15) and (8, 24) has a constant rate of 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q5/5

948

QUESTION ID

S27-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q5/5

SET 27 • 8.F.B.4

BACKWARD 2

Subtract integers • Grade 7 • The Number System • 7.NS.A.1.c

Subtract.

$6 - (-2)$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q1/5

950

QUESTION ID

S27-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q1/5

Subtract.

$-3 - 5$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q2/5

951

QUESTION ID

S27-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q2/5

Subtract.

$-4 - (-9)$

A. 5

B. 13

C. -13

D. -5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q3/5

952

QUESTION ID

S27-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q3/5

True or false?

$2 - 7 = -5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q4/5

953

QUESTION ID

S27-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q4/5

True or false?

$-6 - (-6) = -12$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q5/5

954

QUESTION ID

S27-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q5/5

SET 27 • 8.F.B.4

FORWARD 1

Construct linear and exponential functions from two points • High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

Write the linear function whose graph passes through (1, 4) and (3, 10).

f(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q1

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q1/5

956

QUESTION ID

S27-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3x + 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q1

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q1/5

Write the exponential function f(x) = a · b^x whose graph passes through (0, 5) and (1, 15).

f(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q2

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q2/5

957

QUESTION ID

S27-F1-Q2

TYPE

Short answer

CORRECT ANSWER

5 · 3^x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F1-Q2**

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q2/5

Which exponential function passes through (0, 2) and (2, 18)?

A. f(x) = 8x + 2

B. f(x) = 2 · 3^x

C. f(x) = 18 · 2^x

D. f(x) = 2 · 9^x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q3

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q3/5

958

QUESTION ID

S27-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. f(x) = 2 · 3^x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q3

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q3/5

True or false?

The linear function through (0, −1) and (4, 7) is f(x) = 2x − 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q4

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q4/5

959

QUESTION ID

S27-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q4

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q4/5

True or false?

The exponential function through (0, 4) and (1, 2) is f(x) = 4 · 2^x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q5

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q5/5

960

QUESTION ID

S27-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The function is f(x) = 4 · (0.5)^x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q5

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q5/5

SET 27 • 8.F.B.4

FORWARD 2

Write arithmetic and geometric sequences recursively and explicitly • High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2

Write an explicit formula for the arithmetic sequence 7, 11, 15, ..., starting with n = 1.

a(n) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q1/5

962

QUESTION ID

S27-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4n + 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q1/5

Write a recursive formula for the sequence 7, 11, 15, ...

Formula: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q2/5

963

QUESTION ID

S27-F2-Q2

TYPE

Short answer

CORRECT ANSWER

a(1) = 7;

a(n) = a(n − 1) + 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q2/5

Which explicit formula gives 3, 6, 12, 24, ..., starting with n = 1?

A. a(n) = 2 · 3^(n − 1)

B. a(n) = 3 · 2^(n − 1)

C. a(n) = 3 + 2n

D. a(n) = 3n

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q3/5

964

QUESTION ID

S27-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. a(n) = 3 · 2^(n − 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q3/5

True or false?

An arithmetic sequence is a linear function whose domain is the counting numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q4/5

965

QUESTION ID

S27-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q4/5

True or false?

The sequence 2, 5, 8, 11, ... has a common ratio of 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q5/5

966

QUESTION ID

S27-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has a common difference of 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q5/5

SET 28 • 8.F.B.4 • FUNCTIONS

Construct a linear function to model a situation

MAIN	Grade 8 main question	**8.F.B.4**
BACKWARD 1	Write an equation for a situation	**6.EE.C.9**
BACKWARD 2	Solve a two-step equation from a model	**7.EE.B.4.a**
FORWARD 1	Write functions for situations with fees, thresholds, and restricted.	**HSF-BF.A.1.a**
FORWARD 2	Rearrange a formula to solve for a chosen quantity	**HSA-CED.A.4**

SET 28 • 8.F.B.4

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.B.4

A plumber charges $65 for a visit plus $40 per hour.

Write a function for the cost C after h hours.

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q1

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q1/5

969

QUESTION ID

S28-M-Q1

TYPE

Short answer

CORRECT ANSWER

40h + 65

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q1

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A candle is 30 cm tall and burns down 2.5 cm each hour.

Write a function for its height h after t hours.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q2

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q2/5

970

QUESTION ID

S28-M-Q2

TYPE

Short answer

CORRECT ANSWER

30 − 2.5t

GRADING NOTE −2.5t + 30 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q2

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q2/5

A phone plan costs $20 per month after a $50 activation fee.

Which function gives the total cost C after m months?

A. C = 20m − 50

B. C = 20m + 50

C. C = 50m + 20

D. C = 70m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q3

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q3/5

971

QUESTION ID

S28-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. C = 20m + 50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q3

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q3/5

The cost of a taxi ride is a linear function of the miles m. A 3-mile ride costs $11 and a 7-mile ride costs $21.

Write the function.

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q4

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q4/5

QUESTION ID

S28-M-Q4

TYPE

Short answer

CORRECT ANSWER

2.5m + 3.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q4

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q4/5

True or false?

A tank holds 500 liters and drains 20 liters each minute. The function V = 500 − 20t gives the liters V after t minutes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q5

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q5/5

973

QUESTION ID

S28-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q5

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q5/5

SET 28 • 8.F.B.4

BACKWARD 1

Write an equation for a situation • Grade 6 • Expressions & Equations • 6.EE.C.9

Tickets cost $9 each.

Write an equation for the cost c of t tickets.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q1/5

975

QUESTION ID

S28-B1-Q1

TYPE

Short answer

CORRECT ANSWER

c = 9t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q1/5

Mia has $15 and saves $5 each week.

Write an equation for her savings s after w weeks.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q2/5

976

QUESTION ID

S28-B1-Q2

TYPE

Short answer

CORRECT ANSWER

s = 5w + 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q2/5

A pool holds 100 gallons and fills at 10 gallons per minute.

Which equation gives the gallons g after m minutes?

A. g = 110m

B. g = 100m + 10

C. g = 10m + 100

D. g = 10m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q3/5

977

QUESTION ID

S28-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. g = 10m + 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q3/5

True or false?

Apples cost $3 per pound. The equation c = 3p gives the cost c of p pounds.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q4/5

978

QUESTION ID

S28-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q4/5

True or false?

A game costs a $10 fee plus $2 per game. The equation c = 10g + 2 gives the cost c of g games.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q5/5

979

QUESTION ID

S28-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The equation is c = 2g + 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q5/5

SET 28 • 8.F.B.4

BACKWARD 2

Solve a two-step equation from a model • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for h.

40h + 65 = 185

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q1/5

981

QUESTION ID

S28-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q1/5

Solve for t.

$30 - 2.5t = 10$

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q2/5

982

QUESTION ID

S28-B2-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q2/5

Solve for m.

20m + 50 = 170

A. m = 3

B. m = 11

C. m = 8.5

D. m = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q3/5

983

QUESTION ID

S28-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. m = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q3/5

True or false?

t = 4 is the solution of 500 − 20t = 420.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q4/5

984

QUESTION ID

S28-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q4/5

True or false?

h = 5 is the solution of 12h + 8 = 50.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q5/5

985

QUESTION ID

S28-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is h = 3.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q5/5

SET 28 • 8.F.B.4

FORWARD 1

Write functions for situations with fees, thresholds, and restricted domains • High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a

ANSWER KEY Reduced copy of the question page, with its answer

A phone plan costs $30 per month for up to 5 GB of data, plus $8 for each GB over 5.

Write a function C(g) for the monthly cost when g > 5 GB, and find C(9).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q1

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q1/5

987

QUESTION ID

S28-F1-Q1

TYPE

Short answer

CORRECT ANSWER

C(g) = 30 + 8(g − 5);
C(9) = $62

GRADING NOTE Equivalent forms such as C(g) = 8g − 10 are correct. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q1

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q1/5

A pool holds 12,000 gallons and drains at 400 gallons per hour.

Write a function V(t) for the water left after t hours, and give the domain that makes sense.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q2

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q2/5

988

QUESTION ID
S28-F1-Q2

TYPE
Short answer

CORRECT ANSWER
V(t) = 12,000 − 400t, for 0 ≤ t ≤ 30

GRADING NOTE Both the function and the domain are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q2

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q2/5

A taxi charges $3 for the first mile and $2 for each mile after the first.

Which function gives the cost for m miles, where m ≥ 1?

A. C(m) = 3 + 2(m − 1)

B. C(m) = 3 + 2m

C. C(m) = 5m

D. C(m) = 2 + 3(m − 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q3

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q3/5

989

QUESTION ID

S28-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A.

C(m) = 3 + 2(m − 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q3

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q3/5

True or false?

A gym charges a $50 fee and then $20 per month. For $C(m) = 50 + 20m$, $C(m + 1) - C(m) = 20$ for every m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q4

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q4/5

990

QUESTION ID

S28-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q4

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q4/5

True or false?

A store takes $10 off any purchase over $50. For a purchase of p dollars, where p > 50, the cost is c(p) = 10 − p.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q5

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q5/5

991

QUESTION ID

S28-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The cost is c(p) = p − 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q5

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q5/5

SET 28 • 8.F.B.4

FORWARD 2

Rearrange a formula to solve for a chosen quantity • High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve C = 40h + 65 for h.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q1

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q1/5

993

QUESTION ID

S28-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(C − 65)/40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q1

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $F = \frac{9}{5}C + 32$ for C.

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q2

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q2/5

994

QUESTION ID

S28-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(5/9)(F − 32)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q2

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q2/5

Solve $y = mx + b$ for b.

A. $b = \frac{y}{mx}$

B. $b = mx - y$

C. $b = y + mx$

D. $b = y - mx$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q3

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q3/5

995

QUESTION ID

S28-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. b = y − mx

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q3

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q3/5

True or false?

For $t \neq 0$, solving $d = rt$ for r gives $r = \frac{d}{t}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q4

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q4/5

996

QUESTION ID

S28-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q4

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

For $l \neq 0$, solving $A = lw$ for w gives $w = A - l$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q5

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q5/5

997

QUESTION ID
S28-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $w = A/l$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q5

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q5/5

SET 29 • 8.F.B.4 • FUNCTIONS

Interpret the rate of change and initial value of a linear function in context

MAIN	Grade 8 main question	**8.F.B.4**
BACKWARD 1	Explain points on the graph of a proportional relationship	**7.RP.A.2.d**
BACKWARD 2	Identify the independent and dependent variables	**6.EE.C.9**
FORWARD 1	Interpret the slope and intercept of a linear model when the units…	**HSS-ID.C.7**
FORWARD 2	Interpret the parameters of linear and exponential functions	**HSF-LE.B.5**

SET 29 • 8.F.B.4

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Renting a bike costs C = 4h + 10 dollars for h hours.

What do the 4 and the 10 mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q1

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q1/5

1000

QUESTION ID

S29-M-Q1

TYPE

Short answer

CORRECT ANSWER

4: the cost goes up $4 for each hour. 10: a $10 fee (the cost at 0 hours).

GRADING NOTE Grade both numbers.

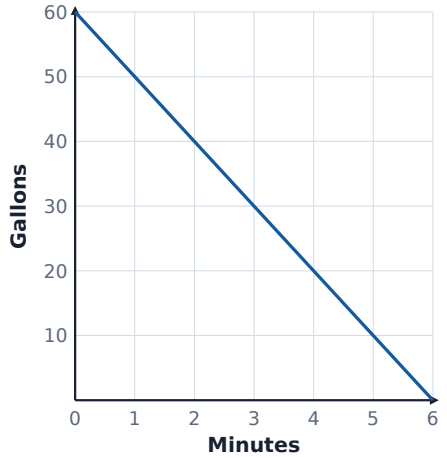
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q1

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q1/5

The graph shows the water in a tank.

What do the rate of change and the initial value mean?

Meaning: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q2

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q2/5

1001

QUESTION ID

S29-M-Q2

TYPE

Short answer

CORRECT ANSWER

The tank starts with 60 gallons (initial value) and drains 10 gallons per minute (rate of change −10).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q2

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q2/5

A plant's height is h = 0.5w + 12 centimeters after w weeks.

What does 0.5 mean?

A. The plant grows for 0.5 week.

B. The plant grows 0.5 cm each week.

C. The plant was 0.5 cm tall at the start.

D. The plant grows 12 cm each week.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q3

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q3/5

1002

QUESTION ID

S29-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. The plant grows 0.5 cm each week.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q3

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The table shows the money in a savings account.

What does the rate of change mean?

Weeks	0	4	8
Dollars	150	210	270

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q4

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q4/5

1003

QUESTION ID

S29-M-Q4

TYPE

Short answer

CORRECT ANSWER

$15 is added each week.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q4

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q4/5

True or false?

For C = 4h + 10, the initial value means the bike costs $4 before any hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q5

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q5/5

1004

QUESTION ID

S29-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The initial value is $10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q5

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q5/5

SET 29 • 8.F.B.4

BACKWARD 1

Explain points on the graph of a proportional relationship • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d

A proportional graph of gallons (x) and miles (y) passes through (1, 28).

What does this point mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q1/5

1006

QUESTION ID

S29-B1-Q1

TYPE

Short answer

CORRECT ANSWER

The car travels 28 miles on 1 gallon.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q1/5

A proportional graph shows hours worked (x) and pay (y).

What does the point (0, 0) mean?

A. 0 hours of work earn $0.

B. The worker starts with $0 per hour.

C. The pay is $0 per hour.

D. There is no graph.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q2/5

1007

QUESTION ID

S29-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 0 hours of work earn $0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q2/5

A proportional graph of minutes (x) and laps (y) passes through (6, 3).

What does this point mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q3/5

1008

QUESTION ID

S29-B1-Q3

TYPE

Short answer

CORRECT ANSWER

3 laps are swum in 6 minutes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q3/5

True or false?

On the graph of a proportional relationship, the point (1, r) shows the unit rate r.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q4/5

1009

QUESTION ID

S29-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q4/5

True or false?

On a graph of tickets (x) and cost (y), the point (2, 18) means each ticket costs $18.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q5/5

1010

QUESTION ID

S29-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It means 2 tickets cost $18 ($9 each).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q5/5

SET 29 • 8.F.B.4

BACKWARD 2

Identify the independent and dependent variables • Grade 6 • Expressions & Equations • 6.EE.C.9

ANSWER KEY Reduced copy of the question page, with its answer

In C = 4h + 10, which variable is the independent variable?

Variable: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q1/5

1012

QUESTION ID

S29-B2-Q1

TYPE

Short answer

CORRECT ANSWER

h

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

"Pay depends on the hours worked." Which quantity is the dependent variable?

Quantity: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q2/5

1013

QUESTION ID

S29-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Pay

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q2/5

In d = 60t, what is d?

A. The rate

B. The independent variable

C. A constant

D. The dependent variable

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q3/5

1014

QUESTION ID

S29-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The dependent variable

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q3/5

True or false?

In a table of hours worked and pay, the hours are usually the independent variable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q4/5

1015

QUESTION ID

S29-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q4/5

True or false?

In y = 3x + 2, x depends on y.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q5/5

1016

QUESTION ID

S29-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q5/5

SET 29 • 8.F.B.4

FORWARD 1

Interpret the slope and intercept of a linear model when the units or the starting point change • High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7

ANSWER KEY Reduced copy of the question page, with its answer

A model gives a city's population, in thousands, as $P = 2.4t + 85$, where t is the number of years after 2010.

What do 2.4 and 85 mean? Include units.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q1/5

1018

QUESTION ID

S29-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2.4: the population grows by about 2,400 people per year. 85: the model gives about 85,000 people in 2010.

GRADING NOTE Both meanings, with units, are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q1/5

A phone plan is modeled by C = 0.05m + 20 dollars for m minutes of calls.

Rewrite the model for h hours of calls, and say what the new slope means.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q2/5

1019

QUESTION ID

S29-F1-Q2

TYPE

Short answer

CORRECT ANSWER

C = 3h + 20; the plan costs $3 for each hour of calls.

GRADING NOTE Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F1-Q2**

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q2/5

A model gives the air temperature as $T = -6.5h + 15$ °C at a height of h kilometers.

What does -6.5 mean?

A. The air is 6.5 km high.

B. The temperature drops about 6.5°C for each kilometer of height.

C. The temperature at the ground is −6.5°C.

D. The temperature rises 6.5°C for each kilometer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q3/5

1020

QUESTION ID

S29-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. The temperature drops about 6.5°C for each kilometer of height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q3/5

True or false?

In the model $y = 4(x - 2000) + 150$ for the sales y in year x, 150 is the predicted sales in the year 2000.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q4/5

1021

QUESTION ID

S29-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q4/5

True or false?

A model for x years since 2000 has the intercept 40. This means the model predicts 40 in the year 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q5/5

1022

QUESTION ID

S29-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE x = 0 is the year 2000, so 40 is the prediction for 2000.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q5/5

SET 29 • 8.F.B.4

FORWARD 2

Interpret the parameters of linear and exponential functions • High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5

P(t) = 2,000(1.05)^t gives an account balance after t years.

What do 2,000 and 1.05 mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q1

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q1/5

1024

QUESTION ID

S29-F2-Q1

TYPE

Short answer

CORRECT ANSWER

2,000 is the starting amount; 1.05 means the balance grows 5% each year.

GRADING NOTE Grade both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q1

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q1/5

V(t) = 25,000(0.85)^t gives a car's value after t years.

What is the yearly percent change?

Change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q2

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q2/5

1025

QUESTION ID

S29-F2-Q2

TYPE

Short answer

CORRECT ANSWER

A 15% decrease each year

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q2

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q2/5

A phone plan costs C(m) = 0.12m + 9.99 dollars for m minutes.

What does 9.99 represent?

A. A fixed monthly charge

B. The number of minutes

C. The total cost

D. The cost per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q3

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q3/5

1026

QUESTION ID

S29-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A fixed monthly charge

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F2-Q3**

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q3/5

True or false?

In f(t) = 100(2)^t, the quantity doubles each time t increases by 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q4

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q4/5

1027

QUESTION ID

S29-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q4

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q4/5

True or false?

In f(t) = 50(0.9)^t, the quantity grows by 90% each period.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q5

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q5/5

1028

QUESTION ID

S29-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It decreases by 10% each period.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q5

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q5/5

SET 30 • 8.F.B.5 • FUNCTIONS

Describe the relationship shown by a graph

MAIN	Grade 8 main question	**8.F.B.5**
BACKWARD 1	Interpret points on a graph of a situation	**5.G.A.2**
BACKWARD 2	Compare speeds as unit rates	**6.RP.A.3.b**
FORWARD 1	Interpret the key features of a nonlinear graph in context	**HSF-IF.B.4**
FORWARD 2	Graph and evaluate piecewise-defined functions	**HSF-IF.C.7.b**

SET 30 • 8.F.B.5

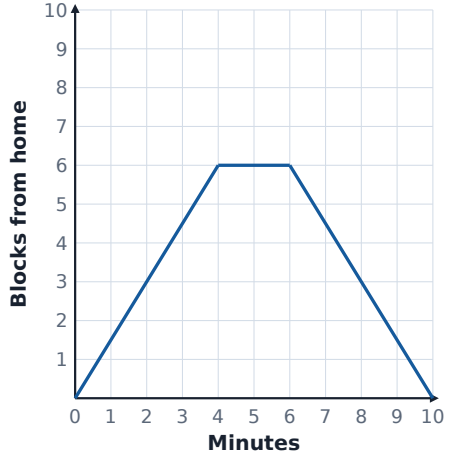
MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.B.5

The graph shows Ana's walk.

Describe what happens from 4 to 6 minutes.

Description: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q1

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q1/5

1031

QUESTION ID

S30-M-Q1

TYPE

Short answer

CORRECT ANSWER

She stays 6 blocks from home; she stops.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q1

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q1/5

The graph shows Ana's walk.

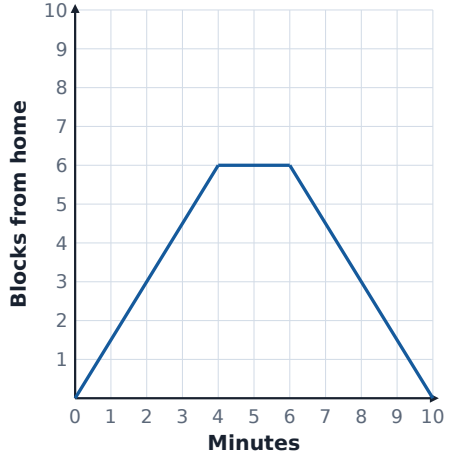
During which interval is Ana moving toward home?

A. 0 to 4 minutes

B. 6 to 10 minutes

C. 4 to 6 minutes

D. 0 to 6 minutes



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q2

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q2/5

1032

QUESTION ID

S30-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 6 to 10 minutes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

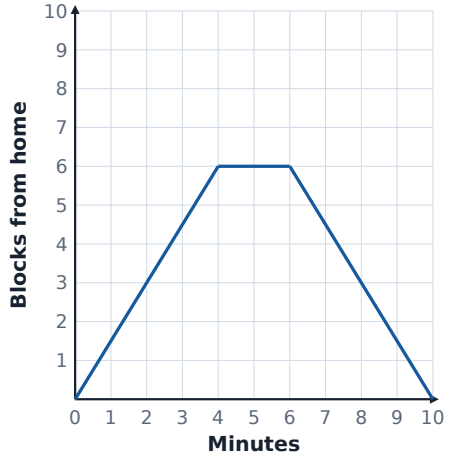
ID S30-M-Q2

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q2/5

The graph shows Ana's walk.

Is the relationship linear or nonlinear over the whole walk? Explain.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q3

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q3/5

1033

QUESTION ID

S30-M-Q3

TYPE

Short answer

CORRECT ANSWER

Nonlinear. The graph is not one straight line; the rate of change is different on different parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

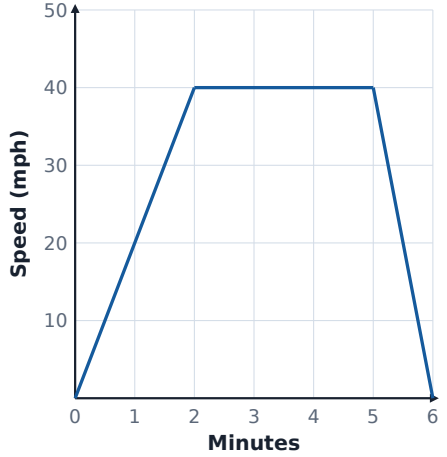
ID S30-M-Q3

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q3/5

The graph shows a car's speed.

Describe how the speed changes over the 6 minutes.

Description: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q4

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q4/5

1034

QUESTION ID

S30-M-Q4

TYPE

Short answer

CORRECT ANSWER

The car speeds up for 2 minutes, travels at a steady 40 mph until minute 5, then slows to a stop.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q4

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q4/5

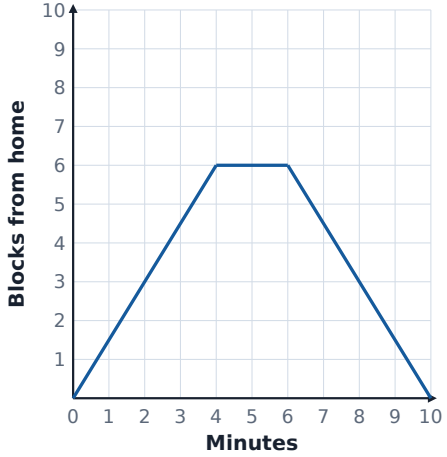
ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In the graph of Ana's walk, she walks faster going home than going away from home.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q5

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q5/5

1035

QUESTION ID
S30-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Both trips cover 6 blocks in 4 minutes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q5

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q5/5

SET 30 • 8.F.B.5

BACKWARD 1

Interpret points on a graph of a situation • Grade 5 • Geometry • 5.G.A.2

A graph shows minutes (x) and blocks from home (y).
What does the point (3, 5) mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q1/5

1037

QUESTION ID

S30-B1-Q1

TYPE

Short answer

CORRECT ANSWER

After 3 minutes, the walker is 5 blocks from home.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q1/5

A graph shows days (x) and plant height in centimeters (y). What does the point (10, 4) mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q2/5

1038

QUESTION ID

S30-B1-Q2

TYPE

Short answer

CORRECT ANSWER

After 10 days, the plant is 4 cm tall.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q2/5

A graph shows hours (x) and miles driven (y).

Which point shows 120 miles after 2 hours?

A. (2, 120)

B. (60, 2)

C. (2, 60)

D. (120, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q3/5

1039

QUESTION ID

S30-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (2, 120)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q3/5

True or false?

On a graph of time (x) and distance (y), the point (0, 0) can mean the start of a trip.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q4/5

1040

QUESTION ID

S30-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q4/5

True or false?

The point (4, 1) on a graph of hours (x) and miles (y) means 4 miles in 1 hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q5/5

1041

QUESTION ID

S30-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q5/5

SET 30 • 8.F.B.5

BACKWARD 2

Compare speeds as unit rates • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b

Ana walks 6 blocks in 4 minutes at a steady pace.

What is her speed in blocks per minute?

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q1/5

1043

QUESTION ID

S30-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1.5 blocks per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q1/5

Which is faster: 12 miles in 3 hours, or 10 miles in 2 hours?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q2/5

1044

QUESTION ID

S30-B2-Q2

TYPE

Short answer

CORRECT ANSWER

10 miles in 2 hours (5 mph, compared with 4 mph)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q2/5

Which rate is the greatest?

A. 60 miles in 1 hour

B. 100 miles in 2 hours

C. 20 miles in 0.5 hour

D. 150 miles in 3 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q3/5

1045

QUESTION ID

S30-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 60 miles in 1 hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q3/5

True or false?

Walking 3 miles in 1 hour is faster than walking 5 miles in 2 hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q4/5

1046

QUESTION ID

S30-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q4/5

True or false?

Running 8 km in 40 minutes is the same rate as running 2 km in 20 minutes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q5/5

1047

QUESTION ID

S30-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q5/5

SET 30 • 8.F.B.5

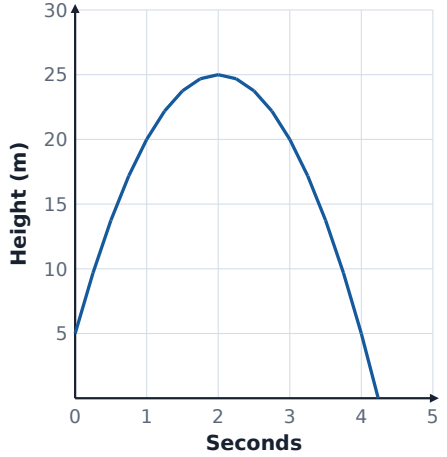
FORWARD 1

Interpret the key features of a nonlinear graph in context • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows a ball's height after it is thrown.

What is the greatest height, and when does it happen?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q1/5

1049

QUESTION ID

S30-F1-Q1

TYPE

Short answer

CORRECT ANSWER

25 m, at 2 seconds

GRADING NOTE Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q1

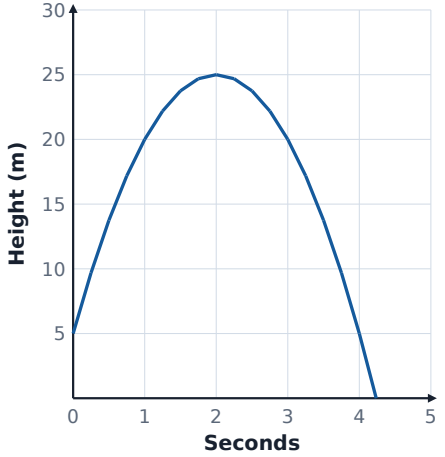
High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows a ball's height after it is thrown.

On what interval is the height increasing, and on what interval is it decreasing?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q2/5

1050

QUESTION ID

S30-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Increasing from 0 to 2 seconds; decreasing from 2 seconds until it lands (a little after 4 seconds).

GRADING NOTE Both intervals are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows a ball's height after it is thrown.

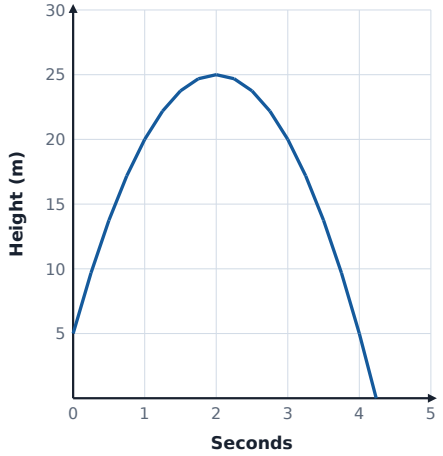
What does the point where the graph meets the vertical axis mean?

A. The ball lands after 5 seconds.

B. The ball was 5 m above the ground when it was thrown.

C. The ball rises 5 m each second.

D. The ball's greatest height is 5 m.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q3/5

1051

QUESTION ID

S30-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. The ball was 5 m above the ground when it was thrown.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q3

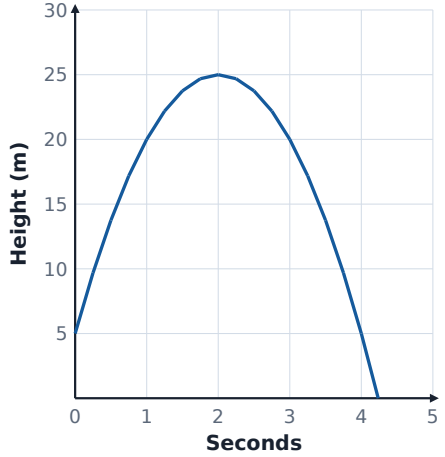
High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q3/5

True or false?

In the graph, the ball hits the ground a little after 4 seconds.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q4/5

1052

QUESTION ID

S30-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q4/5

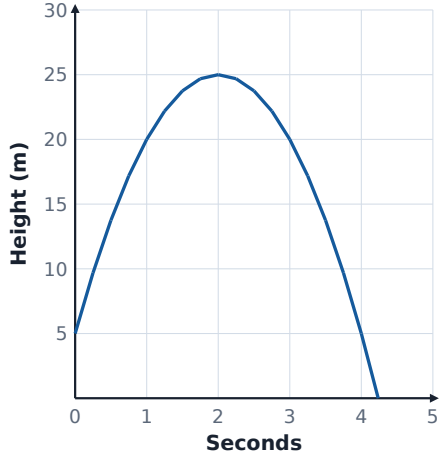
ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In the graph, the ball's height changes by the same amount each second.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q5/5

1053

QUESTION ID
S30-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The graph is curved, so the rate of change is not constant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q5/5

SET 30 • 8.F.B.5

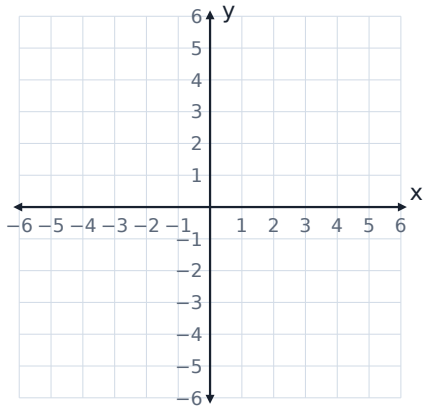
FORWARD 2

Graph and evaluate piecewise-defined functions • High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b

Graph the function on the coordinate plane.

$f(x) = x + 2$ for $x < 0$

$f(x) = 2$ for $x \geq 0$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q1/5

1055

QUESTION ID

S30-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The line $y = x + 2$ for $x < 0$ (an open end at (0, 2) is acceptable) and the horizontal line $y = 2$ for $x \geq 0$, with a closed point at (0, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q1/5

$f(x) = 3$ for $x \leq 2$, and $f(x) = x + 1$ for $x > 2$.

Find $f(5)$.

$f(5) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q2/5

1056

QUESTION ID

S30-F2-Q2

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q2/5

A taxi charges $3 for the first mile and $2 for each mile after that.

Which function gives the cost for m miles?

A. C(m) = 2m + 3 for all m

B. C(m) = 3m for all m

C. C(m) = 3 for 0 < m ≤ 1; C(m) = 3 + 2(m − 1) for m > 1

D. C(m) = 3 + 2m for m ≤ 1 only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q3/5

1057

QUESTION ID

S30-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. C(m) = 3 for 0 < m ≤ 1; C(m) = 3 + 2(m − 1) for m > 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q3/5

True or false?

For f(x) = |x|, f(−4) = 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q4/5

1058

QUESTION ID

S30-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q4/5

True or false?

A piecewise-defined function can never have an unbroken graph.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q5/5

1059

QUESTION ID

S30-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE For example, f(x) = |x| is piecewise-defined and unbroken.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q5/5

SET 31 • 8.F.B.5 • FUNCTIONS

Sketch a graph from a description of a relationship

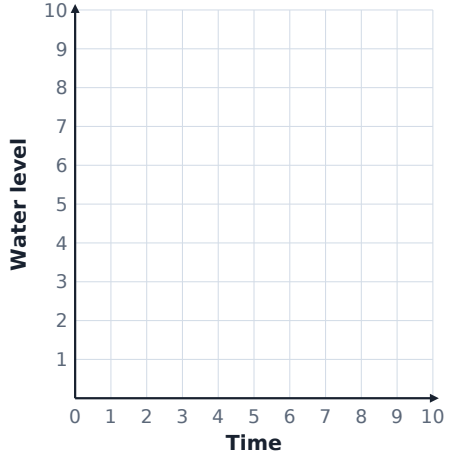
MAIN	Grade 8 main question	**8.F.B.5**
BACKWARD 1	Graph a relationship from a table	**6.EE.C.9**
BACKWARD 2	Plot points that represent a situation	**5.G.A.2**
FORWARD 1	Sketch graphs that show given key features	**HSF-IF.B.4**
FORWARD 2	Graph exponential functions and describe their end behavior	**HSF-IF.C.7.e**

SET 31 • 8.F.B.5

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.B.5

Sketch a graph of the water level in a bathtub over time: it fills at a steady rate, stays the same while someone bathes, and then drains quickly.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q1

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q1/5

1062

QUESTION ID

S31-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

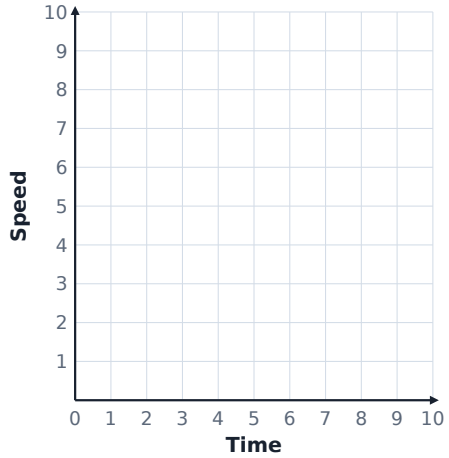
A rising straight segment from 0, then a horizontal segment, then a steeper falling segment back to 0. Any lengths are accepted if the order and steepness are right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q1

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q1/5

Sketch a graph of a car's speed over time: it speeds up, travels at a steady speed, and then slows down to a stop.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q2

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q2/5

1063

QUESTION ID

S31-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

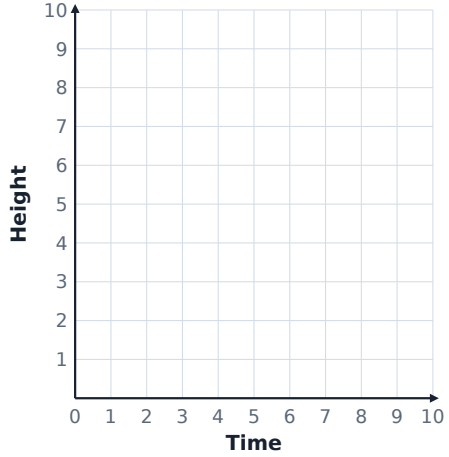
A rising segment from 0, a horizontal segment, then a falling segment back to 0. Any lengths are accepted if the order is right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q2

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q2/5

Sketch the height of a ball thrown straight up over time: it rises, slows down, reaches a highest point, and falls back to the ground.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q3

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q3/5

1064

QUESTION ID

S31-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

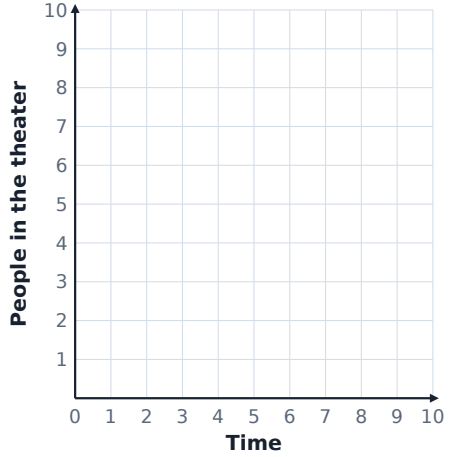
A curve that rises, levels off at one highest point, and falls back to height 0, like an upside-down U. Starting at ground level or at the throwing height is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q3

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q3/5

Sketch the number of people in a theater from 6:00 to 10:00. It fills quickly before a 7:00 show, stays full, and empties after the show ends at 9:30.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q4

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q4/5

1065

QUESTION ID

S31-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

A steep rise ending at about 7:00, a horizontal segment from 7:00 to 9:30, then a steep fall after 9:30 back to about 0 by 10:00. The rise, level part, and fall must be placed at these clock times; exact heights are not required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q4

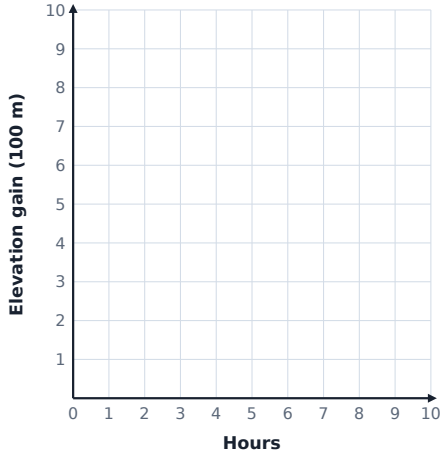
Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Sketch a graph of a hiker's elevation gain over time: she gains elevation at a steady 200 meters per hour for 2 hours, then at a steady 80 meters per hour for 1 hour, then rests for 1 hour.

Is elevation gain a linear function of time for the whole hike? Explain.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q5

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q5/5

1066

QUESTION ID

S31-M-Q5

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A rising segment from (0, 0) to (2, 4) (400 m), a less steep rising segment to (3, 4.8) (480 m), and a horizontal segment to (4, 4.8)

Written: No. The rate of change is different on each part (200, then 80, then 0 meters per hour), so the whole graph is not one line.

GRADING NOTE Grade both: the three-part graph with the right order and steepness, and "no" with a reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q5

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q5/5

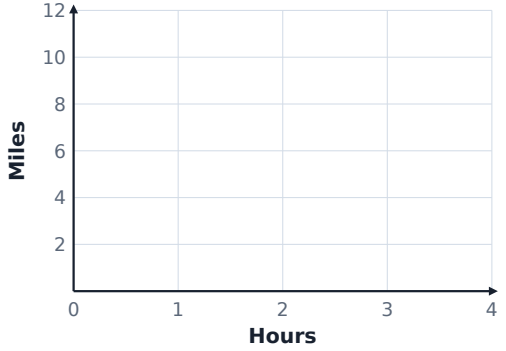
SET 31 • 8.F.B.5

BACKWARD 1

Graph a relationship from a table • Grade 6 • Expressions & Equations • 6.EE.C.9

Graph the pairs in the table.

Hours	0	1	2	3
Miles	0	4	8	12



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q1/5

1068

QUESTION ID

S31-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Points at (0, 0), (1, 4), (2, 8), and (3, 12)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q1/5

A hiker walks 4 miles each hour, so miles = 4 × hours.

How many miles does she walk in 2.5 hours?

Miles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q2/5

1069

QUESTION ID

S31-B1-Q2

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q2/5

Which table shows y increasing by the same amount each time?

A. x: 0, 1, 2 and y: 0, 3, 4

B. x: 0, 1, 2 and y: 5, 5, 6

C. x: 0, 1, 2 and y: 6, 3, 9

D. x: 0, 1, 2 and y: 0, 3, 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q3/5

1070

QUESTION ID

S31-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x: 0, 1, 2 and y: 0, 3, 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q3/5

True or false?

A table where y is 5 for every x shows y not changing.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q4/5

1071

QUESTION ID

S31-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q4/5

True or false?

A table with y-values 2, 4, 3 shows y always increasing.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q5/5

1072

QUESTION ID

S31-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q5/5

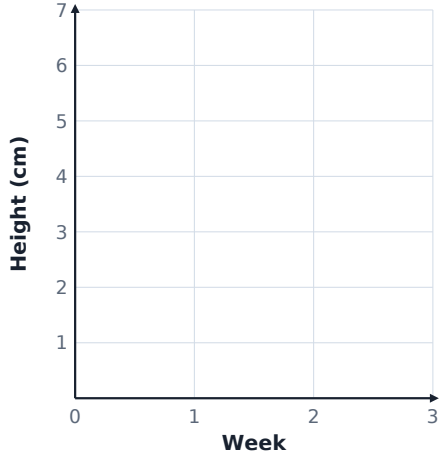
SET 31 • 8.F.B.5

BACKWARD 2

Plot points that represent a situation • Grade 5 • Geometry • 5.G.A.2

A plant is 2 cm tall at week 0, 4 cm at week 1, and 6 cm at week 2.

Plot the points (week, height).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q1/5

1074

QUESTION ID

S31-B2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Points at (0, 2), (1, 4), and (2, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q1/5

A plant is 2 cm tall at week 0 and grows 2 cm each week.

Which point shows its height at week 3?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q2/5

1075

QUESTION ID

S31-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(3, 8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q2/5

On a graph of weeks (x) and plant height in cm (y), what does the point (2, 6) mean?

A. The plant grows 2 cm.

B. At week 6 the plant is 2 cm tall.

C. The plant grows 6 cm each week.

D. At week 2 the plant is 6 cm tall.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q3/5

1076

QUESTION ID

S31-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. At week 2 the plant is 6 cm tall.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q3/5

True or false?

On a graph of weeks (x) and height (y), the point (0, 2) shows the height at the start.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q4/5

1077

QUESTION ID

S31-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q4/5

True or false?

On a graph of weeks (x) and height in cm (y), the point (1, 4) means the plant is 1 cm tall at week 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q5/5

1078

QUESTION ID

S31-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q5/5

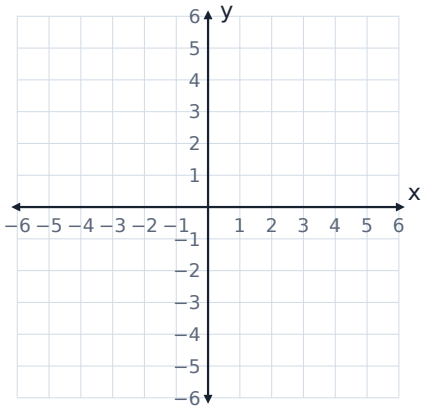
SET 31 • 8.F.B.5

FORWARD 1

Sketch graphs that show given key features • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Sketch a function that increases for $0 < x < 3$, has a maximum value of 5 at $x = 3$, and decreases for $x > 3$.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q1/5

1080

QUESTION ID

S31-F1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

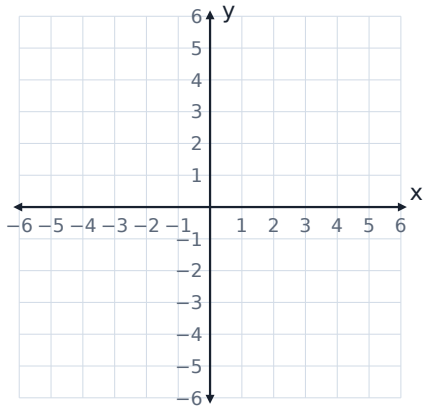
Any graph that rises up to the point (3, 5) and falls after it, with 5 as its highest value

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Sketch a function whose graph crosses the y-axis at 2 and that increases for all x shown.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q2/5

1081

QUESTION ID

S31-F1-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Any always-increasing graph through (0, 2); a line or a curve is accepted

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q2/5

Which description matches $y = -(x - 2)^2 + 4$?

A. A maximum of 4 at $x = 2$; increasing for $x < 2$

B. A minimum of 4 at $x = 2$

C. A maximum of 2 at $x = 4$

D. Increasing for every x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q3/5

1082

QUESTION ID

S31-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A maximum of 4 at x = 2; increasing for x < 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q3/5

True or false?

The function $f(x) = -(x - 3)^2 + 5$ has its maximum at $x = 3$ and decreases for every x greater than 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q4/5

1083

QUESTION ID

S31-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q4/5

True or false?

A function that is always decreasing can have its maximum in the middle of its domain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q5/5

1084

QUESTION ID

S31-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q5

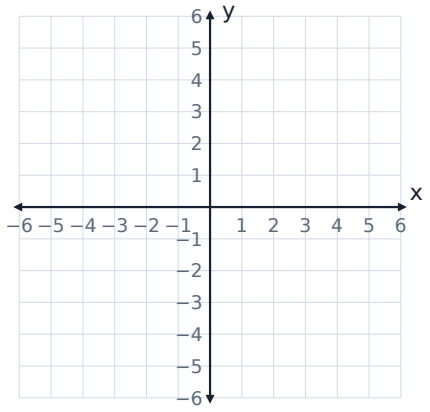
High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q5/5

SET 31 • 8.F.B.5

FORWARD 2

Graph exponential functions and describe their end behavior • High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e

Graph y = 2^x for x from −2 to 2 on the coordinate plane.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q1/5

1086

QUESTION ID

S31-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The points (−2, 1/4), (−1, 1/2), (0, 1), (1, 2), and (2, 4) joined by a smooth increasing curve that gets close to the x-axis on the left

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q1/5

What is the y-intercept of the graph of y = 5(0.5)^x?

y-intercept: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q2/5

1087

QUESTION ID

S31-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(0, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q2/5

For y = 3^x, what happens to y as x decreases without bound?

A. y gets closer to 3.

B. y decreases without bound.

C. y increases.

D. y gets closer to 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q3/5

1088

QUESTION ID

S31-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. y gets closer to 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q3/5

True or false?

The graph of y = 2^x never touches the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q4/5

1089

QUESTION ID

S31-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q4/5

True or false?

The function y = (0.5)^x is increasing.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q5/5

1090

QUESTION ID

S31-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q5/5

SET 32 • 8.G.A.1.a • GEOMETRY

Verify that rigid motions take lines to lines and segments to segments of the same length

MAIN	Grade 8 main question	**8.G.A.1.a**
BACKWARD 1	Reflect points across the axes	**6.NS.C.6.b**
BACKWARD 2	Find the distance between points with the same x- or y-coordinate	**6.NS.C.8**
FORWARD 1	Define rotations, reflections, and translations precisely	**HSG-CO.A.4**
FORWARD 2	Use rigid motions to decide whether figures are congruent	**HSG-CO.B.6**

SET 32 • 8.G.A.1.a

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.1.a

Segment AB has endpoints A(1, 2) and B(5, 2). It is translated 3 units down.

What are the endpoints of the image, and how long is the image?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q1

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q1/5

1093

QUESTION ID

S32-M-Q1

TYPE

Short answer

CORRECT ANSWER

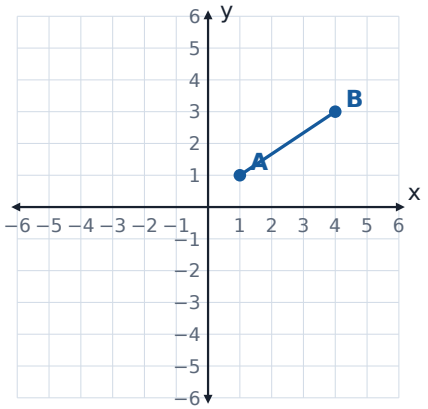
A'(1, −1) and B'(5, −1); it is 4 units long, the same as AB.

GRADING NOTE Grade both the endpoints and the length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q1

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q1/5

Reflect segment AB across the y-axis. Draw the image A'B'. Then find the lengths of AB and A'B' and compare them.



Lengths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q2

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q2/5

1094

QUESTION ID

S32-M-Q2

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: The segment from A'(−1, 1) to B'(−4, 3)

Written: AB = A'B' = √13 units (about 3.6 units); the reflection kept the length.

GRADING NOTE Grade both: the image segment, and the two equal lengths found from the grid (run 3, rise 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q2

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q2/5

A(1, 2) and B(4, 6) are reflected across the x-axis to A' and B'.

How long is A'B'?

A. 4 units

B. 5 units

C. 3 units

D. 8 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q3

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q3/5

1095

QUESTION ID

S32-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 5 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q3

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q3/5

True or false?

The segment from (0, 0) to (3, 4) is rotated 90° counterclockwise about the origin to the segment from (0, 0) to (−4, 3). Both segments are 5 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q4

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q4/5

1096

QUESTION ID

S32-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q4

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q4/5

True or false?

Segment PQ from P(2, 1) to Q(2, 5) is translated 3 units left. Its image is 7 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q5

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q5/5

1097

QUESTION ID

S32-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The image runs from (−1, 1) to (−1, 5), so it is 4 units long, the same as PQ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q5

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q5/5

SET 32 • 8.G.A.1.a

BACKWARD 1

Reflect points across the axes • Grade 6 • The Number System • 6.NS.C.6.b

The point (3, −4) is reflected across the x-axis.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q1/5

1099

QUESTION ID

S32-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q1/5

The point (−2, 5) is reflected across the y-axis.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q2/5

1100

QUESTION ID

S32-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q2/5

Which point is the reflection of (6, 1) across the y-axis?

A. (1, 6)

B. (6, −1)

C. (−6, −1)

D. (−6, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q3/5

1101

QUESTION ID

S32-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (−6, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q3/5

True or false?

The reflection of (−3, −3) across the x-axis is (−3, 3).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q4/5

1102

QUESTION ID

S32-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q4/5

True or false?

The reflection of (4, 2) across the x-axis is (−4, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q5/5

1103

QUESTION ID

S32-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q5/5

SET 32 • 8.G.A.1.a

BACKWARD 2

Find the distance between points with the same x- or y-coordinate • Grade 6 • The Number System • 6.NS.C.8

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance between (1, 2) and (5, 2).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q1/5

1105

QUESTION ID

S32-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q1/5

Find the distance between (1, −1) and (5, −1).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q2/5

1106

QUESTION ID

S32-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q2/5

Which two points are 6 units apart?

A. (−2, 3) and (2, 3)

B. (0, 1) and (0, 5)

C. (3, −2) and (3, 2)

D. (−2, 3) and (4, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q3/5

1107

QUESTION ID

S32-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (−2, 3) and (4, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q3/5

True or false?

(0, −3) and (0, 4) are 7 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q4/5

1108

QUESTION ID

S32-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

(−5, 1) and (2, 1) are 3 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q5/5

1109

QUESTION ID

S32-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are 7 units apart.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q5/5

SET 32 • 8.G.A.1.a

FORWARD 1

Define rotations, reflections, and translations precisely • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4

A reflection across line m maps a point P that is not on m to the point P'.

How is line m related to segment PP'?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q1/5

1111

QUESTION ID

S32-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Line m is the perpendicular bisector of PP'.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q1/5

A rotation of 90° about point O maps a point P (P ≠ O) to P'.

What is true about OP, OP', and angle POP'?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q2/5

1112

QUESTION ID

S32-F1-Q2

TYPE

Short answer

CORRECT ANSWER

OP = OP', and angle POP' measures 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q2/5

In a translation by a nonzero distance, the segments that join each point to its image are...

A. perpendicular to each other

B. all through one point

C. different lengths

D. parallel and the same length

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q3/5

1113

QUESTION ID

S32-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. parallel and the same length

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q3/5

True or false?

A rotation keeps every point the same distance from the center of rotation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q4/5

1114

QUESTION ID

S32-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q4/5

True or false?

In a reflection, a point (not on the line) and its image are on the same side of the line of reflection.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q5/5

1115

QUESTION ID

S32-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q5/5

SET 32 • 8.G.A.1.a

FORWARD 2

Use rigid motions to decide whether figures are congruent • High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6

Triangle ABC is mapped onto triangle DEF by a rotation followed by a translation.

What can you conclude about the triangles?

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q1/5

1117

QUESTION ID

S32-F2-Q1

TYPE

Short answer

CORRECT ANSWER

They are congruent; their corresponding sides and angles are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q1/5

Explain why two figures are congruent when a sequence of rigid motions maps one onto the other.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q2/5

1118

QUESTION ID

S32-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Rigid motions keep all lengths and angle measures, so every part of one figure matches the corresponding part of the other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q2/5

Which transformation does NOT always map a figure onto a congruent figure?

A. A translation

B. A rotation

C. A dilation with a scale factor of 2

D. A reflection

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q3/5

1119

QUESTION ID

S32-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. A dilation with a scale factor of 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q3/5

True or false?

If a translation maps segment AB onto segment CD, then AB = CD.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q4/5

1120

QUESTION ID

S32-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q4/5

True or false?

Two triangles with the same area must be congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q5/5

1121

QUESTION ID

S32-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q5/5

SET 33 • 8.G.A.1.b • GEOMETRY

Verify that rigid motions take angles to angles of the same measure

MAIN	Grade 8 main question	**8.G.A.1.b**
BACKWARD 1	Find unknown angles with supplementary, complementary, and v…	**7.G.B.5**
BACKWARD 2	Classify triangles by their angles	**4.G.A.2**
FORWARD 1	Specify a sequence of transformations that maps a figure onto a…	**HSG-CO.A.5**
FORWARD 2	Prove theorems about parallelograms — nearest related (no direc…	**HSG-CO.C.11**

SET 33 • 8.G.A.1.b

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.1.b

Angle ABC has A(4, 0), B(0, 0), and C(3, 3). It is reflected across the y-axis.

Find A', B', and C', and explain why angle A'B'C' has the same measure as angle ABC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q1

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q1/5

1124

QUESTION ID

S33-M-Q1

TYPE

Short answer

CORRECT ANSWER

A'(−4, 0), B'(0, 0), C'(−3, 3). Ray BC and ray B'C' each run along a diagonal of the grid squares, so each angle is half of a right angle: both measure 45°.

GRADING NOTE Grade both: the three image points, and a check that both angles measure 45°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q1

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The right triangle with vertices (0, 0), (4, 0), and (0, 4) is translated 2 units right and 1 unit up.

Find the image vertices, and explain why the angle at the image of (0, 0) is still a right angle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q2

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q2/5

1125

QUESTION ID

S33-M-Q2

TYPE

Short answer

CORRECT ANSWER

(2, 1), (6, 1), and (2, 5). The sides from (2, 1) are horizontal and vertical, so they are perpendicular: the angle is 90°.

GRADING NOTE Grade both: the vertices and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q2

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q2/5

Angle DEF has D(0, 3), E(0, 0), and F(3, 0). It is reflected across the line y = x.

What does the image angle measure?

A. 90°

B. 30°

C. 180°

D. 45°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q3

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q3/5

1126

QUESTION ID

S33-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 90°

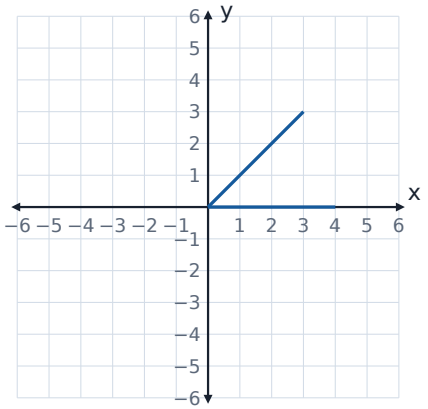
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q3

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The angle has its vertex at the origin. Rotate it 90° counterclockwise about the origin and draw the image. Then compare the measures of the two angles.



Measures: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q4

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q4/5

1127

QUESTION ID

S33-M-Q4

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: An angle with its vertex at the origin and rays through (0, 4) and (−3, 3)

Written: Both angles measure 45°.

GRADING NOTE Grade both: the rotated angle, and the two equal measures (from a protractor or the grid diagonals).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q4

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q4/5

True or false?

Angle ABC with A(2, 0), B(0, 0), and C(0, 2) is translated 4 units right. The image angle measures 45°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q5

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q5/5

1128

QUESTION ID

S33-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The image is still a right angle (90°).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q5

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q5/5

SET 33 • 8.G.A.1.b

BACKWARD 1

Find unknown angles with supplementary, complementary, and vertical angles • Grade 7 • Geometry • 7.G.B.5

Two angles form a straight line. One measures 65°.

What does the other measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q1/5

1130

QUESTION ID

S33-B1-Q1

TYPE

Short answer

CORRECT ANSWER

115°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q1/5

Two lines intersect. One angle measures 40°.

What does the vertical angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q2/5

1131

QUESTION ID

S33-B1-Q2

TYPE

Short answer

CORRECT ANSWER

40°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q2/5

What is the complement of a 25° angle?

A. 65°

B. 75°

C. 155°

D. 25°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q3/5

1132

QUESTION ID

S33-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 65°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q3/5

True or false?

Vertical angles have equal measures.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q4/5

1133

QUESTION ID

S33-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q4/5

True or false?

Supplementary angles add to 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q5/5

1134

QUESTION ID

S33-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q5/5

SET 33 • 8.G.A.1.b

BACKWARD 2

Classify triangles by their angles • Grade 4 • Geometry • 4.G.A.2

A triangle has angles of 40°, 60°, and 80°.

Classify it by its angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q1/5

1136

QUESTION ID

S33-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Acute triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q1/5

A triangle has a 90° angle.

Classify it by its angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q2/5

1137

QUESTION ID

S33-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Right triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q2/5

Which angle measures make an obtuse triangle?

A. 60°, 60°, 60°

B. 30°, 40°, 110°

C. 45°, 45°, 90°

D. 50°, 60°, 70°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q3/5

1138

QUESTION ID

S33-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 30°, 40°, 110°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q3/5

True or false?

An acute triangle has three angles that each measure less than 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q4/5

1139

QUESTION ID

S33-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q4/5

True or false?

A right triangle can have an obtuse angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q5/5

1140

QUESTION ID

S33-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q5/5

SET 33 • 8.G.A.1.b

FORWARD 1

Specify a sequence of transformations that maps a figure onto another • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle A(1, 1), B(3, 1), C(1, 4) is mapped onto A'(−1, −1), B'(−3, −1), C'(−1, −4).

Describe a transformation or sequence of transformations that does this.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q1/5

1142

QUESTION ID

S33-F1-Q1

TYPE

Short answer

CORRECT ANSWER

A 180° rotation about the origin.

GRADING NOTE A reflection across the x-axis followed by a reflection across the y-axis (or the reverse) is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q1/5

Which single transformation maps (2, 3) to (2, −3) and (5, 1) to (5, −1)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q2/5

1143

QUESTION ID

S33-F1-Q2

TYPE

Short answer

CORRECT ANSWER

A reflection across the x-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q2/5

Which rotation about the origin maps (4, 0) to (0, 4)?

A. 270° counterclockwise

B. 180°

C. 90° clockwise

D. 90° counterclockwise

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q3/5

1144

QUESTION ID

S33-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 90° counterclockwise

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q3/5

True or false?

Reflecting a figure across a line and then reflecting the image across the same line gives back the original figure.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q4/5

1145

QUESTION ID

S33-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q4/5

True or false?

A translation 3 units right followed by a translation 3 units left moves the figure.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q5/5

1146

QUESTION ID
S33-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The figure returns to where it started.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q5/5

SET 33 • 8.G.A.1.b

FORWARD 2

Prove theorems about parallelograms • High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11

⚑ No high-school standard directly continues this skill at this step. This branch uses the nearest related high-school standard (placed at Grade 10 in this bank).

ANSWER KEY Reduced copy of the question page, with its answer

ABCD is a parallelogram. Prove that angle A and angle B are supplementary. Give a reason for each step.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q1/5 • NEAREST RELATED

1148

QUESTION ID

S33-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

Angle A + angle B = 180°

GRADING NOTE REQUIRED WORK: A valid chain of reasons. Other valid orders are accepted. A conclusion without reasons is not full credit. Example: AD ∥ BC (definition of a parallelogram), and AB is a transversal; angles A and B are same-side interior angles, so they are supplementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q1/5 • NEAREST RELATED

In parallelogram ABCD, angle A = 70°.

Find the other three angles and give a reason for each.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q2/5 • NEAREST RELATED

1149

QUESTION ID

S33-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Angle B = 110° and angle D = 110° (consecutive angles are supplementary); angle C = 70° (opposite angles are congruent).

GRADING NOTE Grade the angles and the reasons.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F2-Q2**

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q2/5 • NEAREST RELATED

Which statement about every parallelogram can be proved?

A. All of its angles are right angles.

B. Its diagonals are congruent.

C. Its diagonals bisect each other.

D. Its diagonals are perpendicular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q3/5 • NEAREST RELATED

1150

QUESTION ID

S33-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Its diagonals bisect each other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

Opposite sides of a parallelogram are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q4/5 • NEAREST RELATED

1151

QUESTION ID

S33-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

A quadrilateral with one pair of parallel sides must be a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q5/5 • NEAREST RELATED

1152

QUESTION ID

S33-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q5/5 • NEAREST RELATED

SET 34 • 8.G.A.1.c • GEOMETRY

Verify that rigid motions take parallel lines to parallel lines

MAIN	Grade 8 main question	**8.G.A.1.c**
BACKWARD 1	Identify parallel lines and segments	**4.G.A.1**
BACKWARD 2	Draw polygons in the coordinate plane and find side lengths	**6.G.A.3**
FORWARD 1	Prove theorems about angles formed by parallel lines	**HSG-CO.C.9**
FORWARD 2	Use coordinates to prove simple geometric theorems	**HSG-GPE.B.4**

SET 34 • 8.G.A.1.c

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.1.c

The parallel lines y = 2x and y = 2x + 3 are reflected across the x-axis.

Write the equations of the images and check whether the images are parallel.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q1

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q1/5

1155

QUESTION ID

S34-M-Q1

TYPE

Short answer

CORRECT ANSWER

y = −2x and y = −2x − 3; both have slope −2 (and different y-intercepts), so the images are parallel.

GRADING NOTE Grade both: the two equations and the slope check.

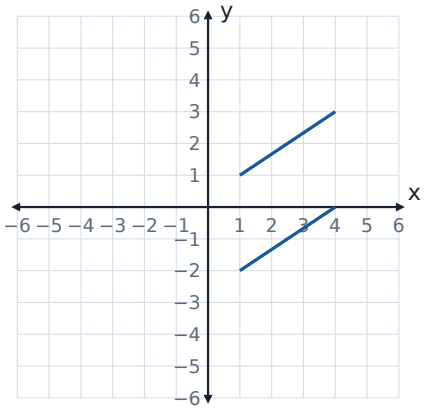
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q1

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The two segments are parallel. Translate both segments 4 units to the left and draw the images. Then find the slope of each image and compare.

Slopes: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q2

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q2/5

1156

GRADING NOTE Grade both: the two image segments, and the two equal slopes.

QUESTION ID

S34-M-Q2

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: Segments from (−3, 1) to (0, 3) and from (−3, −2) to (0, 0)

Written: Both images have slope 2/3, so they are still parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q2

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q2/5

The parallel segments from (0, 0) to (2, 1) and from (0, 3) to (2, 4) are rotated 90° counterclockwise about the origin.

What are the slopes of the images?

A. -2 and $\frac{1}{2}$

B. Both 2

C. Both -2

D. Both $\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q3

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q3/5

1157

QUESTION ID

S34-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Both −2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q3

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q3/5

The line y = 2x + 1 is translated 3 units up.

What is the equation of the image? Is the image parallel to the original line?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q4

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q4/5

1158

QUESTION ID

S34-M-Q4

TYPE

Short answer

CORRECT ANSWER

y = 2x + 4; yes, the slope is still 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q4

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q4/5

True or false?

The parallel segments from (1, 1) to (3, 2) and from (1, 4) to (3, 5) are translated 2 units down. The images have different slopes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q5

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q5/5

1159

QUESTION ID

S34-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Both images have slope 1/2, so they are still parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q5

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q5/5

SET 34 • 8.G.A.1.c

BACKWARD 1

Identify parallel lines and segments • Grade 4 • Geometry • 4.G.A.1

Name a quadrilateral that has two pairs of parallel sides.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q1/5

1161

QUESTION ID

S34-B1-Q1

TYPE

Short answer

CORRECT ANSWER

A rectangle

GRADING NOTE Any parallelogram (square, rhombus, rectangle) is correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q1/5

How many pairs of parallel sides does a trapezoid that is not a parallelogram have?

Pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q2/5

1162

QUESTION ID

S34-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q2/5

Which capital letter contains a pair of parallel segments?

A. V

B. H

C. T

D. L

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q3/5

1163

QUESTION ID

S34-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. H

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q3/5

True or false?

Parallel lines never meet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q4/5

1164

QUESTION ID

S34-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q4/5

True or false?

Perpendicular lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q5/5

1165

QUESTION ID

S34-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q5

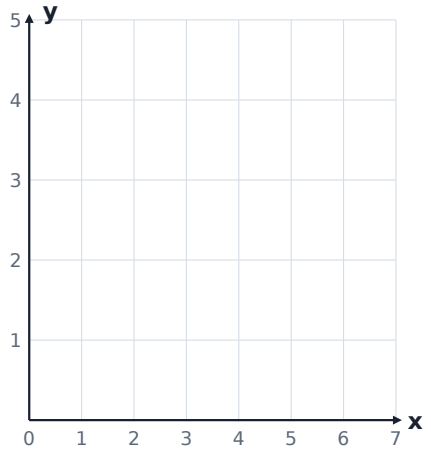
Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q5/5

SET 34 • 8.G.A.1.c

BACKWARD 2

Draw polygons in the coordinate plane and find side lengths • Grade 6 • Geometry • 6.G.A.3

Draw the quadrilateral with vertices (1, 1), (5, 1), (6, 4), and (2, 4).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q1/5

1167

QUESTION ID

S34-B2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The quadrilateral with those four vertices (a parallelogram)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q1/5

A quadrilateral has vertices (1, 1), (5, 1), (6, 4), and (2, 4).

How long is the side from (1, 1) to (5, 1)?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q2/5

1168

QUESTION ID

S34-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q2/5

A quadrilateral has vertices (1, 1), (5, 1), (6, 4), and (2, 4).

Which side is parallel to the side from (1, 1) to (5, 1)?

A. From (1, 1) to (6, 4)

B. From (2, 4) to (6, 4)

C. From (1, 1) to (2, 4)

D. From (5, 1) to (6, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q3/5

1169

QUESTION ID

S34-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. From (2, 4) to (6, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q3/5

True or false?

The side from (2, 4) to (6, 4) is 4 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q4/5

1170

QUESTION ID

S34-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The segment from (1, 1) to (1, 4) is 4 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q5/5

1171

QUESTION ID

S34-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 3 units long.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q5/5

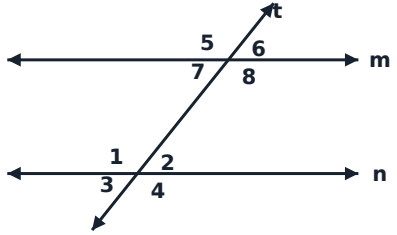
SET 34 • 8.G.A.1.c

FORWARD 1

Prove theorems about angles formed by parallel lines • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9

ANSWER KEY Reduced copy of the question page, with its answer

In the figure, lines m and n are parallel. Prove that angle 2 = angle 7. Give a reason for each step.



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q1/5

1173

QUESTION ID

S34-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

Angle 2 = angle 7

GRADING NOTE REQUIRED WORK: A valid chain of reasons. Other valid orders are accepted. A conclusion without reasons is not full credit. Example: angle 2 = angle 6 (corresponding angles, m ∥ n); angle 6 = angle 7 (vertical angles); so angle 2 = angle 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q1/5

Two parallel lines are cut by a transversal. One of the interior angles measures 72°.

What does the other interior angle on the same side of the transversal measure? Give the reason.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q2/5

1174

QUESTION ID

S34-F1-Q2

TYPE

Short answer

CORRECT ANSWER

108°; same-side interior angles are supplementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q2/5

Which statement can be proved from "corresponding angles formed by parallel lines are congruent" and "vertical angles are congruent"?

A. Alternate interior angles are congruent.

B. The transversal is parallel to the lines.

C. The lines are perpendicular.

D. All the angles are right angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q3/5

1175

QUESTION ID

S34-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Alternate interior angles are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q3/5

True or false?

If alternate interior angles formed by a transversal are congruent, then the two lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q4/5

1176

QUESTION ID

S34-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q4/5

True or false?

Same-side interior angles formed by parallel lines are always congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q5/5

1177

QUESTION ID

S34-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are supplementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q5/5

SET 34 • 8.G.A.1.c

FORWARD 2

Use coordinates to prove simple geometric theorems • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Show that the quadrilateral with vertices (0, 0), (4, 0), (5, 3), and (1, 3) is a parallelogram.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q1/5

1179

QUESTION ID

S34-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The sides from (0, 0) to (4, 0) and from (1, 3) to (5, 3) both have slope 0; the sides from (0, 0) to (1, 3) and from (4, 0) to (5, 3) both have slope 3. Both pairs of opposite sides are parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q1/5

Is the triangle with vertices (0, 0), (4, 0), and (0, 3) a right triangle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q2/5

1180

QUESTION ID

S34-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. One side is horizontal and one is vertical, so they are perpendicular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-F2-Q2**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q2/5

In order, (0, 0), (2, 0), (3, 2), and a fourth point form a parallelogram.

Which is the fourth point?

A. (0, 3)

B. (3, 0)

C. (1, 2)

D. (2, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q3/5

1181

QUESTION ID

S34-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (1, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q3/5

True or false?

A quadrilateral whose opposite sides have equal slopes is a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q4/5

1182

QUESTION ID

S34-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q4/5

True or false?

The points (0, 0), (1, 1), and (2, 3) lie on one line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q5/5

1183

QUESTION ID

S34-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q5/5

SET 35 • 8.G.A.2 • GEOMETRY

Describe a sequence of rigid motions that shows two figures are congruent

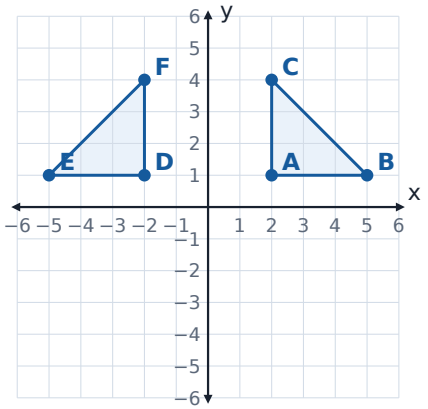
MAIN	Grade 8 main question	**8.G.A.2**
BACKWARD 1	Draw a polygon from its vertices	**6.G.A.3**
BACKWARD 2	Use signs to locate points across the axes	**6.NS.C.6.b**
FORWARD 1	Use corresponding parts to explain triangle congruence	**HSG-CO.B.7**
FORWARD 2	Explain the SAS, ASA, and SSS criteria with rigid motions	**HSG-CO.B.8**

SET 35 • 8.G.A.2

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.2

Describe a rigid motion that maps triangle ABC onto triangle DEF.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q1/5

1186

QUESTION ID

S35-M-Q1

TYPE

Short answer

CORRECT ANSWER

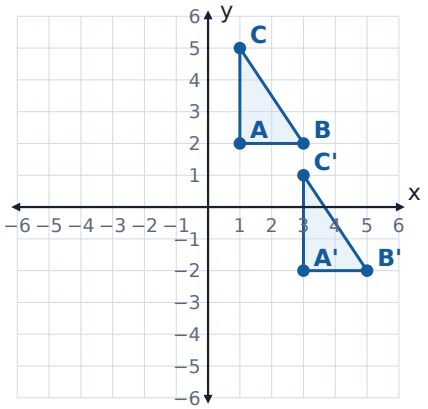
A reflection across the y-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q1/5

Describe a rigid motion that maps triangle ABC onto triangle A'B'C'.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q2/5

1187

QUESTION ID

S35-M-Q2

TYPE

Short answer

CORRECT ANSWER

A translation 2 units right and 4 units down

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q2/5

Figure P is mapped onto figure Q by a 90° rotation followed by a translation.

Which statement is true?

A. P and Q are congruent.

B. Q is larger than P.

C. P and Q are not congruent because they face different directions.

D. P and Q have different angle measures.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q3/5

1188

QUESTION ID

S35-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. P and Q are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q3/5

Explain why a triangle and its image after a dilation with a scale factor of 2 are not congruent.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q4/5

1189

QUESTION ID

S35-M-Q4

TYPE

Short answer

CORRECT ANSWER

The dilation doubles every side length, but rigid motions (which show congruence) keep lengths the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q4/5

True or false?

Two figures are congruent if a sequence of rotations, reflections, and translations maps one onto the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q5/5

1190

QUESTION ID

S35-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q5

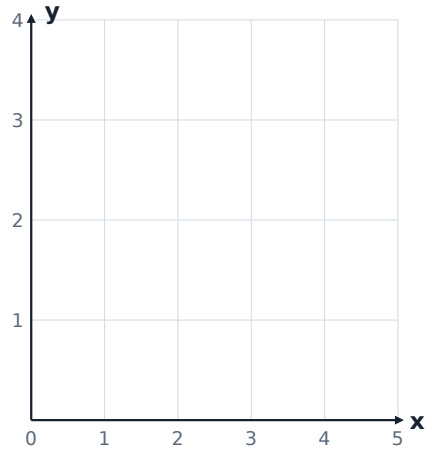
Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q5/5

SET 35 • 8.G.A.2

BACKWARD 1

Draw a polygon from its vertices • Grade 6 • Geometry • 6.G.A.3

Draw the triangle with vertices (1, 1), (4, 1), and (1, 3).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q1/5

1192

QUESTION ID

S35-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The triangle with those three vertices

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q1/5

A triangle has vertices (1, 1), (4, 1), and (1, 3).

How long is the side from (1, 1) to (4, 1)?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q2/5

1193

QUESTION ID

S35-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q2/5

A rectangle has vertices (0, 0), (5, 0), and (5, 2).

What is the fourth vertex?

A. (0, 2)

B. (2, 0)

C. (5, 5)

D. (2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q3/5

1194

QUESTION ID

S35-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (0, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q3/5

True or false?

The side from (1, 1) to (1, 3) is 2 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q4/5

1195

QUESTION ID

S35-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q4/5

True or false?

The points (2, 1), (2, 5), and (6, 1) form a triangle with a side 3 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q5/5

1196

QUESTION ID

S35-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its sides along the grid lines are 4 units long.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q5/5

SET 35 • 8.G.A.2

BACKWARD 2

Use signs to locate points across the axes • Grade 6 • The Number System • 6.NS.C.6.b

The point (5, 2) is reflected across the y-axis.

What is its image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q1/5

1198

QUESTION ID

S35-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(−5, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q1/5

The point $(-1, -4)$ is reflected across the x-axis.

What is its image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q2/5

1199

QUESTION ID

S35-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$(-1, 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q2/5

Points (3, 2) and (3, −2) differ only in the sign of the y-coordinate.

How are they related?

A. They are reflections across the x-axis.

B. They are reflections across the y-axis.

C. They are the same point.

D. They are 3 units apart.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q3/5

1200

QUESTION ID

S35-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. They are reflections across the x-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q3/5

True or false?

(−4, 1) and (4, 1) are reflections of each other across the y-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q4/5

1201

QUESTION ID

S35-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q4/5

True or false?

(2, −6) and (−2, 6) are reflections of each other across the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q5/5

1202

QUESTION ID

S35-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q5/5

SET 35 • 8.G.A.2

FORWARD 1

Use corresponding parts to explain triangle congruence • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7

A sequence of rigid motions maps triangle ABC onto triangle DEF, taking A to D, B to E, and C to F. AB = 5 and angle B = 40°.

Which side and which angle of DEF must equal these? Explain using rigid motions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q1/5

1204

QUESTION ID

S35-F1-Q1

TYPE

Short answer

CORRECT ANSWER

DE = 5 and angle E = 40°: the rigid motions map segment AB onto DE and angle B onto angle E, and rigid motions keep lengths and angle measures.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q1/5

Two triangles have all corresponding sides and all corresponding angles congruent.

What can you conclude? Explain with rigid motions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q2/5

1205

QUESTION ID

S35-F1-Q2

TYPE

Short answer

CORRECT ANSWER

The triangles are congruent; a sequence of rigid motions maps one onto the other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q2/5

Triangle PQR is congruent to triangle XYZ.

Which statement must be true?

A. PQ = YZ

B. QR = YZ

C. Angle P = angle Z

D. PR = XY

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q3/5

1206

QUESTION ID

S35-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. QR = YZ

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q3/5

True or false?

Corresponding parts of congruent triangles are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q4/5

1207

QUESTION ID

S35-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If triangle ABC is congruent to triangle DEF, then AC = EF.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q5/5

1208

QUESTION ID

S35-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE AC = DF.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q5/5

SET 35 • 8.G.A.2

FORWARD 2

Explain the SAS, ASA, and SSS criteria with rigid motions • High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Triangles ABC and DEF have AB = DE, angle A = angle D, and AC = DF (SAS).

Explain the rigid motions that map triangle ABC onto triangle DEF, and why C must land on F.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q1/5

1210

QUESTION ID

S35-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

A rigid motion maps A to D, B to E, and C to F, so the triangles are congruent.

GRADING NOTE REQUIRED WORK: A complete argument: translate A to D; rotate so ray AB lies along ray DE, and B lands on E because AB = DE; if C is on the other side of DE from F, reflect across DE. Since angle A = angle D, ray AC lies along ray DF, and since AC = DF, C lands on F. Any equivalent order is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangles ABC and DEF have angle A = angle D, AB = DE, and angle B = angle E (ASA).

After rigid motions map segment AB onto segment DE (with C and F on the same side), explain why C must land on F.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q2/5

1211

QUESTION ID

S35-F2-Q2

TYPE

Answer with required work

CORRECT ANSWER

C lands on F, so the triangles are congruent.

GRADING NOTE REQUIRED WORK: Because angle A = angle D, the image of ray AC lies along ray DF; because angle B = angle E, the image of ray BC lies along ray EF. C is on both rays, and the two rays meet at only one point, F.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q2/5

Triangles ABC and DEF have AB = DE, BC = EF, and AC = DF (SSS). Rigid motions map AB onto DE, with C and F on the same side of DE.

Explain why C must land on F.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q3/5

1212

QUESTION ID

S35-F2-Q3

TYPE

Short answer

CORRECT ANSWER

C is at distance AC from D and BC from E. On that side of DE, only one point (F) is at those two distances (the two circles meet there once), so C lands on F.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q3/5

Why do three pairs of equal angles (AAA) NOT prove that two triangles are congruent?

A. A triangle has only two angles that matter.

B. Angles cannot be measured exactly.

C. A dilation keeps every angle but changes the side lengths.

D. AAA always proves congruence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q4/5

1213

QUESTION ID

S35-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. A dilation keeps every angle but changes the side lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q4/5

True or false?

In the rigid-motion explanation of SAS, a reflection may be needed to put C on the same side of DE as F.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q5/5

1214

QUESTION ID

S35-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q5/5

SET 36 • 8.G.A.3 • GEOMETRY

Translate figures on the coordinate plane

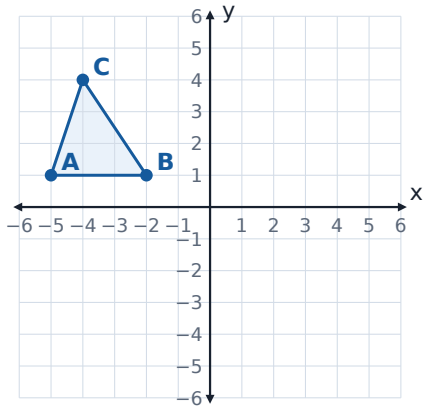
MAIN	Grade 8 main question	**8.G.A.3**
BACKWARD 1	Find and plot points in all four quadrants	**6.NS.C.6.c**
BACKWARD 2	Add integers	**7.NS.A.1.b**
FORWARD 1	Describe transformations as functions and decide which preserve…	**HSG-CO.A.2**
FORWARD 2	Find the point that divides a segment in a given ratio — nearest r…	**HSG-GPE.B.6**

SET 36 • 8.G.A.3

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.3

Translate triangle ABC 4 units right and 2 units down. Draw the image.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q1/5

1217

QUESTION ID

S36-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The triangle with vertices A'(−1, −1), B'(2, −1), and C'(0, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q1/5

The point P(3, −2) is translated 5 units left and 4 units up.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q2/5

1218

QUESTION ID

S36-M-Q2

TYPE

Short answer

CORRECT ANSWER

(−2, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q2/5

Which rule describes a translation 6 units down?

A. $(x, y) \to (x, y - 6)$

B. $(x, y) \to (x, y + 6)$

C. $(x, y) \to (x + 6, y)$

D. $(x, y) \to (x - 6, y)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q3/5

1219

QUESTION ID

S36-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $(x, y) \to (x, y - 6)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q3/5

A translation maps A(2, 5) to A'(−1, 7).

Where does it map B(4, 0)?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q4/5

1220

QUESTION ID

S36-M-Q4

TYPE

Short answer

CORRECT ANSWER

(1, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q4/5

True or false?

The rule (x, y) → (x + 3, y − 1) moves every point 3 units right and 1 unit down.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q5/5

1221

QUESTION ID

S36-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q5

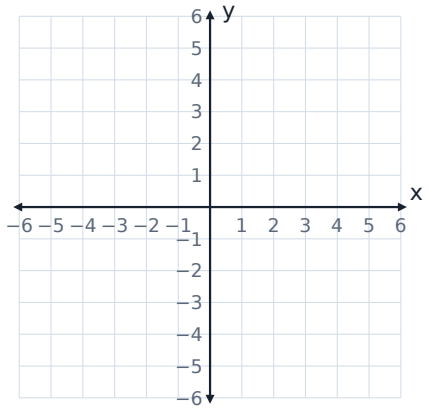
Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q5/5

SET 36 • 8.G.A.3

BACKWARD 1

Find and plot points in all four quadrants • Grade 6 • The Number System • 6.NS.C.6.c

Plot the points (−3, 2) and (4, −5).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q1/5

1223

QUESTION ID

S36-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

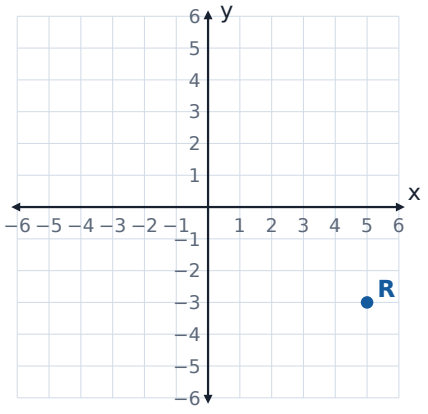
Points at (−3, 2) and (4, −5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q1/5

What are the coordinates of point R?



Coordinates: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q2/5

1224

QUESTION ID

S36-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(5, −3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q2/5

Which point is 2 units to the left of (1, 4)?

A. (1, 6)

B. (−1, 4)

C. (1, 2)

D. (3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q3/5

1225

QUESTION ID

S36-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (−1, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q3/5

True or false?

Moving 3 units up from (2, −1) gives (2, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q4/5

1226

QUESTION ID

S36-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q4/5

True or false?

Moving 4 units left from (0, 0) gives (0, −4).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q5/5

1227

QUESTION ID

S36-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q5/5

SET 36 • 8.G.A.3

BACKWARD 2

Add integers • Grade 7 • The Number System • 7.NS.A.1.b

ANSWER KEY Reduced copy of the question page, with its answer

Add.

$-4 + 4$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q1/5

1229

QUESTION ID

S36-B2-Q1

TYPE

Short answer

CORRECT ANSWER

0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q1/5

Add.

$3 + (-5)$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q2/5

1230

QUESTION ID

S36-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Add.

$-1 + 6$

A. 5

B. -5

C. 7

D. -7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q3/5

1231

QUESTION ID

S36-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q3/5

True or false?

$-2 + (-3) = -5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q4/5

1232

QUESTION ID

S36-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q4/5

True or false?

$5 + (-8) = 3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q5/5

1233

QUESTION ID

S36-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q5/5

SET 36 • 8.G.A.3

FORWARD 1

Describe transformations as functions and decide which preserve distance • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2

Write the rule for a translation 2 units left and 7 units up.

Rule: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q1/5

1235

QUESTION ID

S36-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$(x, y) \rightarrow (x - 2, y + 7)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q1/5

Does the rule (x, y) → (2x, y) keep distances the same? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q2/5

1236

QUESTION ID

S36-F1-Q2

TYPE

Short answer

CORRECT ANSWER

No. It doubles horizontal distances; the segment from (0, 0) to (1, 0) becomes 2 units long.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q2/5

Which transformation keeps all distances and angles the same?

A. $(x, y) \to (x, 2y)$

B. $(x, y) \to (x^2, y)$

C. $(x, y) \to (3x, 3y)$

D. $(x, y) \to (x + 4, y - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q3/5

1237

QUESTION ID

S36-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (x, y) → (x + 4, y − 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q3/5

True or false?

A translation is a function whose inputs and outputs are points of the plane.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q4/5

1238

QUESTION ID

S36-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q4/5

True or false?

The rule (x, y) → (x + 1, y + 1) changes the length of every segment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q5/5

1239

QUESTION ID

S36-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q5/5

SET 36 • 8.G.A.3

FORWARD 2

Find the point that divides a segment in a given ratio • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6

⚑ No high-school standard directly continues this skill at this step. This branch uses the nearest related high-school standard (placed at Grade 10 in this bank).

Find the midpoint of the segment from (2, 4) to (8, 10).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q1/5 • NEAREST RELATED

1241

QUESTION ID

S36-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(5, 7)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-F2-Q1**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q1/5 • NEAREST RELATED

Find the point that is $\frac{1}{3}$ of the way from A(0, 0) to B(9, 6).

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q2/5 • NEAREST RELATED

1242

QUESTION ID

S36-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(3, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-F2-Q2**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q2/5 • NEAREST RELATED

Point P is on segment AB from A(1, 1) to B(7, 4), with AP : PB = 2 : 1.

What are the coordinates of P?

A. (3, 2)

B. (4, 2.5)

C. (6, 3)

D. (5, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q3/5 • NEAREST RELATED

1243

QUESTION ID

S36-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (5, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

The midpoint of the segment from (−4, 2) to (6, −2) is (1, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q4/5 • NEAREST RELATED

1244

QUESTION ID

S36-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

The point halfway from (0, 0) to (10, 4) is (10, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q5/5 • NEAREST RELATED

1245

QUESTION ID

S36-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The midpoint is (5, 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-F2-Q5**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q5/5 • NEAREST RELATED

SET 37 • 8.G.A.3 • GEOMETRY

Reflect figures across the x-axis and the y-axis

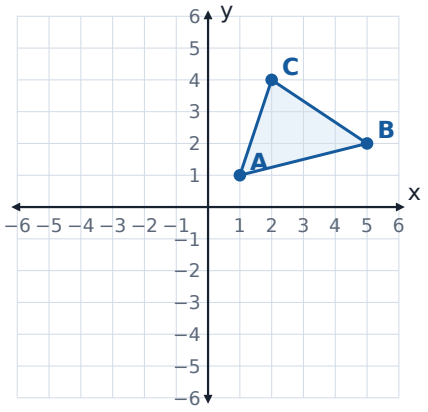
MAIN	Grade 8 main question	**8.G.A.3**
BACKWARD 1	Locate points that are reflections across the axes	**6.NS.C.6.b**
BACKWARD 2	Find the opposite of a number	**6.NS.C.6.a**
FORWARD 1	Describe the rotations and reflections that carry a polygon onto it…	**HSG-CO.A.3**
FORWARD 2	Draw the image of a figure after a sequence of transformations	**HSG-CO.B.6**

SET 37 • 8.G.A.3

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.3

Reflect triangle ABC across the x-axis. Draw the image.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q1/5

1248

QUESTION ID

S37-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The triangle with vertices A'(1, −1), B'(5, −2), and C'(2, −4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q1/5

The point $(-3, 5)$ is reflected across the y-axis.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q2/5

1249

QUESTION ID

S37-M-Q2

TYPE

Short answer

CORRECT ANSWER

(3, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q2/5

Which rule describes a reflection across the x-axis?

A. (x, y) → (−x, y)

B. (x, y) → (−x, −y)

C. (x, y) → (y, x)

D. (x, y) → (x, −y)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q3/5

1250

QUESTION ID

S37-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (x, y) → (x, −y)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q3/5

A segment has endpoints (2, 3) and (5, 1). It is reflected across the y-axis.

What are the endpoints of the image?

Endpoints: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q4/5

1251

QUESTION ID

S37-M-Q4

TYPE

Short answer

CORRECT ANSWER

(−2, 3) and (−5, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q4/5

True or false?

Reflecting (4, −6) across the x-axis gives (−4, −6).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q5/5

1252

QUESTION ID

S37-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The image is (4, 6).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q5/5

SET 37 • 8.G.A.3

BACKWARD 1

Locate points that are reflections across the axes • Grade 6 • The Number System • 6.NS.C.6.b

The point (7, −2) is reflected across the x-axis.

What is its image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q1/5

1254

QUESTION ID

S37-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(7, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q1/5

The point (−6, −3) is reflected across the y-axis.

What is its image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q2/5

1255

QUESTION ID

S37-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(6, −3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q2/5

Reflecting a point across the y-axis changes...

A. the sign of the y-coordinate

B. neither sign

C. the sign of the x-coordinate

D. both signs

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q3/5

1256

QUESTION ID

S37-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. the sign of the x-coordinate

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q3/5

True or false?

(0, 5) is its own reflection across the y-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q4/5

1257

QUESTION ID

S37-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q4/5

True or false?

The reflection of (−1, 8) across the y-axis is (−1, −8).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q5/5

1258

QUESTION ID

S37-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q5/5

SET 37 • 8.G.A.3

BACKWARD 2

Find the opposite of a number • Grade 6 • The Number System • 6.NS.C.6.a

What is the opposite of −7?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q1/5

1260

QUESTION ID

S37-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the opposite of 2.5?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q2/5

1261

QUESTION ID

S37-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q2/5

Which pair are opposites?

A. −4 and 4

B. 4 and $\frac{1}{4}$

C. 0 and 1

D. −4 and −4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q3/5

1262

QUESTION ID

S37-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. −4 and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q3/5

True or false?

The opposite of the opposite of 3 is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q4/5

1263

QUESTION ID

S37-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q4/5

True or false?

The opposite of 0 is −1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q5/5

1264

QUESTION ID

S37-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q5/5

SET 37 • 8.G.A.3

FORWARD 1

Describe the rotations and reflections that carry a polygon onto itself • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3

How many lines of symmetry does a square have?

Lines: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q1/5

1266

QUESTION ID

S37-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q1/5

Which rotations (between 0° and 360°) about its center carry a regular hexagon onto itself?

Rotations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q2/5

1267

QUESTION ID

S37-F1-Q2

TYPE

Short answer

CORRECT ANSWER

60°, 120°, 180°, 240°, and 300°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q2/5

Which figure is carried onto itself by a 180° rotation but has no line of symmetry?

A. A parallelogram that is not a rectangle or a rhombus

B. A square

C. An equilateral triangle

D. An isosceles trapezoid

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q3/5

1268

QUESTION ID

S37-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A parallelogram that is not a rectangle or a rhombus

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q3/5

True or false?

A rectangle that is not a square has exactly 2 lines of symmetry.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q4/5

1269

QUESTION ID

S37-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q4/5

True or false?

An isosceles trapezoid is carried onto itself by a 90° rotation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q5/5

1270

QUESTION ID

S37-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q5

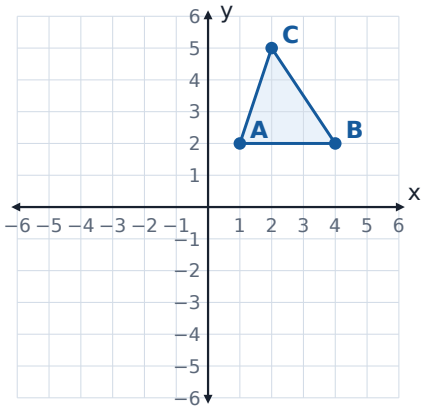
High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q5/5

SET 37 • 8.G.A.3

FORWARD 2

Draw the image of a figure after a sequence of transformations • High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6

Reflect triangle ABC across the x-axis, then reflect that image across the y-axis. Draw the final image.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q1/5

1272

QUESTION ID

S37-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The triangle with vertices A''(−1, −2), B''(−4, −2), and C''(−2, −5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q1/5

What single transformation has the same effect as a reflection across the x-axis followed by a reflection across the y-axis?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q2/5

1273

QUESTION ID

S37-F2-Q2

TYPE

Short answer

CORRECT ANSWER

A 180° rotation about the origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q2/5

The point (3, 1) is reflected across the line y = x.

What is its image?

A. (3, −1)

B. (−1, −3)

C. (−3, 1)

D. (1, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q3/5

1274

QUESTION ID

S37-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (1, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q3/5

True or false?

Reflecting a figure across one line and then across a parallel line has the same effect as a translation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q4/5

1275

QUESTION ID

S37-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q4/5

True or false?

Doing a translation and a reflection in the other order always gives the same image.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q5/5

1276

QUESTION ID

S37-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q5/5

SET 38 • 8.G.A.3 • GEOMETRY

Rotate figures 90° and 180° about the origin

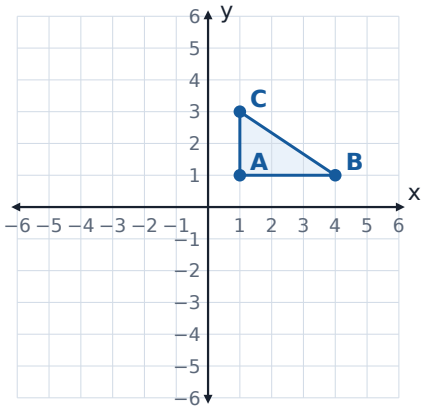
MAIN	Grade 8 main question	**8.G.A.3**
BACKWARD 1	Relate turns to angle measures	**4.MD.C.5.a**
BACKWARD 2	Locate a point reflected across both axes	**6.NS.C.6.b**
FORWARD 1	Describe a rotation in terms of distance and angle	**HSG-CO.A.4**
FORWARD 2	Use relationships among central angles, inscribed angles, and rad…	**HSG-C.A.2**

SET 38 • 8.G.A.3

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.3

Rotate triangle ABC 90° counterclockwise about the origin. Draw the image.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q1/5

1279

QUESTION ID

S38-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The triangle with vertices A'(−1, 1), B'(−1, 4), and C'(−3, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q1/5

The point (5, −2) is rotated 180° about the origin.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q2/5

1280

QUESTION ID

S38-M-Q2

TYPE

Short answer

CORRECT ANSWER

(−5, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q2/5

Which rule describes a 90° clockwise rotation about the origin?

A. (x, y) → (y, x)

B. (x, y) → (y, −x)

C. (x, y) → (−y, x)

D. (x, y) → (−x, −y)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q3/5

1281

QUESTION ID

S38-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (x, y) → (y, −x)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q3/5

The point (2, 3) is rotated 90° counterclockwise about the origin.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q4/5

1282

QUESTION ID

S38-M-Q4

TYPE

Short answer

CORRECT ANSWER

(−3, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q4/5

True or false?

Rotating $(-4, 1)$ by $180°$ about the origin gives $(4, -1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q5/5

1283

QUESTION ID

S38-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q5/5

SET 38 • 8.G.A.3

BACKWARD 1

Relate turns to angle measures • Grade 4 • Measurement & Data • 4.MD.C.5.a

ANSWER KEY Reduced copy of the question page, with its answer

How many degrees are in a quarter turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q1/5

1285

QUESTION ID

S38-B1-Q1

TYPE

Short answer

CORRECT ANSWER

90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q1/5

How many degrees are in a half turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q2/5

1286

QUESTION ID

S38-B1-Q2

TYPE

Short answer

CORRECT ANSWER

180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q2/5

How many degrees are in a three-quarter turn?

A. 360°

B. 180°

C. 270°

D. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q3/5

1287

QUESTION ID

S38-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 270°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q3/5

True or false?

A full turn is 360°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q4/5

1288

QUESTION ID

S38-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q4/5

True or false?

A half turn is 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q5/5

1289

QUESTION ID

S38-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q5/5

SET 38 • 8.G.A.3

BACKWARD 2

Locate a point reflected across both axes • Grade 6 • The Number System • 6.NS.C.6.b

The point (3, 4) is reflected across the x-axis and then across the y-axis.

What is the final image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q1/5

1291

QUESTION ID

S38-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(−3, −4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q1/5

The point (−2, 5) is reflected across both axes.

What is the final image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q2/5

1292

QUESTION ID

S38-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(2, −5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q2/5

Reflecting a point across both axes changes...

A. the signs of both coordinates

B. only the y-coordinate

C. neither coordinate

D. only the x-coordinate

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q3/5

1293

QUESTION ID

S38-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. the signs of both coordinates

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q3/5

True or false?

Reflecting (1, −6) across both axes gives (−1, 6).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q4/5

1294

QUESTION ID

S38-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q4/5

True or false?

Reflecting (4, 4) across both axes gives (4, 4).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q5/5

1295

QUESTION ID

S38-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q5/5

SET 38 • 8.G.A.3

FORWARD 1

Describe a rotation in terms of distance and angle • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4

A rotation of 120° about point O maps a point A (A ≠ O) to A'.

Which lengths and which angle are determined by the rotation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q1/5

1297

QUESTION ID

S38-F1-Q1

TYPE

Short answer

CORRECT ANSWER

OA = OA', and angle AOA' = 120°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q1/5

P(6, 0) is rotated 60° counterclockwise about the origin.

How far is the image from the origin?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q2/5

1298

QUESTION ID

S38-F1-Q2

TYPE

Short answer

CORRECT ANSWER

6 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q2/5

A rotation about point O through an angle θ, with $0° < θ ≤ 180°$, maps each point P other than O to a point P' so that...

A. P' is on the line of reflection

B. P' is twice as far from O as P

C. OP = OP' and angle POP' = θ

D. PP' is parallel to one fixed line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q3/5

1299

QUESTION ID

S38-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. OP = OP' and angle POP' = θ

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q3/5

True or false?

In a rotation by a nonzero angle less than 360°, the center is the only point that does not move.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q4/5

1300

QUESTION ID

S38-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q4/5

True or false?

A 90° rotation maps every horizontal segment to another horizontal segment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q5/5

1301

QUESTION ID
S38-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It maps horizontal segments to vertical segments.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q5/5

SET 38 • 8.G.A.3

FORWARD 2

Use relationships among central angles, inscribed angles, and radii • High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2

⚑ No high-school standard directly continues this skill at this step. This branch uses the nearest related high-school standard (placed at Grade 10 in this bank).

A central angle of a circle measures 80°.

What does an inscribed angle that intercepts the same arc measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q1

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q1/5 • NEAREST RELATED

1303

QUESTION ID

S38-F2-Q1

TYPE

Short answer

CORRECT ANSWER

40°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-F2-Q1**

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q1/5 • NEAREST RELATED

An inscribed angle intercepts a semicircle.

What is its measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q2

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q2/5 • NEAREST RELATED

1304

QUESTION ID

S38-F2-Q2

TYPE

Short answer

CORRECT ANSWER

90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-F2-Q2**

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q2/5 • NEAREST RELATED

A radius is drawn to the point where a tangent line touches a circle.

What angle do the radius and the tangent line form?

A. 90°

B. 45°

C. 180°

D. 60°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q3

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q3/5 • NEAREST RELATED

1305

QUESTION ID

S38-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q3

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

Rotating a circle about its center by any angle carries the circle onto itself.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q4

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q4/5 • NEAREST RELATED

1306

QUESTION ID

S38-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q4

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

An inscribed angle measures twice its intercepted arc.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q5

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q5/5 • NEAREST RELATED

1307

QUESTION ID

S38-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It measures half of its intercepted arc.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q5

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q5/5 • NEAREST RELATED

SET 39 • 8.G.A.3 • GEOMETRY

Dilate figures on the coordinate plane with the origin as the center

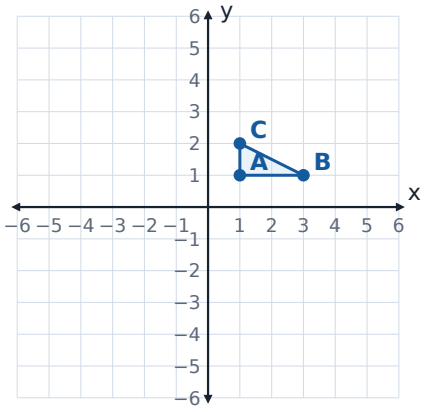
MAIN	Grade 8 main question	**8.G.A.3**
BACKWARD 1	Use a scale factor to find lengths in a scale drawing	**7.G.A.1**
BACKWARD 2	Multiply rational numbers	**7.NS.A.2.a**
FORWARD 1	Verify the properties of dilations	**HSG-SRT.A.1**
FORWARD 2	Use similar triangles to solve problems	**HSG-SRT.B.5**

SET 39 • 8.G.A.3

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.3

Dilate triangle ABC by a scale factor of 2 with the origin as the center. Draw the image.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q1/5

1310

QUESTION ID

S39-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The triangle with vertices A'(2, 2), B'(6, 2), and C'(2, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q1/5

The point $(6, -4)$ is dilated by a scale factor of $\frac{1}{2}$ with the origin as the center.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q2/5

1311

QUESTION ID

S39-M-Q2

TYPE

Short answer

CORRECT ANSWER

(3, −2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q2/5

Which rule describes a dilation with a scale factor of 3 and the origin as the center?

A. $(x, y) \rightarrow (3x, y)$

B. $(x, y) \rightarrow (x + 3, y + 3)$

C. $(x, y) \rightarrow (\frac{x}{3}, \frac{y}{3})$

D. $(x, y) \rightarrow (3x, 3y)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q3/5

1312

QUESTION ID

S39-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (x, y) → (3x, 3y)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q3/5

A dilation with the origin as the center maps (4, 2) to (10, 5).

What is the scale factor?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q4/5

1313

GRADING NOTE 5/2 is also correct.

QUESTION ID

S39-M-Q4

TYPE

Short answer

CORRECT ANSWER

2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q4/5

True or false?

A dilation with a scale factor of $\frac{1}{3}$ and the origin as the center maps (9, 6) to (3, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q5/5

1314

QUESTION ID

S39-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q5/5

SET 39 • 8.G.A.3

BACKWARD 1

Use a scale factor to find lengths in a scale drawing • Grade 7 • Geometry • 7.G.A.1

A 4 cm segment is enlarged by a scale factor of 3.

How long is the new segment?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q1/5

1316

QUESTION ID

S39-B1-Q1

TYPE

Short answer

CORRECT ANSWER

12 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q1/5

A 10 cm segment is reduced by a scale factor of $\frac{1}{2}$.

How long is the new segment?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q2/5

1317

QUESTION ID

S39-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q2/5

A rectangle 2 cm by 5 cm is enlarged by a scale factor of 4.

What are the new dimensions?

A. 8 cm by 20 cm

B. 8 cm by 5 cm

C. 6 cm by 9 cm

D. 2 cm by 20 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q3/5

1318

QUESTION ID

S39-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 8 cm by 20 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q3/5

True or false?

A scale factor greater than 1 makes a figure larger.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q4/5

1319

QUESTION ID

S39-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q4/5

True or false?

A scale factor of $\frac{1}{4}$ makes each length 4 times as long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q5/5

1320

QUESTION ID

S39-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q5/5

SET 39 • 8.G.A.3

BACKWARD 2

Multiply rational numbers • Grade 7 • The Number System • 7.NS.A.2.a

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

$2 \times (-4)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q1/5

1322

QUESTION ID

S39-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

$\frac{1}{2} \times (-6)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q2/5

1323

QUESTION ID

S39-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q2/5

Multiply.

3.5×4

A. 12

B. 0.875

C. 14

D. 7.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q3/5

1324

QUESTION ID

S39-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q3/5

True or false?

$3 \times (-2) = -6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q4/5

1325

QUESTION ID

S39-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q4/5

True or false?

$\frac{1}{3} \times 9 = 27$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q5/5

1326

QUESTION ID

S39-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q5/5

SET 39 • 8.G.A.3

FORWARD 1

Verify the properties of dilations • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1

A dilation with the origin as the center and a scale factor of 3 is applied to the line y = x + 1.

What is the equation of the image?

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q1/5

1328

QUESTION ID

S39-F1-Q1

TYPE

Short answer

CORRECT ANSWER

y = x + 3 (a line parallel to the original)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q1/5

A segment 4 units long is dilated by a scale factor of 2.5.

How long is the image?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q2/5

1329

QUESTION ID

S39-F1-Q2

TYPE

Short answer

CORRECT ANSWER

10 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F1-Q2**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q2/5

A dilation is applied to a line that passes through the center of dilation.

What is the image?

A. A parallel line

B. A single point

C. A perpendicular line

D. The same line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q3/5

1330

QUESTION ID

S39-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The same line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q3/5

True or false?

A dilation maps a line that does not pass through the center to a parallel line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q4/5

1331

QUESTION ID

S39-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q4/5

True or false?

A dilation with a scale factor of 2 doubles every angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q5/5

1332

QUESTION ID

S39-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Dilations keep angle measures the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q5/5

SET 39 • 8.G.A.3

FORWARD 2

Use similar triangles to solve problems • High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

A 6-foot person casts a 4-foot shadow. At the same time, a tree casts a 22-foot shadow.

How tall is the tree?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q1

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q1/5

1334

QUESTION ID

S39-F2-Q1

TYPE

Short answer

CORRECT ANSWER

33 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q1

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q1/5

Triangle ABC is similar to triangle DEF, with AB = 8, DE = 12, and BC = 10.

Find EF.

EF = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q2

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q2/5

1335

QUESTION ID

S39-F2-Q2

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F2-Q2**

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q2/5

In triangle ABC, D is on AB and E is on AC, and segment DE is parallel to BC. AD = 3, DB = 6, and AE = 4.

How long is EC?

A. 2

B. 12

C. 8

D. 4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q3

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q3/5

1336

QUESTION ID

S39-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q3/5

True or false?

In similar triangles, the ratio of the perimeters equals the scale factor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q4

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q4/5

1337

QUESTION ID

S39-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q4

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q4/5

True or false?

Triangles with sides 3, 4, 5 and 6, 8, 12 are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q5

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q5/5

1338

QUESTION ID

S39-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q5

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q5/5

SET 40 • 8.G.A.4 • GEOMETRY

Describe a sequence of transformations that shows two figures are similar

MAIN	Grade 8 main question	**8.G.A.4**
BACKWARD 1	Find a scale factor between scale copies	**7.G.A.1**
BACKWARD 2	Decide from a table whether two quantities are in a constant ratio	**7.RP.A.2.a**
FORWARD 1	Use the definition of similarity transformations to prove equal ang…	**HSG-SRT.A.2**
FORWARD 2	Use a translation and a dilation to show that any two circles are s…	**HSG-C.A.1**

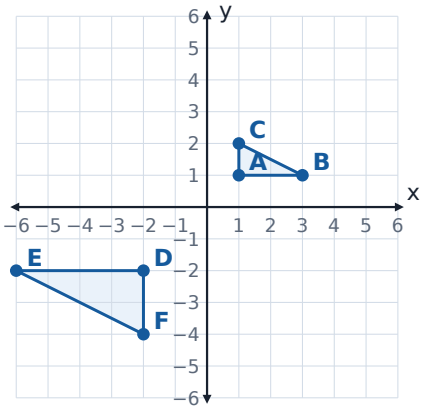
SET 40 • 8.G.A.4

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Describe a sequence of transformations that maps triangle ABC onto triangle DEF.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q1

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q1/5

1341

QUESTION ID

S40-M-Q1

TYPE

Short answer

CORRECT ANSWER

A dilation with a scale factor of 2 and the origin as the center, followed by a 180° rotation about the origin.

GRADING NOTE Any correct sequence is accepted, for example the rotation first, then the dilation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q1

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q1/5

Explain why a 2-by-3 rectangle and a 4-by-6 rectangle are similar.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q2

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q2/5

1342

QUESTION ID

S40-M-Q2

TYPE

Short answer

CORRECT ANSWER

A dilation with a scale factor of 2 (followed by rigid motions to move it into place) maps the first rectangle onto the second.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q2

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q2/5

Which sequence of transformations maps a figure onto a figure that is similar but not congruent?

A. A rotation, then a translation

B. A translation only

C. Two reflections

D. A dilation with a scale factor of 3, then a translation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q3

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q3/5

1343

QUESTION ID

S40-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A dilation with a scale factor of 3, then a translation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q3

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q3/5

Are a 3-4-5 triangle and a 5-12-13 triangle similar? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q4

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q4/5

1344

QUESTION ID

S40-M-Q4

TYPE

Short answer

CORRECT ANSWER

No. The ratios of corresponding sides (3/5, 4/12, 5/13) are not equal, so no dilation and rigid motions can map one onto the other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q4

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q4/5

True or false?

Two figures are similar if a sequence of rotations, reflections, translations, and dilations maps one onto the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q5

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q5/5

1345

QUESTION ID

S40-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q5

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q5/5

SET 40 • 8.G.A.4

BACKWARD 1

Find a scale factor between scale copies • Grade 7 • Geometry • 7.G.A.1

A 3 cm side becomes 12 cm in a scale copy.

What is the scale factor?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q1/5

1347

QUESTION ID

S40-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q1/5

A 15 m side becomes 5 m in a scale copy.

What is the scale factor?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q2/5

1348

QUESTION ID

S40-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q2/5

A triangle's sides are 2, 3, and 4. Which side lengths are a scale copy?

A. 6, 9, 12

B. 4, 5, 6

C. 2, 6, 8

D. 3, 4, 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q3/5

1349

QUESTION ID

S40-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 6, 9, 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q3/5

True or false?

A scale copy with a scale factor of 1 is the same size as the original.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q4/5

1350

QUESTION ID

S40-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q4/5

True or false?

A rectangle 2 by 3 and a rectangle 3 by 4 are scale copies of each other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q5/5

1351

QUESTION ID

S40-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q5/5

SET 40 • 8.G.A.4

BACKWARD 2

Decide from a table whether two quantities are in a constant ratio • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

The table shows side lengths of a figure and of its copy.

Are the pairs in a constant ratio? Explain.

Figure	4	6	8
Copy	10	15	20

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q1/5

1353

QUESTION ID

S40-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. Each copy length is 2.5 times the figure length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q1/5

The table shows side lengths of a figure and of another figure.

Are the pairs in a constant ratio? Explain.

Figure	3	5
Other	9	12

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q2/5

1354

QUESTION ID

S40-B2-Q2

TYPE

Short answer

CORRECT ANSWER

No. 9 ÷ 3 = 3, but 12 ÷ 5 = 2.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q2/5

Which table shows y in a constant ratio to x?

A. x: 2, 4, 6; y: 1, 3, 5

B. x: 2, 4, 6; y: 5, 10, 15

C. x: 2, 4, 6; y: 4, 16, 36

D. x: 2, 4, 6; y: 5, 7, 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q3/5

1355

QUESTION ID

S40-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. x: 2, 4, 6; y: 5, 10, 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q3/5

True or false?

For the pairs (2, 7), (4, 14), and (6, 21), y is always 3.5 times x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q4/5

1356

QUESTION ID

S40-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q4/5

True or false?

The pairs (4, 5) and (5, 6) are in the same ratio.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q5/5

1357

QUESTION ID

S40-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $4 \div 5 = 0.8$, but $5 \div 6 \approx 0.83$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q5/5

SET 40 • 8.G.A.4

FORWARD 1

Use the definition of similarity transformations to prove equal angles and proportional sides • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC is dilated by a scale factor of 3 about point O, then rotated 90° about O, giving triangle A'B'C'. AB = 4 and angle A = 50°.

Find A'B' and angle A'. Explain using the properties of the transformations.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q1/5

1359

QUESTION ID

S40-F1-Q1

TYPE

Short answer

CORRECT ANSWER

A'B' = 12 and angle A' = 50°: the dilation multiplies lengths by 3 and the rotation keeps lengths; both keep angle measures.

GRADING NOTE Both values and the reasons are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Use the definition of similarity (a sequence of rigid motions and dilations) to explain why similar triangles have equal corresponding angles and proportional corresponding sides.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q2/5

1360

QUESTION ID

S40-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

Corresponding angles are equal, and corresponding sides are proportional.

GRADING NOTE REQUIRED WORK: Each rigid motion and each dilation keeps angle measures, so corresponding angles are equal. Rigid motions keep lengths, and a dilation multiplies every length by the same scale factor, so every pair of corresponding sides has the same ratio.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q2/5

Rectangle P is 2 by 6. Rectangle Q is 3 by 8.

Which statement is true?

A. They are not similar, because $3 \div 2 \neq 8 \div 6$.

B. They are similar, because all their angles are 90°.

C. They are similar with a scale factor of 1.5.

D. They are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q3/5

1361

QUESTION ID

S40-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. They are not similar, because $3 \div 2 \neq 8 \div 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q3/5

True or false?

If triangle DEF is the image of triangle ABC under a similarity transformation, then
DE ÷ AB = EF ÷ BC = DF ÷ AC.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q4/5

1362

QUESTION ID
S40-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q4/5

True or false?

Two quadrilaterals whose four pairs of corresponding angles are equal must be similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q5/5

1363

QUESTION ID

S40-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A square and a long rectangle have equal angles but are not similar.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q5/5

SET 40 • 8.G.A.4

FORWARD 2

Use a translation and a dilation to show that any two circles are similar • High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1

Circle A has center (0, 0) and radius 2. Circle B has center (5, 1) and radius 6.

Describe a translation followed by a dilation that maps circle A onto circle B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q1

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q1/5

1365

QUESTION ID

S40-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Translate 5 units right and 1 unit up, then dilate about (5, 1) with a scale factor of 3.

GRADING NOTE Any correct sequence is accepted, for example a dilation about (0, 0) with scale factor 3 followed by a translation 5 units right and 1 unit up.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q1

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q1/5

What scale factor of a dilation about the center maps a circle with a radius of 4 onto a circle with the same center and a radius of 10?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q2

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q2/5

1366

QUESTION ID

S40-F2-Q2

TYPE

Short answer

CORRECT ANSWER

2.5

GRADING NOTE $\frac{5}{2}$ is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q2

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q2/5

Which statement explains why any two circles are similar?

A. A rotation alone maps any circle onto any other circle.

B. All circles have the same radius.

C. A translation moves one center onto the other, and then a dilation about that center makes the radii equal.

D. All circles have the same area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q3

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q3/5

1367

QUESTION ID

S40-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. A translation moves one center onto the other, and then a dilation about that center makes the radii equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q3

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q3/5

True or false?

Any two circles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q4

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q4/5

1368

QUESTION ID

S40-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q4

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q4/5

True or false?

A circle with a radius of 3 can be mapped onto a circle with a radius of 6 by a translation alone.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q5

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q5/5

1369

QUESTION ID

S40-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A translation keeps the radius 3; a dilation is also needed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q5

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q5/5

SET 41 • 8.G.A.5 • GEOMETRY

Use the angle sum and exterior angles of a triangle

MAIN	Grade 8 main question	**8.G.A.5**
BACKWARD 1	Write and solve equations for supplementary and complementary…	**7.G.B.5**
BACKWARD 2	Find an unknown angle in a sum of angles	**4.MD.C.7**
FORWARD 1	Prove theorems about triangles	**HSG-CO.C.10**
FORWARD 2	Define trigonometric ratios from similar right triangles — nearest r	**HSG-SRT.C.6**

SET 41 • 8.G.A.5

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.5

Two angles of a triangle measure 52° and 71°.

What does the third angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q1/5

1372

QUESTION ID

S41-M-Q1

TYPE

Short answer

CORRECT ANSWER

57°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q1/5

The angles of a triangle measure x°, (x + 10)°, and (2x − 30)°.

Write and solve an equation to find x.

Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q2/5

1373

QUESTION ID

S41-M-Q2

TYPE

Short answer

CORRECT ANSWER

The equation x + (x + 10) + (2x − 30) = 180 and the answer x = 50

GRADING NOTE Both parts are required: a valid equation and the correct solution. Any valid equation is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q2/5

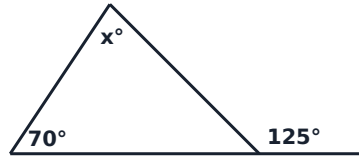
In the figure, what is the value of x?

A. 55

B. 70

C. 125

D. 35



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q3/5

1374

QUESTION ID

S41-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 55

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q3/5

Explain why an exterior angle of a triangle equals the sum of the two remote interior angles. Give a reason for each step.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q4/5

1375

QUESTION ID

S41-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

The exterior angle = the sum of the two remote interior angles

GRADING NOTE REQUIRED WORK: A valid chain of reasons. Other valid orders are accepted. A conclusion without reasons is not full credit. Example: the exterior angle and the adjacent interior angle form a linear pair, so they add to 180°; the three interior angles also add to 180°; so the exterior angle equals the sum of the other two interior angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q4/5

True or false?

A triangle can have two obtuse angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q5/5

1376

QUESTION ID

S41-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Two obtuse angles alone add to more than 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q5/5

SET 41 • 8.G.A.5

BACKWARD 1

Write and solve equations for supplementary and complementary angles • Grade 7 • Geometry • 7.G.B.5

Two supplementary angles measure x° and 130°.

What is x?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q1/5

1378

QUESTION ID

S41-B1-Q1

TYPE

Short answer

CORRECT ANSWER

50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q1/5

Two complementary angles measure 2x° and 40°.

What is x?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q2/5

1379

QUESTION ID

S41-B1-Q2

TYPE

Short answer

CORRECT ANSWER

25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q2/5

Angles of x° and (x + 30)° are supplementary.

What is x?

A. 75

B. 150

C. 105

D. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q3/5

1380

QUESTION ID

S41-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q3/5

True or false?

Angles of 47° and 43° are complementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q4/5

1381

QUESTION ID

S41-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q4/5

True or false?

Angles of 100° and 90° are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q5/5

1382

QUESTION ID

S41-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q5/5

SET 41 • 8.G.A.5

BACKWARD 2

Find an unknown angle in a sum of angles • Grade 4 • Measurement & Data • 4.MD.C.7

A 90° angle is split into two angles. One is 35°.

What is the other?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q1/5

1384

QUESTION ID

S41-B2-Q1

TYPE

Short answer

CORRECT ANSWER

55°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q1/5

A straight angle (180°) is split into two angles. One is 110°.

What is the other?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q2/5

1385

QUESTION ID

S41-B2-Q2

TYPE

Short answer

CORRECT ANSWER

70°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q2/5

An angle of 120° is split into angles of 75° and x°.

What is x?

A. 195

B. 55

C. 45

D. 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q3/5

1386

QUESTION ID

S41-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q3/5

True or false?

Angles of 30° and 60° together make a 90° angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q4/5

1387

QUESTION ID

S41-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q4/5

True or false?

Angles of 80° and 80° together make a 180° angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q5/5

1388

QUESTION ID

S41-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q5/5

SET 41 • 8.G.A.5

FORWARD 1

Prove theorems about triangles • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10

ANSWER KEY Reduced copy of the question page, with its answer

Prove that the angles of triangle ABC add to 180°. Use a line through C that is parallel to AB. Give a reason for each step.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q1/5

1390

QUESTION ID

S41-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

Angle A + angle B + angle C = 180°

GRADING NOTE REQUIRED WORK: A valid chain of reasons. Other valid orders are accepted. A conclusion without reasons is not full credit. Example: draw line l through C parallel to AB. The angles at C along l add to 180° (straight angle). The two outer ones equal angles A and B (alternate interior angles, l ∥ AB). So angle A + angle B + angle C = 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q1/5

The angles of a triangle are 2x°, 3x°, and 4x°.

What is x?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q2/5

1391

QUESTION ID

S41-F1-Q2

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q2/5

An isosceles triangle has base angles of 40°.

What does its third angle measure?

A. 100°

B. 40°

C. 80°

D. 140°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q3/5

1392

QUESTION ID

S41-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 100°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q3/5

True or false?

The base angles of an isosceles triangle are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q4/5

1393

QUESTION ID

S41-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q4/5

True or false?

An exterior angle of a triangle equals the sum of all three interior angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q5/5

1394

QUESTION ID

S41-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q5/5

SET 41 • 8.G.A.5

FORWARD 2

Define trigonometric ratios from similar right triangles • High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

⚑ No high-school standard directly continues this skill at this step. This branch uses the nearest related high-school standard (placed at Grade 10 in this bank).

A right triangle has a leg of 3 opposite angle A, a leg of 4 next to angle A, and a hypotenuse of 5.

What is sin A?

sin A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q1

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q1/5 • NEAREST RELATED

1396

QUESTION ID

S41-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q1

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q1/5 • NEAREST RELATED

A right triangle has a leg of 3 opposite angle A, a leg of 4 next to angle A, and a hypotenuse of 5.

What is tan A?

tan A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q2

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q2/5 • NEAREST RELATED

1397

QUESTION ID

S41-F2-Q2

TYPE

Short answer

CORRECT ANSWER

3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q2

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q2/5 • NEAREST RELATED

A right triangle has a leg of 3 opposite angle A, a leg of 4 next to angle A, and a hypotenuse of 5.

What is cos A?

A. $\frac{3}{5}$

B. $\frac{5}{4}$

C. $\frac{4}{5}$

D. $\frac{3}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S41-F2-Q3

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q3/5 • NEAREST RELATED

1398

QUESTION ID

S41-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 4/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S41-F2-Q3

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

In every right triangle with a 30° angle, the side opposite the 30° angle is half the hypotenuse, so $\sin 30° = \frac{1}{2}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q4

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q4/5 • NEAREST RELATED

1399

QUESTION ID

S41-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q4

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q4/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

sin A changes when a right triangle is enlarged but its angles stay the same.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q5

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q5/5 • NEAREST RELATED

1400

QUESTION ID
S41-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Similar right triangles have equal ratios.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q5

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q5/5 • NEAREST RELATED

SET 63 • 8.G.A.5 • GEOMETRY

Give informal arguments for the triangle angle sum and the exterior angle of a triangle

MAIN	Grade 8 main question	**8.G.A.5**
BACKWARD 1	Use linear pairs and straight angles to find unknown angles	**7.G.B.5**
BACKWARD 2	Find an unknown part of an angle that is split into parts	**4.MD.C.7**
FORWARD 1	Prove theorems about isosceles and right triangles	**HSG-CO.C.10**
FORWARD 2	Use the triangle angle sum to prove angle facts about parallelogra	**HSG-CO.C.11**

SET 63 • 8.G.A.5

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.5

Three copies of the same triangle are placed side by side so that its three different angles meet at one point, with no gaps or overlaps, along a straight line.

Explain why this shows that the angles of the triangle add to 180°.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 63 • MAIN • Q1/5

1403

QUESTION ID

S63-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

The three angles fill a straight angle, so they add to 180°.

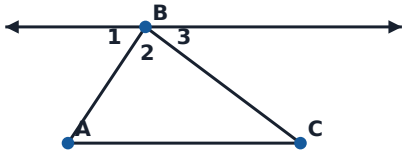
GRADING NOTE REQUIRED WORK: The three angles, one of each kind, together fill a straight angle, which measures 180°. So angle A + angle B + angle C = 180°. Any equivalent explanation is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 63 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In the figure, the line through B is parallel to AC. Use angles 1, 2, and 3 to explain why the angles of triangle ABC add to 180°.



Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 63 • MAIN • Q2/5

1404

QUESTION ID

S63-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

Angle A + angle B + angle C = 180°

GRADING NOTE REQUIRED WORK: Angle 1 = angle A and angle 3 = angle C (alternate interior angles, because the line is parallel to AC). Angles 1, 2, and 3 together form a straight angle, so angle A + angle B + angle C = 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 63 • MAIN • Q2/5

Use the triangle angle sum to explain why the exterior angle at C of triangle ABC equals angle A + angle B.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 63 • MAIN • Q3/5

1405

QUESTION ID

S63-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

The exterior angle at C = angle A + angle B

GRADING NOTE REQUIRED WORK: The exterior angle and angle C form a linear pair, so exterior angle = 180° − angle C. The triangle's angles add to 180°, so angle A + angle B = 180° − angle C. Therefore exterior angle = angle A + angle B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 63 • MAIN • Q3/5

A student tears off the three corners of a paper triangle and places them together, point to point, with no gaps.

What does she see?

A. The three angles form a full turn (360°).

B. The three angles form a straight angle (180°).

C. The angles cannot fit together.

D. The three angles form a right angle (90°).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 63 • MAIN • Q4/5

1406

QUESTION ID

S63-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

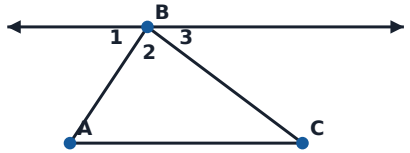
B. The three angles form a straight angle (180°).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 63 • MAIN • Q4/5

True or false?

In the figure, the argument for the triangle angle sum uses the fact that alternate interior angles formed by parallel lines are equal.



A. True **B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 63 • MAIN • Q5/5

1407

QUESTION ID

S63-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 63 • MAIN • Q5/5

SET 63 • 8.G.A.5

BACKWARD 1

Use linear pairs and straight angles to find unknown angles • Grade 7 • Geometry • 7.G.B.5

Two angles form a straight line. One measures 47°.

What does the other measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 63 • BACKWARD 1 • Q1/5

1409

QUESTION ID

S63-B1-Q1

TYPE

Short answer

CORRECT ANSWER

133°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 63 • BACKWARD 1 • Q1/5

Three angles together form a straight line. Two of them measure 50° and 60°.

What does the third measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 63 • BACKWARD 1 • Q2/5

1410

QUESTION ID

S63-B1-Q2

TYPE

Short answer

CORRECT ANSWER

70°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 63 • BACKWARD 1 • Q2/5

Angles of 2x° and 3x° together form a straight angle.

What is x?

A. 45

B. 18

C. 72

D. 36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 63 • BACKWARD 1 • Q3/5

1411

QUESTION ID

S63-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 63 • BACKWARD 1 • Q3/5

True or false?

Two angles that form a linear pair are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 63 • BACKWARD 1 • Q4/5

1412

QUESTION ID

S63-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 63 • BACKWARD 1 • Q4/5

True or false?

Angles of 95° and 95° together form a straight angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 63 • BACKWARD 1 • Q5/5

1413

QUESTION ID

S63-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They add to 190°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 63 • BACKWARD 1 • Q5/5

SET 63 • 8.G.A.5

BACKWARD 2

Find an unknown part of an angle that is split into parts • Grade 4 • Measurement & Data • 4.MD.C.7

A 180° angle is split into three angles of 40°, 75°, and x°.

What is x?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 63 • BACKWARD 2 • Q1/5

1415

QUESTION ID

S63-B2-Q1

TYPE

Short answer

CORRECT ANSWER

65

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 63 • BACKWARD 2 • Q1/5

A 90° angle is split into angles of 28° and x°.

What is x?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 63 • BACKWARD 2 • Q2/5

1416

QUESTION ID

S63-B2-Q2

TYPE

Short answer

CORRECT ANSWER

62

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 63 • BACKWARD 2 • Q2/5

A 150° angle is split into angles of 90° and x°.

What is x?

A. 240

B. 70

C. 60

D. 50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 63 • BACKWARD 2 • Q3/5

1417

QUESTION ID

S63-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 63 • BACKWARD 2 • Q3/5

True or false?

Angles of 45° and 45° together make a 90° angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 63 • BACKWARD 2 • Q4/5

1418

QUESTION ID

S63-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 63 • BACKWARD 2 • Q4/5

True or false?

Angles of 70°, 50°, and 50° together make a 180° angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 63 • BACKWARD 2 • Q5/5

1419

QUESTION ID

S63-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They make 170°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 63 • BACKWARD 2 • Q5/5

SET 63 • 8.G.A.5

FORWARD 1

Prove theorems about isosceles and right triangles • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has AB = AC. Prove that angle B = angle C. (Use the reflection across the line that bisects angle A.)

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 63 • FORWARD 1 • Q1/5

1421

QUESTION ID

S63-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

Angle B = angle C

GRADING NOTE REQUIRED WORK: A valid chain of reasons. Other valid orders are accepted. A conclusion without reasons is not full credit. Example: the reflection across the bisector of angle A maps ray AB onto ray AC; since AB = AC, it maps B to C (and C to B). So it maps triangle ABC onto itself and angle B onto angle C; reflections keep angle measures, so angle B = angle C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 63 • FORWARD 1 • Q1/5

An isosceles triangle has a vertex angle of 50°.

Find each base angle, and name the theorems you used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 63 • FORWARD 1 • Q2/5

1422

QUESTION ID

S63-F1-Q2

TYPE

Short answer

CORRECT ANSWER

65° each: the base angles are congruent, and the three angles add to 180°.

GRADING NOTE Both the measure and the theorems are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 63 • FORWARD 1 • Q2/5

Which statement can be proved from the triangle angle sum theorem and the fact that an equilateral triangle's angles are equal?

A. Opposite sides of a parallelogram are congruent.

B. The diagonals of a rectangle are congruent.

C. Vertical angles are congruent.

D. Each angle of an equilateral triangle measures 60°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S63-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 63 • FORWARD 1 • Q3/5

1423

QUESTION ID

S63-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Each angle of an equilateral triangle measures 60°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S63-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 63 • FORWARD 1 • Q3/5

True or false?

In a right triangle, the two acute angles are complementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 63 • FORWARD 1 • Q4/5

1424

QUESTION ID

S63-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 63 • FORWARD 1 • Q4/5

True or false?

A triangle can have two right angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 63 • FORWARD 1 • Q5/5

1425

QUESTION ID

S63-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Two right angles alone add to 180°, leaving 0° for the third angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 63 • FORWARD 1 • Q5/5

SET 63 • 8.G.A.5

FORWARD 2

Use the triangle angle sum to prove angle facts about parallelograms • High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11

In parallelogram ABCD, diagonal AC is drawn. Prove that angle B = angle D. Give a reason for each step.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 63 • FORWARD 2 • Q1/5

1427

QUESTION ID

S63-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

Angle B = angle D

GRADING NOTE REQUIRED WORK: A valid chain of reasons. Other valid orders are accepted. A conclusion without reasons is not full credit. Example: angle BAC = angle DCA (alternate interior angles, AB ∥ DC); angle BCA = angle DAC (alternate interior angles, BC ∥ AD); in each triangle the angles add to 180°, so angle B = 180° − angle BAC − angle BCA = 180° − angle DCA − angle DAC = angle D.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 63 • FORWARD 2 • Q1/5

In parallelogram ABCD, angle A = 3x° and angle B = (x + 20)°.

Find x and all four angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 63 • FORWARD 2 • Q2/5

1428

QUESTION ID

S63-F2-Q2

TYPE

Short answer

CORRECT ANSWER

x = 40; angle A = angle C = 120° and angle B = angle D = 60°.

GRADING NOTE Consecutive angles are supplementary: 3x + x + 20 = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 63 • FORWARD 2 • Q2/5

Which fact, with the triangle angle sum, shows that the four angles of a parallelogram add to 360°?

A. Its diagonals are congruent.

B. A diagonal splits it into two triangles whose angles together make up its four angles.

C. It has four right angles.

D. All of its sides are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 63 • FORWARD 2 • Q3/5

1429

QUESTION ID

S63-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. A diagonal splits it into two triangles whose angles together make up its four angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 63 • FORWARD 2 • Q3/5

True or false?

If one angle of a parallelogram is 90°, all four of its angles are 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 63 • FORWARD 2 • Q4/5

1430

QUESTION ID

S63-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 63 • FORWARD 2 • Q4/5

True or false?

The diagonals of every parallelogram are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 63 • FORWARD 2 • Q5/5

1431

QUESTION ID

S63-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only a rectangle's diagonals are always congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 63 • FORWARD 2 • Q5/5

SET 42 • 8.G.A.5 • GEOMETRY

Write and solve equations for angles formed by parallel lines cut by a transversal

MAIN	Grade 8 main question	**8.G.A.5**
BACKWARD 1	Write and solve equations for vertical and adjacent angles	**7.G.B.5**
BACKWARD 2	Solve two-step equations	**7.EE.B.4.a**
FORWARD 1	Prove that alternate interior angles are congruent	**HSG-CO.C.9**
FORWARD 2	Use slopes to show that lines are parallel or perpendicular	**HSG-GPE.B.5**

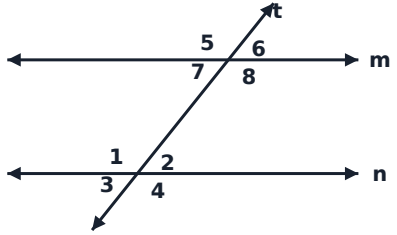
SET 42 • 8.G.A.5

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.5

In the figure, lines m and n are parallel and angle 2 = 52°.

What does angle 6 measure? Give the reason.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q1/5

1434

QUESTION ID

S42-M-Q1

TYPE

Short answer

CORRECT ANSWER

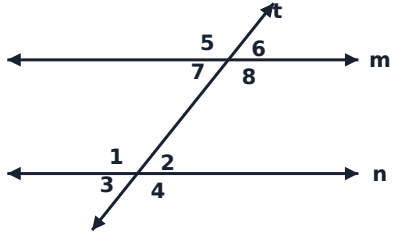
52°; corresponding angles are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q1/5

In the figure, lines m and n are parallel. Angle 1 = (3x + 10)° and angle 8 = 115°.

Write and solve an equation for x.



Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q2/5

1435

QUESTION ID

S42-M-Q2

TYPE

Short answer

CORRECT ANSWER

The equation 3x + 10 = 115 (alternate interior angles are congruent) and the answer x = 35

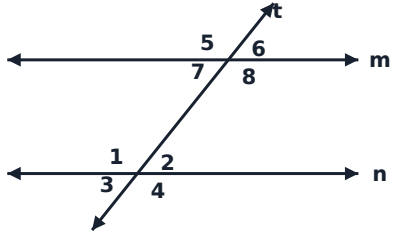
GRADING NOTE Both parts are required: a valid equation and the correct solution. Any valid equation is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q2/5

In the figure, lines m and n are parallel. Angle 2 = (2x + 4)° and angle 8 = (4x − 10)°.

What is x?



A. 93 **B.** 31

C. 43 **D.** 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q3/5

1436

QUESTION ID

S42-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

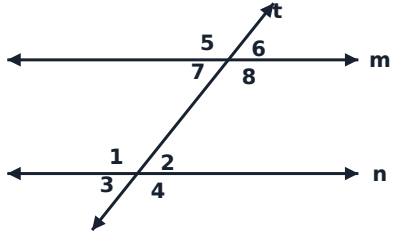
B. 31

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q3/5

In the figure, lines m and n are parallel and angle 4 = 128°.

What does angle 5 measure? Give the reasons.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q4/5

1437

QUESTION ID

S42-M-Q4

TYPE

Short answer

CORRECT ANSWER

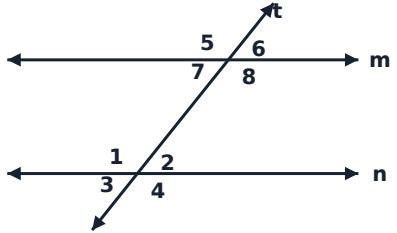
128°: angle 1 = angle 4 (vertical angles), and angle 5 = angle 1 (corresponding angles).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q4/5

True or false?

In the figure, angles 3 and 6 are alternate exterior angles, so they are congruent when m and n are parallel.



A. True **B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q5/5

1438

QUESTION ID

S42-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q5/5

SET 42 • 8.G.A.5

BACKWARD 1

Write and solve equations for vertical and adjacent angles • Grade 7 • Geometry • 7.G.B.5

Two lines intersect. Vertical angles measure (2x + 20)° and 80°.

What is x?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q1/5

1440

QUESTION ID

S42-B1-Q1

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q1/5

Two lines intersect. Adjacent angles measure x° and 3x°.

What is x?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q2/5

1441

QUESTION ID

S42-B1-Q2

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q2/5

Two lines intersect. Adjacent angles measure (x + 40)° and 2x°.

What is x?

A. $\frac{140}{3}$

B. 50

C. 40

D. 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q3/5

1442

QUESTION ID

S42-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 140/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q3/5

True or false?

Vertical angles of 5x° and 75° give x = 15.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q4/5

1443

QUESTION ID

S42-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q4/5

True or false?

Adjacent angles that form a straight line measure (x + 10)° and x°. Then x = 90.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q5/5

1444

QUESTION ID

S42-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE x = 85.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q5/5

SET 42 • 8.G.A.5

BACKWARD 2

Solve two-step equations • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for x.

3x + 10 = 115

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q1/5

1446

QUESTION ID

S42-B2-Q1

TYPE

Short answer

CORRECT ANSWER

35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q1/5

Solve for x.

$2x - 15 = 75$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q2/5

1447

QUESTION ID

S42-B2-Q2

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q2/5

Solve for x.

$6x - 6 = 180$

A. x = 174

B. x = 29

C. x = 31

D. x = 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q3/5

1448

QUESTION ID

S42-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 31

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q3/5

True or false?

x = 20 is the solution of 4x + 20 = 100.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q4/5

1449

QUESTION ID

S42-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x = 25$ is the solution of $5x - 5 = 115$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q5/5

1450

QUESTION ID

S42-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = 24$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q5/5

1450

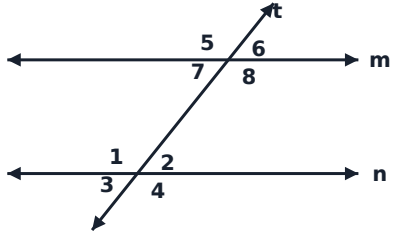
SET 42 • 8.G.A.5

FORWARD 1

Prove that alternate interior angles are congruent • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9

ANSWER KEY Reduced copy of the question page, with its answer

In the figure, lines m and n are parallel. Prove that angle 1 = angle 8. Give a reason for each step.



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q1/5

1452

QUESTION ID

S42-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

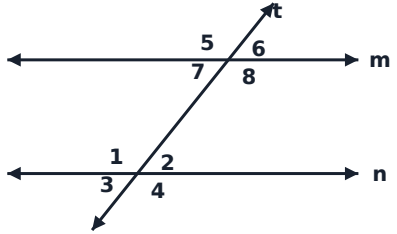
Angle 1 = angle 8

GRADING NOTE REQUIRED WORK: A valid chain of reasons. Other valid orders are accepted. A conclusion without reasons is not full credit. Example: angle 1 = angle 5 (corresponding angles, m ∥ n); angle 5 = angle 8 (vertical angles); so angle 1 = angle 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q1/5

In the figure, lines m and n are parallel. Prove that angle 2 + angle 8 = 180°. Give a reason for each step.



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q2/5

1453

QUESTION ID

S42-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

Angle 2 + angle 8 = 180°

GRADING NOTE REQUIRED WORK: A valid chain of reasons. Other valid orders are accepted. A conclusion without reasons is not full credit. Example: angle 6 + angle 8 = 180° (linear pair); angle 2 = angle 6 (corresponding angles); so angle 2 + angle 8 = 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q2/5

A proof says: angle 1 = angle 5 (corresponding angles). Angle 5 = angle 8 (?). So angle 1 = angle 8.

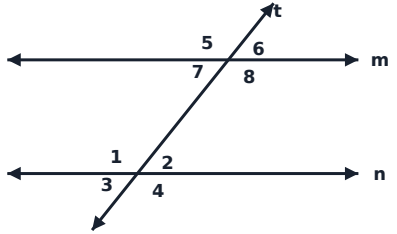
What is the missing reason?

A. Linear pairs are supplementary.

B. Vertical angles are congruent.

C. Alternate interior angles are congruent.

D. The triangle angle sum is 180°.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q3/5

1454

QUESTION ID

S42-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Vertical angles are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q3/5

True or false?

If corresponding angles formed by a transversal are congruent, then the two lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q4/5

1455

QUESTION ID

S42-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q4/5

True or false?

"The angles look equal in the figure" is a valid reason in a proof.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q5/5

1456

QUESTION ID

S42-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q5/5

SET 42 • 8.G.A.5

FORWARD 2

Use slopes to show that lines are parallel or perpendicular • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

A transversal crosses the lines $y = 2x + 1$ and $y = 2x - 3$.

Are the two lines parallel? Explain with slopes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q1/5

1458

QUESTION ID

S42-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes; both lines have slope 2 and different y-intercepts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q1/5

Write an equation of the line through (1, 4) that is perpendicular to $y = -\frac{1}{2}x + 3$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q2/5

1459

QUESTION ID

S42-F2-Q2

TYPE

Short answer

CORRECT ANSWER

y = 2x + 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F2-Q2**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q2/5

Which pair of lines is perpendicular?

A. $y = x$ and $y = x + 2$

B. $y = 3x$ and $y = -3x$

C. $y = 3x + 1$ and $y = 3x - 1$

D. $y = 3x + 1$ and $y = -\frac{1}{3}x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q3/5

1460

QUESTION ID

S42-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $y = 3x + 1$ and $y = -(1/3)x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q3/5

True or false?

Two distinct non-vertical lines are parallel exactly when their slopes are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q4/5

1461

QUESTION ID

S42-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q4/5

True or false?

The lines $y = \frac{2}{5}x$ and $y = \frac{5}{2}x$ are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q5/5

1462

QUESTION ID

S42-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Perpendicular slopes multiply to −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q5/5

SET 64 • 8.G.A.5 • GEOMETRY

Give informal arguments for the angles formed when parallel lines are cut by a transversal

MAIN	Grade 8 main question	**8.G.A.5**
BACKWARD 1	Use vertical angles and linear pairs to find unknown angles	**7.G.B.5**
BACKWARD 2	Recognize parallel and perpendicular lines	**4.G.A.1**
FORWARD 1	Prove that lines are parallel from congruent angles	**HSG-CO.C.9**
FORWARD 2	Construct a line through a point parallel to a given line	**HSG-CO.D.12**

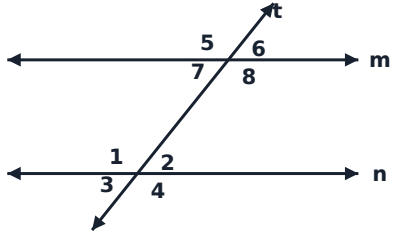
SET 64 • 8.G.A.5

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.5

In the figure, lines m and n are parallel. A translation along the transversal t carries line n onto line m.

Use this translation to explain why angle 2 = angle 6.



Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 64 • MAIN • Q1/5

1465

QUESTION ID

S64-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

Angle 2 = angle 6

GRADING NOTE REQUIRED WORK: The translation moves the point where t meets n to the point where t meets m, keeps t on itself, and carries n onto m (they are parallel). So it carries angle 2 onto angle 6. A translation keeps angle measures, so angle 2 = angle 6.

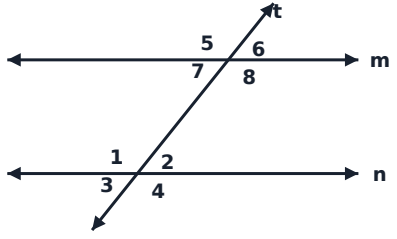
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 64 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In the figure, lines m and n are parallel. A 180° rotation about the midpoint of the part of t between m and n swaps the two lines.

Use this rotation to explain why angle 2 = angle 7.



Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 64 • MAIN • Q2/5

1466

QUESTION ID

S64-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

Angle 2 = angle 7

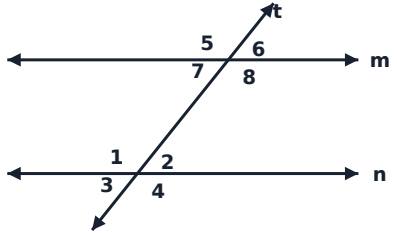
GRADING NOTE REQUIRED WORK: The rotation maps t onto itself and swaps the two intersection points; it maps n onto the parallel line through the other point, which is m. So it carries angle 2 onto angle 7. A rotation keeps angle measures, so angle 2 = angle 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 64 • MAIN • Q2/5

In the figure, lines m and n are parallel and angle 3 = 115°.

Find all eight angles, using only vertical angles, linear pairs, and corresponding angles.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 64 • MAIN • Q3/5

1467

QUESTION ID

S64-M-Q3

TYPE

Short answer

CORRECT ANSWER

Angles 2, 3, 6, and 7 measure 115°; angles 1, 4, 5, and 8 measure 65°.

GRADING NOTE Any correct chain of reasons is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 64 • MAIN • Q3/5

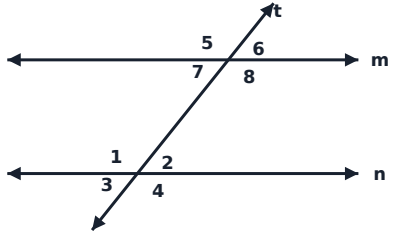
In the figure, which rigid motion carries angle 1 onto angle 5?

A. A translation along t that carries line n onto line m

B. A dilation centered where t meets m

C. A reflection across line m

D. A 90° rotation about the point where t meets n



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 64 • MAIN • Q4/5

1468

QUESTION ID

S64-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. A translation along t that carries line n onto line m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 64 • MAIN • Q4/5

True or false?

If a transversal cuts two lines that are NOT parallel, its corresponding angles are still equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 64 • MAIN • Q5/5

1469

QUESTION ID

S64-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Corresponding angles are equal only when the lines are parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 64 • MAIN • Q5/5

SET 64 • 8.G.A.5

BACKWARD 1

Use vertical angles and linear pairs to find unknown angles • Grade 7 • Geometry • 7.G.B.5

Two lines intersect, and one of the four angles measures 70°.

Find the other three angles.

Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 64 • BACKWARD 1 • Q1/5

1471

QUESTION ID

S64-B1-Q1

TYPE

Short answer

CORRECT ANSWER

110°, 70°, and 110°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 64 • BACKWARD 1 • Q1/5

Two angles form a linear pair. One is 3 times as large as the other.

What are the two angles?

Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 64 • BACKWARD 1 • Q2/5

1472

QUESTION ID

S64-B1-Q2

TYPE

Short answer

CORRECT ANSWER

45° and 135°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 64 • BACKWARD 1 • Q2/5

Two lines intersect. One angle measures 125°.

What does its vertical angle measure?

A. 235°

B. 55°

C. 35°

D. 125°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 64 • BACKWARD 1 • Q3/5

1473

QUESTION ID

S64-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 125°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 64 • BACKWARD 1 • Q3/5

True or false?

Vertical angles are the opposite angles formed by two intersecting lines.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 64 • BACKWARD 1 • Q4/5

1474

QUESTION ID

S64-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 64 • BACKWARD 1 • Q4/5

True or false?

The two angles of a linear pair are always equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 64 • BACKWARD 1 • Q5/5

1475

QUESTION ID

S64-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They add to 180°; they are equal only when both are 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 64 • BACKWARD 1 • Q5/5

SET 64 • 8.G.A.5

BACKWARD 2

Recognize parallel and perpendicular lines • Grade 4 • Geometry • 4.G.A.1

What are lines in a plane called that never meet, however far they are extended?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 64 • BACKWARD 2 • Q1/5

1477

QUESTION ID

S64-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Parallel lines

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 64 • BACKWARD 2 • Q1/5

What are lines called that meet at a right angle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 64 • BACKWARD 2 • Q2/5

1478

QUESTION ID

S64-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Perpendicular lines

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 64 • BACKWARD 2 • Q2/5

Which shape always has two pairs of parallel sides?

A. Kite

B. Triangle

C. Parallelogram

D. Trapezoid

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 64 • BACKWARD 2 • Q3/5

1479

QUESTION ID

S64-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Parallelogram

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 64 • BACKWARD 2 • Q3/5

True or false?

The two rails of a straight railroad track are a model of parallel lines.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 64 • BACKWARD 2 • Q4/5

1480

QUESTION ID

S64-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 64 • BACKWARD 2 • Q4/5

True or false?

Two lines that cross each other can be parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 64 • BACKWARD 2 • Q5/5

1481

QUESTION ID

S64-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 64 • BACKWARD 2 • Q5/5

SET 64 • 8.G.A.5

FORWARD 1

Prove that lines are parallel from congruent angles • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9

ANSWER KEY Reduced copy of the question page, with its answer

Lines m and n are cut by transversal t, and a pair of alternate interior angles is congruent.

Explain why m and n must be parallel. (Suppose they meet at a point P, and look at the triangle formed with t.)

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 64 • FORWARD 1 • Q1/5

1483

QUESTION ID

S64-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

The lines are parallel.

GRADING NOTE REQUIRED WORK: A complete argument: if m and n met at P, they would form a triangle with t. One of the congruent angles would be an exterior angle of that triangle and the other a remote interior angle. But an exterior angle equals the sum of the two remote interior angles, so it is greater than either one. That is a contradiction, so m and n do not meet: they are parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 64 • FORWARD 1 • Q1/5

Lines a and b are cut by a transversal. A pair of corresponding angles measure (4x + 6)° and (5x − 9)°.

For what value of x are the lines parallel?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 64 • FORWARD 1 • Q2/5

1484

QUESTION ID

S64-F1-Q2

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 64 • FORWARD 1 • Q2/5

Two lines are cut by a transversal. Which angle pair, if congruent, proves that the lines are parallel?

A. A pair of alternate interior angles

B. A pair of vertical angles

C. A linear pair

D. Two adjacent angles on the same line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 64 • FORWARD 1 • Q3/5

1485

QUESTION ID

S64-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A pair of alternate interior angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 64 • FORWARD 1 • Q3/5

True or false?

If the same-side interior angles formed by a transversal add to 180°, the two lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 64 • FORWARD 1 • Q4/5

1486

QUESTION ID

S64-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 64 • FORWARD 1 • Q4/5

True or false?

If the vertical angles at one intersection are congruent, the two lines cut by the transversal are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 64 • FORWARD 1 • Q5/5

1487

QUESTION ID

S64-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Vertical angles are always congruent, so they prove nothing about the lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 64 • FORWARD 1 • Q5/5

SET 64 • 8.G.A.5

FORWARD 2

Construct a line through a point parallel to a given line • High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.D.12

Use a compass and straightedge to construct the line through P that is parallel to line k. Draw a transversal through P, then copy the angle it makes with k so that it is a corresponding angle at P. Leave your construction marks.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.D.12 • Set 64 • FORWARD 2 • Q1/5

1489

QUESTION ID

S64-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A transversal through P and k, the arcs used to copy the angle at P, and a line through P parallel to k. Any correct compass-and-straightedge construction with visible marks is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.D.12 • Set 64 • FORWARD 2 • Q1/5

Why does copying the angle in that construction give a line that is parallel to k?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.D.12 • Set 64 • FORWARD 2 • Q2/5

1490

QUESTION ID

S64-F2-Q2

TYPE

Short answer

CORRECT ANSWER

The copied angle and the angle at k are congruent corresponding angles, so the lines are parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.D.12 • Set 64 • FORWARD 2 • Q2/5

When a parallel line is constructed by copying an angle, which angles are made congruent?

A. Vertical angles

B. Corresponding angles formed by the transversal

C. A linear pair

D. Two angles of a triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.D.12 • Set 64 • FORWARD 2 • Q3/5

1491

QUESTION ID

S64-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Corresponding angles formed by the transversal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.D.12 • Set 64 • FORWARD 2 • Q3/5

True or false?

In a compass-and-straightedge construction, the compass is used to copy distances.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.D.12 • Set 64 • FORWARD 2 • Q4/5

1492

QUESTION ID

S64-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.D.12 • Set 64 • FORWARD 2 • Q4/5

True or false?

A compass-and-straightedge construction may use a protractor to measure the angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.D.12 • Set 64 • FORWARD 2 • Q5/5

1493

QUESTION ID

S64-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only a compass and a straightedge are allowed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.D.12 • Set 64 • FORWARD 2 • Q5/5

SET 43 • 8.G.A.5 • GEOMETRY

Use the angle-angle criterion to show that triangles are similar

MAIN	Grade 8 main question	**8.G.A.5**
BACKWARD 1	Find a missing length in a scale copy	**7.G.A.1**
BACKWARD 3	Decide whether three angle measures can make a triangle and w…	**7.G.A.2**
FORWARD 1	Establish the angle-angle criterion with similarity transformations	**HSG-SRT.A.3**
FORWARD 2	Use similar triangles to solve multistep problems, including altitud…	**HSG-SRT.B.5**

SET 43 • 8.G.A.5

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.5

Triangle ABC has angles of 35° and 85°. Triangle DEF has angles of 35° and 60°.

Are the triangles similar? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q1/5

1496

QUESTION ID

S43-M-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. The third angle of DEF is 180° − 35° − 60° = 85°, so two pairs of angles are equal (AA).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q1/5

A 5-foot post casts a 3-foot shadow. At the same time, a tree casts a 12-foot shadow.

Explain why the two triangles are similar, and find the height of the tree.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q2/5

1497

QUESTION ID

S43-M-Q2

TYPE

Short answer

CORRECT ANSWER

Both triangles have a right angle and the same angle of the sun's rays, so they are similar by AA. The tree is 20 feet tall.

GRADING NOTE Grade both: the AA reason and 20 feet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q2/5

Which information shows that two triangles are similar by the angle-angle criterion?

A. One pair of sides is proportional.

B. All the sides are equal.

C. Two pairs of corresponding angles are equal.

D. One pair of corresponding angles is equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q3/5

1498

QUESTION ID

S43-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Two pairs of corresponding angles are equal.

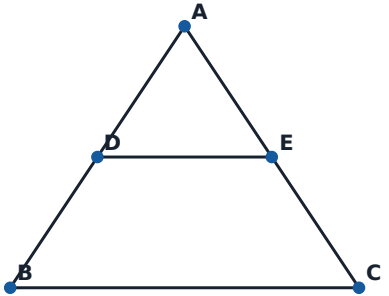
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q3/5

In the figure, segment DE is parallel to BC.

Explain why triangle ADE is similar to triangle ABC.



Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q4/5

1499

QUESTION ID

S43-M-Q4

TYPE

Short answer

CORRECT ANSWER

The triangles share angle A, and angle ADE = angle ABC (corresponding angles, because DE ∥ BC). So they are similar by AA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q4/5

True or false?

Two triangles that each have a 90° angle and a 40° angle are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q5/5

1500

QUESTION ID

S43-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q5/5

SET 43 • 8.G.A.5

BACKWARD 1

Find a missing length in a scale copy • Grade 7 • Geometry • 7.G.A.1

A triangle has sides 4, 6, and 8. In a scale copy, the side of 4 becomes 10.

What does the side of 6 become?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q1/5

1502

QUESTION ID

S43-B1-Q1

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q1/5

A drawing with a scale factor of 3 shows a 2 cm side.

How long is that side in the scale copy?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q2/5

1503

QUESTION ID

S43-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q2/5

A 5-by-8 rectangle is scaled so the 5 becomes 15.

What does the 8 become?

A. 40

B. 24

C. 13

D. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q3/5

1504

QUESTION ID

S43-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q3/5

True or false?

In a scale copy, every length is multiplied by the same scale factor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q4/5

1505

QUESTION ID

S43-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q4/5

True or false?

In a scale copy with a scale factor of 2, every angle is doubled.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q5/5

1506

QUESTION ID

S43-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q5/5

SET 43 • 8.G.A.5

BACKWARD 3

Decide whether three angle measures can make a triangle and whether they determine one triangle • Grade 7 • Geometry • 7.G.A.2

Can angles of 45°, 55°, and 80° be the three angles of a triangle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 43 • BACKWARD 3 • Q1/5

1508

QUESTION ID

S43-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. They add to 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 43 • BACKWARD 3 • Q1/5

A triangle must have angles of 30°, 60°, and 90°.

Do these conditions determine one triangle, more than one triangle, or no triangle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 43 • BACKWARD 3 • Q2/5

1509

QUESTION ID

S43-B3-Q2

TYPE

Short answer

CORRECT ANSWER

More than one triangle: triangles of different sizes have these angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 43 • BACKWARD 3 • Q2/5

Which three angle measures can be the angles of a triangle?

A. 50°, 60°, 70°

B. 90°, 90°, 10°

C. 100°, 50°, 40°

D. 50°, 60°, 80°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 43 • BACKWARD 3 • Q3/5

1510

QUESTION ID

S43-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 50°, 60°, 70°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 43 • BACKWARD 3 • Q3/5

True or false?

Triangles with the same three angle measures can have different side lengths.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 43 • BACKWARD 3 • Q4/5

1511

QUESTION ID

S43-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 43 • BACKWARD 3 • Q4/5

True or false?

Angles of 90°, 60°, and 40° can be the angles of one triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 43 • BACKWARD 3 • Q5/5

1512

QUESTION ID

S43-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They add to 190°, not 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 43 • BACKWARD 3 • Q5/5

SET 43 • 8.G.A.5

FORWARD 1

Establish the angle-angle criterion with similarity transformations • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

Triangles ABC and DEF have angle A = angle D and angle B = angle E.

Which transformations show that the triangles are similar?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q1/5

1514

QUESTION ID

S43-F1-Q1

TYPE

Short answer

CORRECT ANSWER

A dilation that makes AB the same length as DE, followed by rigid motions that move the image onto DEF; the equal angles make the third vertices match.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangles ABC and DEF have angle A = angle D and angle B = angle E.

Explain why a similarity transformation maps triangle ABC onto triangle DEF.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q2/5

1515

QUESTION ID

S43-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Dilate ABC by the scale factor DE ÷ AB, so the image of AB has the same length as DE. Rigid motions then move that image onto DE (A onto D, B onto E), with the third vertex on the same side as F. Because angle A = angle D and angle B = angle E, the image's other two sides lie along rays DF and EF, and two rays meet in only one point, so the third vertex lands on F. (The third angles are then equal too.)

GRADING NOTE Full credit needs the dilation, the rigid motions, and why the third vertex must land on F.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F1-Q2**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q2/5

Triangles with angles 30°, 70°, 80° and 70°, 30°, 80° are...

A. congruent in every case

B. similar

C. not similar

D. impossible to compare

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q3/5

1516

QUESTION ID

S43-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. similar

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F1-Q3**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q3/5

True or false?

Two equilateral triangles are always similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q4/5

1517

QUESTION ID

S43-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q4/5

True or false?

Two isosceles triangles are always similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q5/5

1518

QUESTION ID

S43-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q5/5

SET 43 • 8.G.A.5

FORWARD 2

Use similar triangles to solve multistep problems, including altitudes and parallel segments • High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In right triangle ABC, the right angle is at C, and D is the foot of the altitude from C to hypotenuse AB. AD = 4 and DB = 9.

Find CD, and name the similar triangles you used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q1

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q1/5

1520

QUESTION ID

S43-F2-Q1

TYPE

Short answer

CORRECT ANSWER

CD = 6. Triangles ACD and CBD are similar, so AD ÷ CD = CD ÷ DB and $CD^2 = 4 \cdot 9 = 36$.

GRADING NOTE Both the length and the similar triangles are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q1

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q1/5

In triangle ABC, D is on AB and E is on AC, and DE is parallel to BC. AD = 5, AB = 15, and DE = 4.

Find BC.

BC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q2

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q2/5

1521

QUESTION ID

S43-F2-Q2

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F2-Q2**

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q2/5

A mirror lies on level ground 6 m from the base of a tree. A person stands 2 m from the mirror, on the other side, and sees the top of the tree in it. Her eyes are 1.5 m above the ground. The angles at the mirror are equal.

How tall is the tree?

A. 8 m

B. 0.5 m

C. 4.5 m

D. 9 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q3

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q3/5

1522

QUESTION ID

S43-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 4.5 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q3/5

True or false?

In triangle ABC, D is on AB and E is on AC, and DE is parallel to BC. If AD = 2, DB = 6, and AE = 3, then EC = 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q4

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q4/5

1523

QUESTION ID

S43-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q4

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q4/5

True or false?

In right triangle ABC with the right angle at C, and D the foot of the altitude from C to hypotenuse AB, $AC^2 = AD \cdot DB$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q5

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q5/5

1524

QUESTION ID

S43-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $AC^2 = AD \cdot AB$; it is CD^2 that equals $AD \cdot DB$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q5

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q5/5

SET 65 • 8.G.A.5 • GEOMETRY

Give an informal argument for the angle-angle criterion for similar triangles

MAIN	Grade 8 main question	**8.G.A.5**
BACKWARD 1	Use scale drawings: lengths change, angles stay the same	**7.G.A.1**
BACKWARD 2	Use tables of equivalent ratios	**6.RP.A.3.a**
FORWARD 1	Prove the triangle proportionality theorem with similar triangles	**HSG-SRT.B.4**
FORWARD 2	Explain why trigonometric ratios depend only on the angle (simila.	**HSG-SRT.C.6**

SET 65 • 8.G.A.5

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.5

Triangle ABC has angles of 40° at A and 75° at B. Triangle DEF has angles of 40° at D and 75° at E, and DE = 2 · AB.

Explain, using a dilation and rigid motions, why the triangles are similar.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 65 • MAIN • Q1/5

1527

QUESTION ID

S65-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

The triangles are similar.

GRADING NOTE REQUIRED WORK: Dilate ABC by a scale factor of 2, so the image of AB has the same length as DE. Rigid motions then move that image onto DE, with the third vertex on the same side as F. The 40° and 75° angles make the other two sides lie along rays DF and EF, which meet at only one point, so the third vertex lands on F. A dilation followed by rigid motions maps ABC onto DEF.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 65 • MAIN • Q1/5

Explain why two triangles with two pairs of equal angles must also have the third pair of angles equal.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 65 • MAIN • Q2/5

1528

QUESTION ID

S65-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

The third angles are equal.

GRADING NOTE REQUIRED WORK: In each triangle the three angles add to 180°, so each third angle equals 180° minus the same two measures. The third angles are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 65 • MAIN • Q2/5

Use a ruler and a protractor to draw two triangles of different sizes that each have angles of 50° and 60°. Measure their sides, and compare the ratios of corresponding sides.

Comparison: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 65 • MAIN • Q3/5

1529

QUESTION ID

S65-M-Q3

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: Two triangles of different sizes, each with angles of 50°, 60°, and 70°

Written: The ratios of corresponding sides are all about the same (equal up to measuring error), so the triangles are similar.

GRADING NOTE Grade the two drawings and the ratios computed from the student's own measurements; there is no preset size.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 65 • MAIN • Q3/5

Which argument shows that two triangles with two pairs of equal angles are similar?

A. Rotate one triangle 90°.

B. Dilate one so a pair of corresponding sides match, then use rigid motions; the equal angles force the third vertices to match.

C. Check that the triangles have equal areas.

D. Measure all three sides of each and check that they are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 65 • MAIN • Q4/5

1530

QUESTION ID

S65-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Dilate one so a pair of corresponding sides match, then use rigid motions; the equal angles force the third vertices to match.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 65 • MAIN • Q4/5

True or false?

Knowing that just one pair of angles is equal is enough to show two triangles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 65 • MAIN • Q5/5

1531

QUESTION ID

S65-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Two pairs of equal angles are needed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 65 • MAIN • Q5/5

SET 65 • 8.G.A.5

BACKWARD 1

Use scale drawings: lengths change, angles stay the same • Grade 7 • Geometry • 7.G.A.1

A triangle has angles of 30°, 60°, and 90°, and its longest side is 6 cm. A scale copy has a scale factor of 2.

What are the angles and the longest side of the copy?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 65 • BACKWARD 1 • Q1/5

1533

QUESTION ID

S65-B1-Q1

TYPE

Short answer

CORRECT ANSWER

30°, 60°, and 90°; the longest side is 12 cm.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 65 • BACKWARD 1 • Q1/5

In a scale copy, a 4 cm side becomes 10 cm.

What is the scale factor, and what does a 70° angle become?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 65 • BACKWARD 1 • Q2/5

1534

QUESTION ID

S65-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2.5; the angle is still 70°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 65 • BACKWARD 1 • Q2/5

A scale drawing of a triangle uses a scale factor of $\frac{1}{2}$.

Which statement is true?

A. Each side stays the same length.

B. Each angle is half as large.

C. Each angle doubles.

D. Each angle stays the same, and each side is half as long.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 65 • BACKWARD 1 • Q3/5

1535

QUESTION ID

S65-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Each angle stays the same, and each side is half as long.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 65 • BACKWARD 1 • Q3/5

True or false?

A scale copy of a triangle has the same angle measures as the original.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 65 • BACKWARD 1 • Q4/5

1536

QUESTION ID

S65-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 65 • BACKWARD 1 • Q4/5

True or false?

In a scale copy with a scale factor of 3, a side of 5 cm becomes 8 cm.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 65 • BACKWARD 1 • Q5/5

1537

QUESTION ID

S65-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It becomes 15 cm.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 65 • BACKWARD 1 • Q5/5

SET 65 • 8.G.A.5

BACKWARD 2

Use tables of equivalent ratios • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a

Complete the equivalent ratios.

3 : 5 = 6 : ___ = 9 : ___

Missing values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 65 • BACKWARD 2 • Q1/5

1539

QUESTION ID

S65-B2-Q1

TYPE

Short answer

CORRECT ANSWER

10 and 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 65 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Are 4 : 6 and 6 : 9 equivalent ratios? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 65 • BACKWARD 2 • Q2/5

1540

QUESTION ID

S65-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. Both equal 2 : 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 65 • BACKWARD 2 • Q2/5

Which ratio is equivalent to 8 : 12?

A. 3 : 2

B. 8 : 4

C. 2 : 3

D. 4 : 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 65 • BACKWARD 2 • Q3/5

1541

QUESTION ID

S65-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2 : 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 65 • BACKWARD 2 • Q3/5

True or false?

5 : 7 and 15 : 21 are equivalent ratios.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 65 • BACKWARD 2 • Q4/5

1542

QUESTION ID

S65-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 65 • BACKWARD 2 • Q4/5

True or false?

2 : 5 and 4 : 7 are equivalent ratios.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 65 • BACKWARD 2 • Q5/5

1543

QUESTION ID

S65-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 65 • BACKWARD 2 • Q5/5

SET 65 • 8.G.A.5

FORWARD 1

Prove the triangle proportionality theorem with similar triangles • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, D is on AB and E is on AC, and DE is parallel to BC.

Prove that AD ÷ AB = AE ÷ AC. Give a reason for each step.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 65 • FORWARD 1 • Q1/5

1545

QUESTION ID

S65-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

AD ÷ AB = AE ÷ AC

GRADING NOTE REQUIRED WORK: A valid chain of reasons. Other valid orders are accepted. A conclusion without reasons is not full credit. Example: angle ADE = angle ABC (corresponding angles, DE ∥ BC); angle A is shared; so triangle ADE is similar to triangle ABC (AA), and corresponding sides are proportional: AD ÷ AB = AE ÷ AC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F1-Q1**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 65 • FORWARD 1 • Q1/5

In triangle ABC, D is on AB and E is on AC, and DE is parallel to BC. AD = 4, DB = 6, and AE = 6.

Find EC.

EC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 65 • FORWARD 1 • Q2/5

1546

QUESTION ID

S65-F1-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F1-Q2**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 65 • FORWARD 1 • Q2/5

In triangle ABC, D is on AB and E is on AC, with AD ÷ DB = AE ÷ EC.

What does the converse of the triangle proportionality theorem say?

A. DE is parallel to BC.

B. D and E must be midpoints.

C. DE is perpendicular to BC.

D. DE is as long as BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S65-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 65 • FORWARD 1 • Q3/5

1547

QUESTION ID

S65-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. DE is parallel to BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S65-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 65 • FORWARD 1 • Q3/5

True or false?

The segment joining the midpoints of two sides of a triangle is parallel to the third side and half as long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 65 • FORWARD 1 • Q4/5

1548

QUESTION ID

S65-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 65 • FORWARD 1 • Q4/5

True or false?

If DE is parallel to BC in triangle ABC, with D on AB and E on AC, then DE = BC.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 65 • FORWARD 1 • Q5/5

1549

QUESTION ID

S65-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE DE is shorter: DE ÷ BC = AD ÷ AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 65 • FORWARD 1 • Q5/5

SET 65 • 8.G.A.5

FORWARD 2

Explain why trigonometric ratios depend only on the angle (similar right triangles) • High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

Two right triangles each have a 35° angle.

Explain why the ratio (leg opposite 35°) ÷ (hypotenuse) is the same in both.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q1

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 65 • FORWARD 2 • Q1/5

1551

QUESTION ID

S65-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Both triangles have a 90° angle and a 35° angle, so they are similar by AA, and corresponding sides of similar triangles have equal ratios.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F2-Q1**

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 65 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle with a 35° angle has a hypotenuse of 10 and a leg of 5.74 opposite the 35° angle. A larger right triangle with a 35° angle has a hypotenuse of 25.

How long is its leg opposite the 35° angle?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q2

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 65 • FORWARD 2 • Q2/5

1552

QUESTION ID

S65-F2-Q2

TYPE

Short answer

CORRECT ANSWER

14.35

GRADING NOTE 5.74 ÷ 10 = x ÷ 25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F2-Q2**

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 65 • FORWARD 2 • Q2/5

Why does sin 35° have only one value?

A. All right triangles are congruent.

B. Every hypotenuse has length 1.

C. The opposite leg is always 35 units long.

D. All right triangles with a 35° angle are similar (AA).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q3

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 65 • FORWARD 2 • Q3/5

1553

QUESTION ID

S65-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. All right triangles with a 35° angle are similar (AA).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q3

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 65 • FORWARD 2 • Q3/5

True or false?

In every right triangle with a 20° angle, the ratio (leg next to 20°) ÷ (hypotenuse) is the same.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q4

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 65 • FORWARD 2 • Q4/5

1554

QUESTION ID

S65-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q4

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 65 • FORWARD 2 • Q4/5

True or false?

Of two right triangles with a 50° angle, the one with the longer hypotenuse has the larger sine of 50°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q5

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 65 • FORWARD 2 • Q5/5

1555

QUESTION ID

S65-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The triangles are similar, so sin 50° is the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q5

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 65 • FORWARD 2 • Q5/5

SET 44 • 8.G.B.6 • GEOMETRY

Explain a proof of the Pythagorean theorem and its converse

MAIN	Grade 8 main question	**8.G.B.6**
BACKWARD 1	Find the area of a right triangle	**6.G.A.1**
BACKWARD 2	Find the area of a square	**3.MD.C.7.b**
FORWARD 1	Prove the Pythagorean theorem with similar triangles	**HSG-SRT.B.4**
FORWARD 2	Derive the equation of a circle from the Pythagorean theorem	**HSG-GPE.A.1**

SET 44 • 8.G.B.6

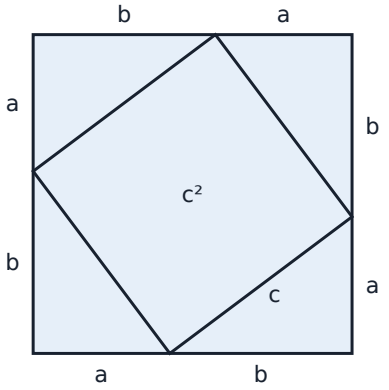
MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.B.6

ANSWER KEY Reduced copy of the question page, with its answer

The figure shows a square with sides a + b, made of four right triangles and a tilted square with side c.

Explain how the figure proves that $a^2 + b^2 = c^2$.



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q1/5

1558

QUESTION ID

S44-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

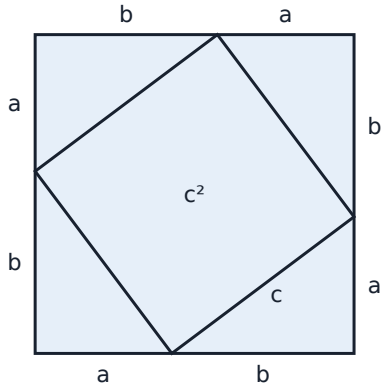
$a^2 + b^2 = c^2$

GRADING NOTE REQUIRED WORK: A complete argument: the large square's area is $(a + b)^2 = a^2 + 2ab + b^2$; it is also 4 triangles ($4 \cdot \frac{1}{2}ab = 2ab$) plus the inner square c^2; so $a^2 + 2ab + b^2 = 2ab + c^2$, and subtracting 2ab gives $a^2 + b^2 = c^2$. Any equivalent wording is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q1/5

In the figure, write the area of the large square as the sum of the areas of the four triangles and the inner square.



Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q2/5

1559

QUESTION ID

S44-M-Q2

TYPE

Short answer

CORRECT ANSWER

4 ·

$$\frac{1}{2}ab + c^2 = 2ab + c^2$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q2/5

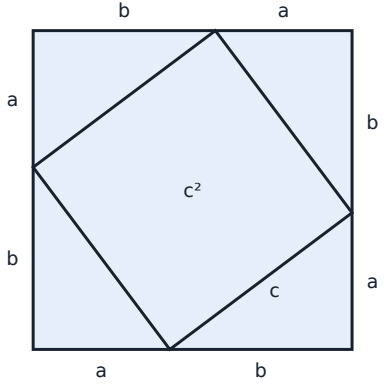
In the proof with the figure, which step comes after writing $(a + b)^2 = 2ab + c^2$?

A. Divide both sides by c

B. Expand to $a^2 + 2ab + b^2 = 2ab + c^2$, then subtract 2ab from both sides

C. Add 2ab to both sides

D. Take the square root of each term



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q3/5

1560

QUESTION ID

S44-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Expand to $a^2 + 2ab + b^2 = 2ab + c^2$, then subtract 2ab from both sides

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q3/5

A triangle has sides a, b, and c with $a^2 + b^2 = c^2$. Explain why it must be a right triangle.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q4/5

1561

QUESTION ID

S44-M-Q4

TYPE

Short answer

CORRECT ANSWER

A right triangle with legs a and b has a hypotenuse of $\sqrt{(a^2 + b^2)} = c$. So the two triangles have the same three sides and are congruent (SSS), and the given triangle has a right angle too.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q4/5

True or false?

The Pythagorean theorem $a^2 + b^2 = c^2$ is true for every triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q5/5

1562

QUESTION ID

S44-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is true only for right triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q5/5

SET 44 • 8.G.B.6

BACKWARD 1

Find the area of a right triangle • Grade 6 • Geometry • 6.G.A.1

A right triangle has legs of 3 and 4.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q1/5

1564

QUESTION ID

S44-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q1/5

A right triangle has legs of 5 and 12.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q2/5

1565

QUESTION ID

S44-B1-Q2

TYPE

Short answer

CORRECT ANSWER

30 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q2/5

What is the total area of four right triangles that each have legs of 2 and 6?

A. 24

B. 12

C. 48

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q3/5

1566

QUESTION ID

S44-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q3/5

True or false?

A right triangle with legs of 8 and 3 has an area of 12 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q4/5

1567

QUESTION ID

S44-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q4/5

True or false?

A right triangle with legs of 6 and 6 has an area of 36 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q5/5

1568

QUESTION ID

S44-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q5/5

SET 44 • 8.G.B.6

BACKWARD 2

Find the area of a square • Grade 3 • Measurement & Data • 3.MD.C.7.b

A square has sides of 7 units.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q1/5

1570

QUESTION ID

S44-B2-Q1

TYPE

Short answer

CORRECT ANSWER

49 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q1/5

A square has sides of 9 units.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q2/5

1571

QUESTION ID

S44-B2-Q2

TYPE

Short answer

CORRECT ANSWER

81 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q2/5

A square has sides of 5 units.

What is its area?

A. 15 square units

B. 25 square units

C. 10 square units

D. 20 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q3/5

1572

QUESTION ID

S44-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 25 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q3/5

True or false?

A square with 4-unit sides has an area of 16 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q4/5

1573

QUESTION ID

S44-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q4/5

True or false?

A square with 6-unit sides has an area of 24 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q5/5

1574

QUESTION ID

S44-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q5/5

SET 44 • 8.G.B.6

FORWARD 1

Prove the Pythagorean theorem with similar triangles • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

In right triangle ABC with the right angle at C, altitude CD is drawn to hypotenuse AB.

Why is $AC^2 = AD \cdot AB$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q1/5

1576

QUESTION ID

S44-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Triangle ACD is similar to triangle ABC (AA), so $AC/AB = AD/AC$, which gives $AC^2 = AD \cdot AB$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q1/5

In right triangle ABC, the right angle is at C, and D is the foot of the altitude from C to hypotenuse AB. Similar triangles give $AC^2 = AD \cdot AB$ and $BC^2 = BD \cdot AB$.

Add these two equations. What do you get?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q2/5

1577

QUESTION ID

S44-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$AC^2 + BC^2 = AD \cdot AB + BD \cdot AB = AB(AD + BD) = AB \cdot AB = AB^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q2/5

In right triangle ABC, the right angle is at C, and D is the foot of the altitude from C to hypotenuse AB.

Which criterion shows that triangle ACD is similar to triangle ABC?

A. ASA

B. AA

C. SAS

D. SSS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q3/5

1578

QUESTION ID

S44-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. AA

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F1-Q3**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q3/5

True or false?

The altitude to the hypotenuse divides a right triangle into two triangles similar to it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q4/5

1579

QUESTION ID

S44-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q4/5

True or false?

The altitude to the hypotenuse always cuts the hypotenuse into two equal parts.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q5/5

1580

QUESTION ID

S44-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q5/5

SET 44 • 8.G.B.6

FORWARD 2

Derive the equation of a circle from the Pythagorean theorem • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

Write the equation of the circle with center (0, 0) and radius 5.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q1/5

1582

QUESTION ID

S44-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 + y^2 = 25$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F2-Q1**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A point (x, y) is on the circle with center (2, −1) and radius 3. Use a right triangle with legs |x − 2| and |y + 1| and the Pythagorean theorem to derive the equation of the circle.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q2/5

1583

QUESTION ID

S44-F2-Q2

TYPE

Answer with required work

CORRECT ANSWER

$(x - 2)^2 + (y + 1)^2 = 9$

GRADING NOTE REQUIRED WORK: The horizontal leg is |x − 2|, the vertical leg is |y + 1|, and the hypotenuse is the radius 3, so $(x - 2)^2 + (y + 1)^2 = 3^2 = 9$. The right triangle and the use of the Pythagorean theorem must be shown.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q2/5

Which point is on the circle $x^2 + y^2 = 100$?

A. (5, 5)

B. (10, 10)

C. (6, 8)

D. (6, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q3/5

1584

QUESTION ID

S44-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (6, 8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q3/5

True or false?

The equation $(x - h)^2 + (y - k)^2 = r^2$ comes from the Pythagorean theorem.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q4/5

1585

QUESTION ID

S44-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q4/5

True or false?

The circle $x^2 + y^2 = 16$ has a radius of 16.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q5/5

1586

QUESTION ID
S44-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Its radius is 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q5/5

SET 45 • 8.G.B.6 • GEOMETRY

Use the converse of the Pythagorean theorem to decide whether a triangle is a right triangle

MAIN	Grade 8 main question	**8.G.B.6**
BACKWARD 1	Find the squares of whole numbers	**6.EE.A.1**
BACKWARD 2	Evaluate an expression with exponents	**6.EE.A.2.c**
FORWARD 1	Prove the converse of the Pythagorean theorem with congruent tri	**HSG-CO.C.10**
FORWARD 2	Use coordinates to show that a triangle is a right triangle	**HSG-GPE.B.4**

SET 45 • 8.G.B.6

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.B.6

Is a triangle with sides 9, 12, and 15 a right triangle? Show your check.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q1/5

1589

QUESTION ID

S45-M-Q1

TYPE

Short answer

CORRECT ANSWER

Yes.

$9^2 + 12^2 = 81 + 144 = 225 = 15^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q1/5

Is a triangle with sides 7, 8, and 11 a right triangle? Show your check.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q2/5

1590

QUESTION ID

S45-M-Q2

TYPE

Short answer

CORRECT ANSWER

No. $7^2 + 8^2 = 49 + 64 = 113$, but $11^2 = 121$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q2/5

Which side lengths form a right triangle?

A. 6, 8, 12

B. 20, 21, 29

C. 5, 7, 9

D. 10, 12, 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q3/5

1591

QUESTION ID

S45-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 20, 21, 29

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q3/5

A carpenter checks a corner. Points 3 feet and 4 feet from the corner along the two edges are 5 feet apart.

Is the corner a right angle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q4/5

1592

QUESTION ID

S45-M-Q4

TYPE

Short answer

CORRECT ANSWER

Yes. $3^2 + 4^2 = 5^2$, so the angle is 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q4/5

True or false?

A triangle with sides 2, 3, and 4 is a right triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q5/5

1593

QUESTION ID

S45-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $4 + 9 = 13$, but $4^2 = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q5/5

SET 45 • 8.G.B.6

BACKWARD 1

Find the squares of whole numbers • Grade 6 • Expressions & Equations • 6.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Find the value.

15^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q1/5

1595

QUESTION ID

S45-B1-Q1

TYPE

Short answer

CORRECT ANSWER

225

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q1/5

Find the value.

11^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q2/5

1596

QUESTION ID

S45-B1-Q2

TYPE

Short answer

CORRECT ANSWER

121

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q2/5

Find the value.

13^2

A. 26

B. 169

C. 139

D. 196

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q3/5

1597

QUESTION ID

S45-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 169

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$20^2 = 400$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q4/5

1598

QUESTION ID

S45-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q4/5

True or false?

$9^2 = 18$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q5/5

1599

QUESTION ID

S45-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q5/5

SET 45 • 8.G.B.6

BACKWARD 2

Evaluate an expression with exponents • Grade 6 • Expressions & Equations • 6.EE.A.2.c